EXAM PACK

EEE337

Power System II



Exam Pack

Session 2021–22 *Solutions and formula sheet of the Final Examination 2024 and Term Tests 1–3*

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Question Paper Solutions 2021–22 — pages 3--81 — 79 pages

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Prepared by Nur Mohammad

October 2026

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Question Paper Solutions 2021–22

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PAST PAPER SOLUTIONS

EEE337

Power System II

Question Paper Solutions

Session 2021–22

Final Examination 2024 (Q1–Q6) and Term Tests 1–3 (Q7–Q9)

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

Q7	Q8	Q9
5 marks	10 marks	10 marks

EACH ANSWER



Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks); a part with its own time (a term test) uses that. Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part-A · Answer any TWO of the following questions				
Q1	(a) Critical Disruptive Voltage	2	4 min	☐
	(b) Locating a Cable Fault by the Murray Loop Test	5	9 min	
	(c) Inductance of a Three-Phase Line with Unsymmetrical Spacing	8	14 min	
	(d) Mutual Inductance between a Power Line and a Telephone Line	10	18 min	
Q2	(a) Monopolar HVDC Link	2	4 min	☐
	(b) Construction of a 3-Core Cable	5	9 min	
	(c) Electric Stress, Capacitance and Charging Current of a Single-Core Cable	8	14 min	
	(d) Grading of Cables and the Three-Layer Capacitance Grading	10	18 min	
Q3	(a) Electrical Breakdown of a Pin-Type Insulator	2	4 min	☐
	(b) Line Supports for Low-Voltage Distribution	5	9 min	
	(c) Voltage Distribution and Efficiency of a Three-Unit String	8	14 min	
	(d) Derivation of String Efficiency and the Effect of K and the Number of Discs	10	18 min	
Part-B · Answer any TWO of the following questions				
Q4	(a) Stringing Charts	2	4 min	☐
	(b) Insulation Thickness from the Insulation Resistance	5	9 min	
	(c) Inductance of Composite Conductor Lines	8	14 min	
	(d) Maximum Stress in a Cable Graded with Two Dielectrics	10	18 min	
Q5	(a) Effect of the Earth on Line Capacitance	2	4 min	☐
	(b) Capacitance to Neutral and Charging Current of a Flat-Spaced Line	5	9 min	
	(c) Three-Disc String from Measured Unit Voltages	8	14 min	
	(d) Capacitance of a Three-Phase Line with Unsymmetrical Spacing	10	18 min	
Q6	(a) Most Economical Size of a Single-Core Cable	2	4 min	☐
	(b) HVDC and the Economics of Power Transmission	5	9 min	
	(c) Sudden Increase in Mechanical Input on the Power-Angle Diagram	8	14 min	
	(d) Transient EMF and Power-Angle Curves for Three Network Conditions	10	18 min	
Term Test 1 · 10 marks · 20 min				
Q7	(a) Inductive Reactance of a Bundled-Conductor Line	5	10 min	☐
	(b) Capacitance with Unsymmetrical Spacing (Same as 5(d))			
Term Test 2 · 10 marks · 20 min				
Q8	(a) Voltage Distribution on a String of Five Insulators	5	10 min	☐
	(b) The Same Single-Core Cable as 2(c)	5	10 min	
Term Test 3 · 10 marks · 20 min				
Q9	(a) Murray Loop Test (Same as 1(b))	5	10 min	☐
	(b) Mutual Inductance when Both Lines Are in One Plane	5	10 min	

Formulas and constants follow V.K. Mehta, Principles of Power System (line constants, corona, insulators, sag, cables, economics) and Grainger and Stevenson, Power System Analysis (stability).

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

B.Sc. (Engg.) 3rd Year, 2nd Semester Final Examination-2024

Course Code: EEE 337 **Session:** 2021-2022

Course Title: Power System II

Total Marks: 100 **Credits: 3.0** **Total Time: 3 Hours**

[The figures in the right margin indicate full marks]

Part-A

*[Answer any **TWO** of the following questions]*

1. (a) Define critical disruptive voltage. Write down it's significance. 2

   (b) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio of 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. 5

   (c) Define transpose transmission line. Derive the expression for the inductance of three phase transmission lines with unsymmetrical spacing. 8

   (d) A single-phase 50 Hz power line is supported on a horizontal cross-arm. The spacing between the conductors is 3 m. A telephone line is supported symmetrically below the power line as shown in Fig. 1. Find the mutual inductance between the two circuits and the voltage induced per kilometer in the telephone line if the current in the power line is 100 A. Assume the telephone line current to be zero. 10

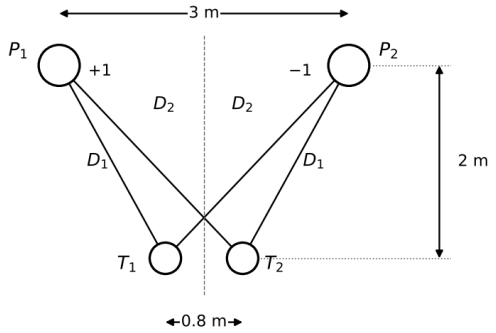


Fig. 1 of the paper (redrawn): figure for Q-1(d)

2. (a) What is monopolar link? 2

   (b) Write down the general construction of a 3-conductor cable with required figure. 5

   (c) A single core cable for use on 11 kV, 50 Hz system has conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find: (i) the maximum electrostatic stress in the cable, (ii) minimum electrostatic stress in the cable, (iii) capacitance of the cable per km length, and (iv) charging current. 8

   (d) Define grading of cables. Why is unequal stress distribution in a cable undesirable? Consider three dielectric layers for capacitance grading and derive the expression of potential difference between core and earthed sheath. 10

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3. (a) How electrical breakdown happen in a pin-type insulator? 2

   (b) What kind of line supports are used for distribution purposes at low voltages (upto 11 kV)? 5

   (c) In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) String efficiency. 8

   (d) Derive the mathematical expression of String efficiency with proper diagram. What will happen if:

   (i) $K = 0.2$ ($K = \frac{shunt\ capacitance}{self\ capacitance}$)
   (ii) $K = 0.8$
   (iii) the number of disc increases in the string. 10

Part-B

*[Answer any **TWO** of the following questions]*

4. (a) What is Stringing charts? 2

   (b) The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2.5 cm and the resistivity of insulation is $4.5 \times 10^{14}\ \Omega - cm$, find the insulation thickness. 5

   (c) Derive the expression of inductance of composite conductor lines. 8

   (d) A 66 kV single-core lead sheathed cable is graded by using two dielectrics of relative permittivity 5 and 3 respectively; thickness of each being 1 cm. The core diameter is 2 cm. Determine the maximum stress in the two dielectrics. 10

5. (a) How Earth affect on transmission line capacitance? 2

   (b) A three phase 50 Hz transmission line has flat horizontal spacing with 3.5 m between adjacent conductors. The conductors are No. 2/0 hard/drawn seven strand copper (outside conductor diameter= 1.05 cm). The voltage of the line is 110 kV. Find the capacitance to neutral and the charging current per kilometre of line. 5

   (c) A 3-phase transmission line is being supported by three-disc insulators. The potentials across top unit (near to the tower) and middle unit are 8 kV and 11 kV, respectively.

   (i) Draw the equivalent circuit
   (ii) Calculate the ratio of capacitance between pin and earth to the self-capacitance of each unit
   (iii) Calculate the line voltage and
   (iv) Calculate string efficiency. 8

   (d) Derive the expression of capacitance of a three phase line with unsymmetrical spacing. 10

6. (a) Find the most economical size of a single-core cable working on a 132 kV, 3- phase system, if a dielectric stress of 60 kV/cm can be allowed. 2

   (b) Define HVDC. Discuss the Economics of Power transmission. 5

   (c) Explain what happens for sudden increase in mechanical input to generator using $Pe - \delta$ diagram. 8

   (d) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus ($|V| = 1.0\ pu$), with the generator terminal voltage ($|V_t| = 1.0\ pu$). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (i) System healthy, (ii) One line shorted (3 phase) in the middle and (iii) One line open. Plot all the three power angle curves. 10

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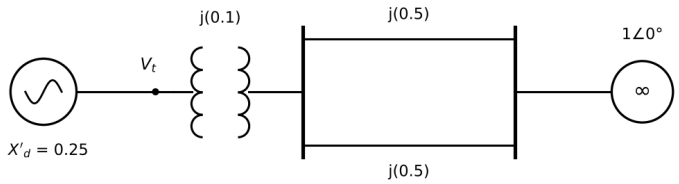


Fig. 2 of the paper (redrawn): figure for Q-6(d)

Term Test 1 (its questions 1 and 2 are 7(a) and 7(b))

[Term Test 1, Total Marks: 10, Time: 20 min]

7. (a) A 500 kV line has a bundling arrangement of 2 conductors per phase as shown in the Figure 1. Compute the inductive reactance per phase of this line at 50 Hz. Each conductor carries 50% of the phase current. 5

   (b) Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing.

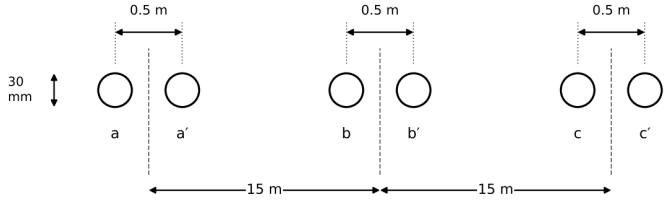


Figure 1 of Term Test 1 (redrawn)

Term Test 2 (its questions 1 and 2 are 8(a) and 8(b))

[Term Test 2, Total Marks: 10, Time: 20 min]

8. (a) A string of 5 insulators is connected across a 100 kV line. If the capacitance of each disc to earth is 0.1 of the capacitance of the insulator, calculate (i) the distribution of voltage on the insulator discs and (ii) the string efficiency. 5

   (b) A single core cable for use on 11 kV, 50 Hz system has a conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find (i) the maximum electrostatic stress in the cable (ii) the minimum electrostatic stress in the cable (iii) the capacitance of the cable per km length (iv) the charging current. 5

Term Test 3 (its questions 1 and 2 are 9(a) and 9(b))

[Term Test 3, Total Marks: 10, Time: 20 min]

9. (a) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. 5

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(b) A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. 5

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1. (a) Define critical disruptive voltage. Write down it's significance. Marks: 2

   (b) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio of 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. Marks: 5

   (c) Define transpose transmission line. Derive the expression for the inductance of three phase transmission lines with unsymmetrical spacing. Marks: 8

   (d) A single-phase 50 Hz power line is supported on a horizontal cross-arm. The spacing between the conductors is 3 m. A telephone line is supported symmetrically below the power line as shown in Fig. 1. Find the mutual inductance between the two circuits and the voltage induced per kilometer in the telephone line if the current in the power line is 100 A. Assume the telephone line current to be zero. Marks: 10

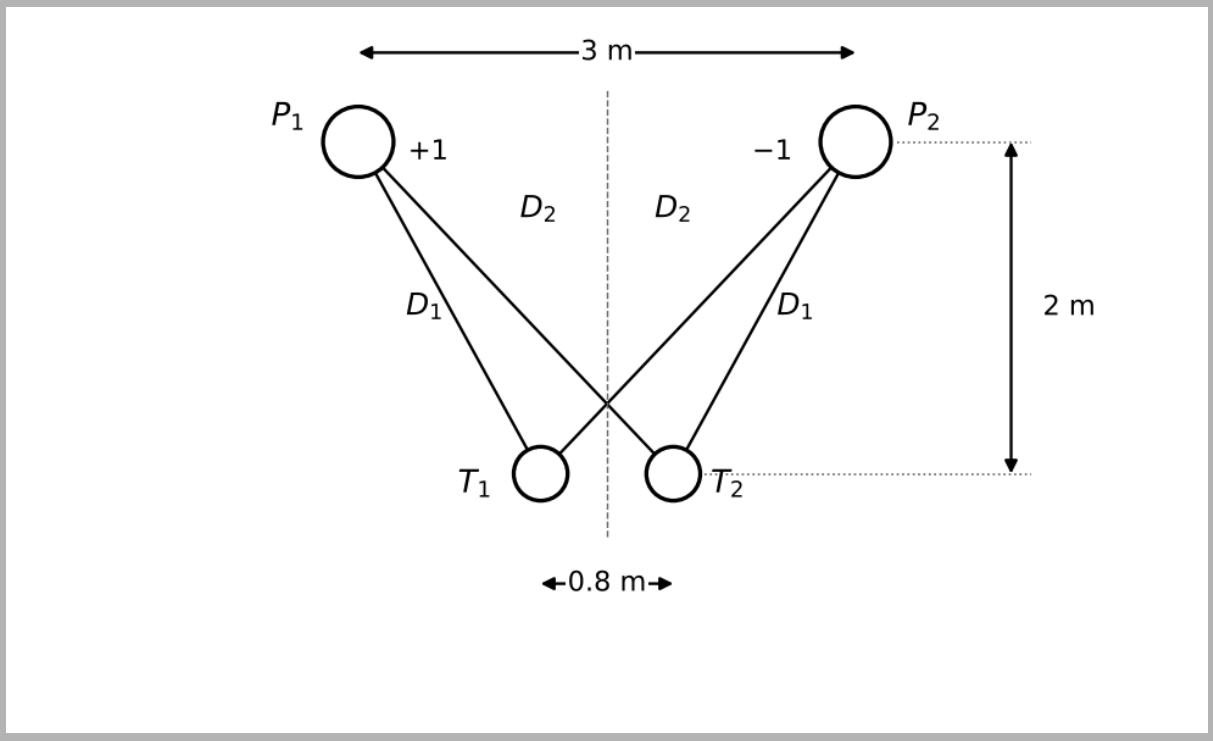


Figure 1: Fig. 1 of the paper (redrawn): figure for Q-1(d)

1.(a) Define critical disruptive voltage. Write down it's significance. Marks: 2

1. Theory

1. What is around an overhead conductor? (Core Idea)

An overhead transmission line is a set of bare metal wires (conductors) hung from towers. There is no rubber or plastic around them. The only insulation between one wire and the next, and between a wire and the earth, is the **air** around it.

When a wire is at a high voltage, it creates an **electric field** in the air around it. The strength of this field is called the **potential gradient** g. It tells us how many volts the voltage drops across each centimetre of air, so its unit is kV/cm.

2. Where is the field strongest?

Field lines start on the wire and spread outwards, like the spokes of a wheel. Close to the wire they are crowded together, so the field is strong. Far away they are spread out, so the field is weak.

For a line with two wires of radius r placed d apart, with voltage V between each wire and the neutral (the middle), the field at a distance x from the centre of one wire is

$$g = \frac{V}{x \ln(d/r)}$$

where

- V is the voltage of the wire to neutral (kV)
- x is the distance from the centre of the wire (cm), with $x \geq r$
- d is the spacing between the wires and r the wire radius (same unit)

The field falls as $1/x$. It is largest at the surface of the wire, where $x = r$:

$$g_{max} = \frac{V}{r \ln(d/r)}$$

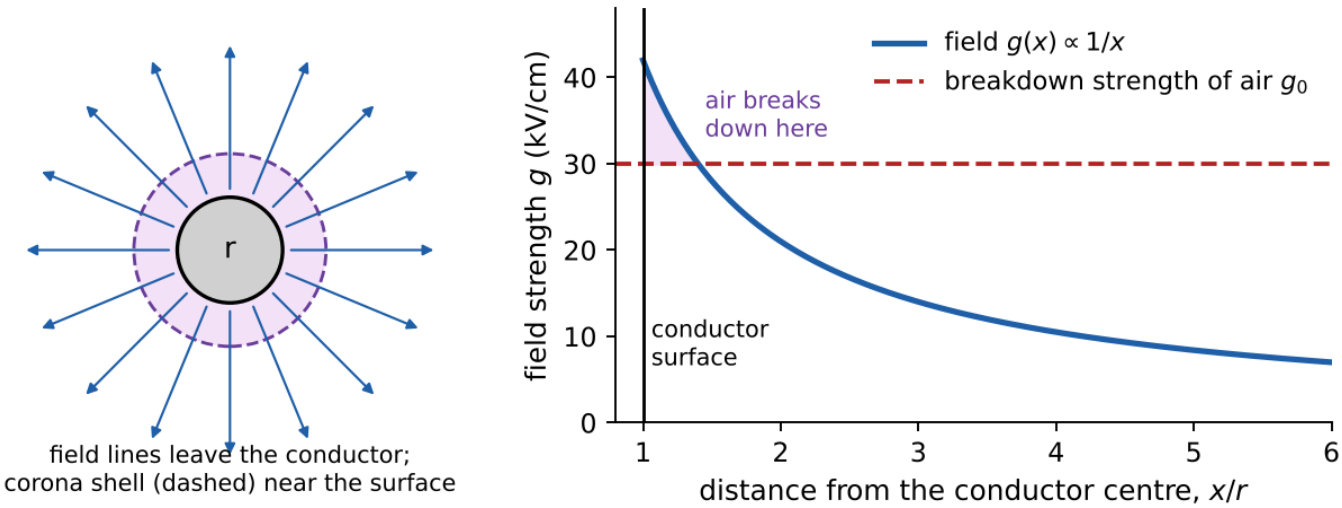


Figure 2: Left: field lines spread out from a conductor, with the thin corona shell near the surface. Right: the field falls as 1/x (solid blue); air breaks down only where it is above the dashed red line g0

Reading the figure. On the left, the arrows are field lines leaving the conductor; the dashed ring is the thin shell of air close to the surface. On the right, the solid blue curve is the field strength. It starts high at the surface ($x/r = 1$) and drops quickly. The dashed red line is the strength that air can stand. Only the small shaded part, very close to the wire, is above that line.

3. When does air stop insulating?

Air always contains a few free electrons. A strong electric field pushes these electrons so hard that they hit air molecules and knock more electrons out of them. The new electrons do the same, and the number grows like an avalanche. The air then becomes a conductor. This is called **ionisation** or **breakdown**.

The field at which this happens is the **dielectric strength of air** g_0. At normal pressure (76 cm of mercury) and 25 °C:

$$g_0 = 30 \text{ kV/cm (peak)} = 21.2 \text{ kV/cm (r.m.s.)}$$

4. What is corona?

When the field at the surface of the wire goes above g_0, the thin shell of air around the wire breaks down. We then see a faint violet glow around the wire, hear a hissing sound and smell ozone. This is **corona**. The breakdown does not spread to the other wire, because a little further out the field is again below g_0 (the figure above).

5. The critical disruptive voltage

Corona starts at the voltage that just makes $g_{max} = g_0$. Putting $g_{max} = g_0$ in the formula above gives this voltage:

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$$V_c = g_0\, r\, \ln \frac{d}{r}$$

Real air and real wires are not ideal, so two correction factors are added (V.K. Mehta):

$$V_c = m_0\, g_0\, \delta\, r\, \ln \frac{d}{r} \text{ kV (phase to neutral)}$$

where

- $g_0 = 21.2$ kV/cm r.m.s. (V.K. Mehta rounds it to 21.1)
- $\delta = \dfrac{3.92\, b}{273 + t}$ is the **air density factor**, with b the pressure in cm of mercury and t the temperature in °C ($\delta = 1$ at 76 cm and 25 °C). Thin air (hills, hot days) breaks down more easily, so $\delta < 1$ lowers V_c.
- m_0 is the **irregularity factor**: 1 for a smooth polished wire, 0.98 to 0.92 for a rough wire, 0.87 to 0.80 for a stranded conductor. Sharp points on the surface raise the local field.
- r and d are in cm.

Unit check: kV/cm × cm = kV, and $\ln(d/r)$ has no unit.

2. Answer

Definition: The **critical disruptive voltage** V_c is the minimum phase-to-neutral voltage of a line at which the electric field at the conductor surface becomes equal to the breakdown strength of air ($g_0 \approx 30$ kV/cm peak, 21.2 kV/cm r.m.s.), so that the air around the conductor starts to ionise and **corona** begins:

$$V_c = m_0\, g_0\, \delta\, r\, \ln \frac{d}{r} \text{ kV}$$

with m_0 the irregularity factor, δ the air density factor, r the conductor radius and d the spacing.

Significance:

- Below V_c there is no corona. Above it, the line has **corona loss** (wasted power), **radio interference**, hissing noise and **ozone**, which corrodes the conductor.
- The line is designed so that its working voltage stays **below** V_c in fair weather. V_c therefore fixes the minimum conductor diameter and spacing for a given voltage; for this reason high-voltage lines use large or bundled conductors.

3. Keyword Explanation

1. Potential gradient

The electric field strength in the air, in kV/cm: how fast the voltage changes with distance.

2. Dielectric strength of air

The largest field that air can stand without breaking down: about 30 kV/cm peak at normal pressure and temperature.

3. Corona

The partial breakdown of the air very close to a high-voltage conductor, seen as a violet glow with a hissing sound.

4. Notes and Remarks

- **Visual critical voltage:** the glow becomes visible at a slightly higher voltage, $V_v = m_v\, g_0\, \delta\, r \left(1 + \dfrac{0.3}{\sqrt{\delta r}}\right) \ln \dfrac{d}{r}$ kV. V_c is where corona starts; V_v is where it can be seen.
- **How designers raise V_c:** a larger radius r (or a hollow conductor), wider spacing d, or **bundled conductors** (two or more sub-conductors per phase, see 7(a)), which act like one conductor of a

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much larger radius.

1.(b) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio of 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. Marks: 5

1. Theory

1. Why do we need a loop test? (Core Idea)

An underground cable is buried, so we cannot see where it is damaged. If the insulation breaks at one point, current leaks from the core to the earth there. This is a **fault to earth**. To repair it we must dig at exactly the right place, so we measure the distance of the fault from one end. The **Murray loop test** does this with a simple bridge circuit.

2. The Wheatstone bridge idea

Take four resistors arranged in a square, with a battery across one diagonal and a sensitive current meter (a **galvanometer**, G) across the other. Change the resistors until G reads zero. Then the two ends of G are at the same voltage, and this can only happen when the resistors are in proportion:

$$\frac{P}{Q} = \frac{R}{S}$$

where P and Q are the two resistors on one side of G, and R and S the two on the other side. If we know P, Q and R, we can find S.

3. The Murray loop: the cable becomes two arms of the bridge

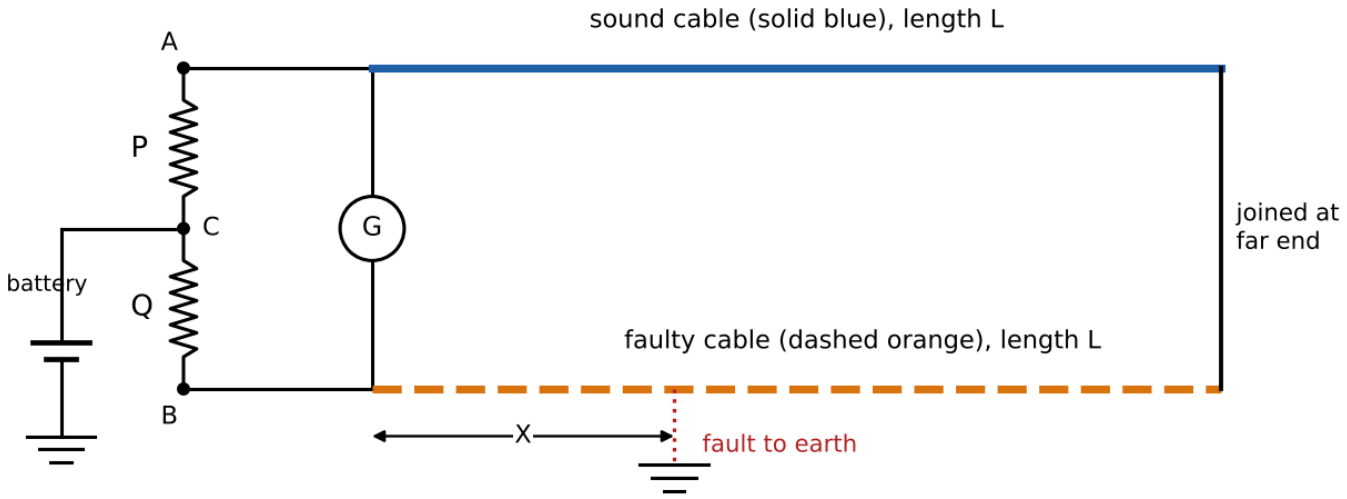


Figure 3: Murray loop test: ratio arms P and Q, the galvanometer G across the two cable ends, the battery from C to earth, and the sound and faulty cables joined at the far end

Reading the figure. The faulty cable (dashed orange) has a fault to earth at distance X from the test end. A good (sound) cable (solid blue) is laid beside it, and the two are joined at the far end, so they form one long **loop**. The ends of the loop, A and B, go to the two adjustable resistors P and Q. The battery is connected between their junction C and the earth, and G between A and B.

The battery current goes through P and Q, along the cables to the fault, and back through the earth. So the fault point (earth) is the fourth corner of the bridge. The four arms are:

- arm C–A: P

- arm C–B: Q
- arm A–fault: the whole sound cable plus the faulty cable from the far end back to the fault, $R_{loop} - R_X$
- arm B–fault: the faulty cable from the test end to the fault, R_X

Here R_{loop} is the resistance of the whole loop (sound cable + faulty cable), and R_X is the resistance of the faulty cable from the test end up to the fault.

4. The balance condition

When G reads zero:

$$\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X}$$

Solving for R_X:

$$R_X = \frac{Q}{P+Q} R_{loop}$$

If both cables have the same resistance per metre, resistance is proportional to length and $X = \frac{Q}{P+Q} \times 2L$. When the two cables are **different** (as in this question), work in ohms first and convert R_X to a length at the end, using the resistance per metre of the **faulty** cable.
The fault resistance itself sits in the battery branch, so it does not upset the balance.

2. Answer

Given:

- Length of each cable: $L = 500$ m
- Faulty cable: 1 Ω per 1000 m = 0.001 Ω/m
- Sound cable: 2.25 Ω per 1000 m = 0.00225 Ω/m
- Ratio arms at balance: $P : Q = 2.75 : 1$, where P is the arm joined to the sound cable and Q the arm joined to the faulty cable

Relationship: at balance

$$\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X} \quad \Rightarrow \quad R_X = \frac{R_{loop}}{P/Q + 1}$$

Step 1: Resistance of the loop

$$R_{faulty} = 0.001 \times 500 = 0.5\ \Omega, \qquad R_{sound} = 0.00225 \times 500 = 1.125\ \Omega$$

$$R_{loop} = 0.5 + 1.125 = 1.625\ \Omega$$

Step 2: Resistance up to the fault

$$R_X = \frac{1.625}{2.75 + 1} = \frac{1.625}{3.75} = 0.4333\ \Omega$$

$$\boxed{R_X = 0.4333\ \Omega}$$

This is the resistance of the faulty cable between the testing end and the fault.

Step 3: Distance of the fault

The faulty cable has 0.001 Ω per metre, so

$$X = \frac{0.4333}{0.001} = 433.3\ \text{m}$$

$$\boxed{X = 433.3\ \text{m from the testing end}}$$

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The fault is 433.3 m along the 500 m faulty cable, which is a possible place, so the answer is sensible. **Final Answer:** the fault is **433.3 m** from the testing end.

3. Notes and Remarks

- **Which arm is 2.75?** If we took the ratio the other way round ($P : Q = 1 : 2.75$), we would get $R_X = 1.625 \times 2.75/3.75 = 1.192\ \Omega$, which is 1192 m: longer than the 500 m cable, so it is impossible. This confirms the larger arm is on the sound-cable side.
- **Same resistance per metre?** Then lengths can be used directly: $X = \dfrac{Q}{P+Q} \times 2L$. That shortcut is wrong here because the two cables differ.
- **Varley loop test** is a related method in which a known variable resistor is put in series with the faulty cable; it can also measure the total loop resistance.

1.(c) Define transpose transmission line. Derive the expression for the inductance of three phase transmission lines with unsymmetrical spacing. Marks: 8

1. Theory

1. What is inductance? (Core Idea)

A current in a wire makes a **magnetic field** around it, in circles around the wire. The circles of field that surround a wire are its **flux**. When the current changes (as alternating current does all the time), the flux changes, and a changing flux makes a voltage that opposes the change. **Inductance** measures how much flux is linked per ampere:

$$L = \frac{\psi}{I}$$

where ψ is the **flux linkage** (flux $\times$ the number of times it surrounds the current, in weber-turns) and I the current. For a line we give L per metre (H/m) or per km (mH/km). The line's inductive reactance is $X_L = 2\pi f L$, and it causes a voltage drop IX_L along the line.

2. Flux linkage of one long round conductor

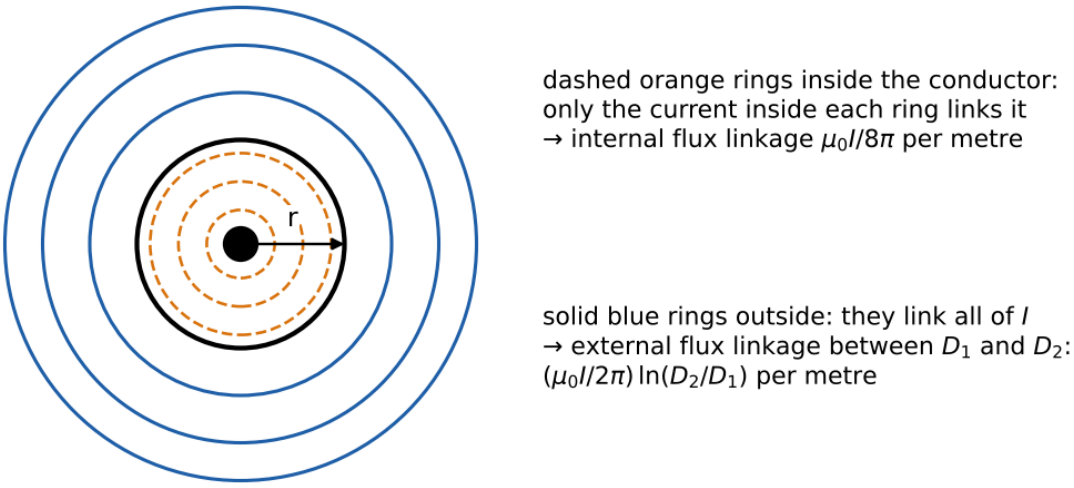


Figure 4: Flux of one conductor carrying current I out of the page: dashed orange rings inside the conductor, solid blue rings outside

Reading the figure. The current flows out of the page (the dot). Rings inside the conductor (dashed orange) surround only part of the current; rings outside (solid blue) surround all of it.

- **Inside** the conductor, a ring of radius x surrounds only the current inside it, Ix^2/r^2 (current spread evenly over the cross-section). Adding up all these partial linkages gives a simple result:

$$\psi_{int} = \frac{\mu_0 I}{8\pi} \text{ Wb-T/m}$$

- **Outside**, the field at distance x is $H = \dfrac{I}{2\pi x}$ and every ring links the full current. The linkage due to the rings between distances D_1 and D_2 from the centre is

$$\psi_{12} = \frac{\mu_0 I}{2\pi} \ln \frac{D_2}{D_1} \text{ Wb-T/m}$$

Here $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of air, so $\dfrac{\mu_0}{2\pi} = 2 \times 10^{-7}$.

3. The geometric mean radius (GMR)

The total linkage of a conductor up to a distance D is

$$\psi = \frac{\mu_0 I}{2\pi}\left(\frac{1}{4} + \ln\frac{D}{r}\right) = \frac{\mu_0 I}{2\pi} \ln \frac{D}{re^{-1/4}} = 2 \times 10^{-7}\, I \ln \frac{D}{r'}$$

where

$$r' = r\, e^{-1/4} = 0.7788\, r$$

r' is the **geometric mean radius** (GMR). It is the radius of an imaginary thin tube with no internal flux that has the same inductance as the solid wire. With r' we no longer have to think about the inside of the wire.

4. Flux linkage of one conductor in a group

In a three-phase line, conductor a is linked by its own flux and also by the flux of b and c. Using the result above for each current (and the fact that $I_a + I_b + I_c = 0$, which removes the far-away reference point):

$$\psi_a = 2 \times 10^{-7}\left[I_a \ln \frac{1}{r'} + I_b \ln \frac{1}{D_{ab}} + I_c \ln \frac{1}{D_{ac}}\right]$$

where D_{ab} and D_{ac} are the distances from a to b and from a to c.

Symmetrical spacing. If the three conductors sit at the corners of an equilateral triangle of side D, then $D_{ab} = D_{ac} = D$ and $I_b + I_c = -I_a$, so $\psi_a = 2 \times 10^{-7} I_a \ln(D/r')$ and

$$L = 2 \times 10^{-7} \ln \frac{D}{r'} \text{ H/m}$$

The same for all three phases.

5. The problem with unsymmetrical spacing, and transposition

In most lines the conductors are not at equal distances (for example, side by side on a cross-arm). Then each phase sees different distances, so the phases have **different inductances**. Their voltage drops differ, the receiving-end voltages become unbalanced, and the line also induces an unbalanced voltage in nearby telephone lines (see 1(d)).

The cure is **transposition**: the conductors swap positions at regular intervals so that, over one cycle, each conductor occupies each position for one third of the length.

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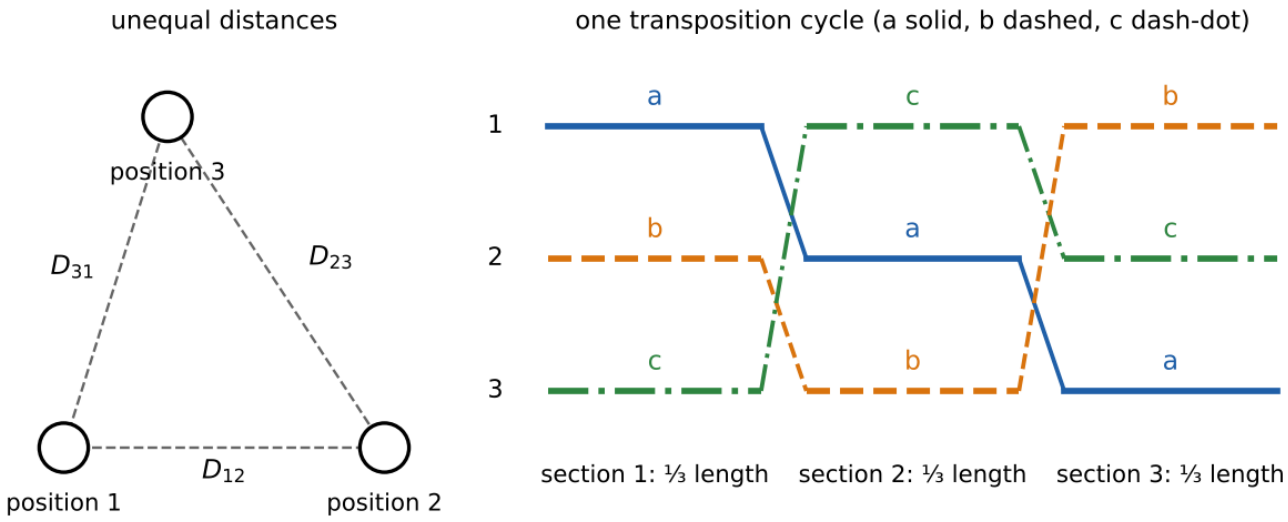


Figure 5: Left: three positions at unequal distances D12, D23, D31. Right: one transposition cycle; conductor a (solid), b (dashed) and c (dash-dot) each take every position for one third of the length

Reading the figure. On the left are the three fixed positions on the towers. On the right, follow conductor a (solid blue): it is at position 1 in section 1, at position 2 in section 2 and at position 3 in section 3. Conductors b (dashed orange) and c (dash-dot green) also visit every position once. So, on average, all three conductors see the same surroundings.

2. Answer

Transposed line: a three-phase line in which the conductors exchange their positions at regular intervals (at transposition towers), so that over one complete cycle each conductor occupies each of the three positions for one third of the length. Its purpose is to make the average inductance (and capacitance) of the three phases equal, so that the phases stay balanced and interference with telephone lines is reduced.

Derivation of the inductance with unsymmetrical spacing (transposed line):

Let the three positions be 1, 2, 3 with distances D_{12}, D_{23} and D_{31} (figure above), conductor radius r and GMR $r' = 0.7788\,r$. The currents are balanced: $I_a + I_b + I_c = 0$. The flux linkage of a conductor is (Theory, item 4)

$$\psi_a = 2 \times 10^{-7} \left[I_a \ln \frac{1}{r'} + I_b \ln \frac{1}{D_{ab}} + I_c \ln \frac{1}{D_{ac}} \right] \quad \text{Wb-T/m}$$

Section 1 (a at 1, b at 2, c at 3):

$$\psi_{a1} = 2 \times 10^{-7} \left[I_a \ln \frac{1}{r'} + I_b \ln \frac{1}{D_{12}} + I_c \ln \frac{1}{D_{31}} \right]$$

Section 2 (a at 2, b at 3, c at 1):

$$\psi_{a2} = 2 \times 10^{-7} \left[I_a \ln \frac{1}{r'} + I_b \ln \frac{1}{D_{23}} + I_c \ln \frac{1}{D_{12}} \right]$$

Section 3 (a at 3, b at 1, c at 2):

$$\psi_{a3} = 2 \times 10^{-7} \left[I_a \ln \frac{1}{r'} + I_b \ln \frac{1}{D_{31}} + I_c \ln \frac{1}{D_{23}} \right]$$

Average over the cycle (each section is one third of the length):

$$\psi_a = \frac{\psi_{a1}+\psi_{a2}+\psi_{a3}}{3} = 2\times10^{-7}\left[I_a \ln\frac{1}{r'} + \frac{I_b+I_c}{3}\ln\frac{1}{D_{12}D_{23}D_{31}}\right]$$

Put $I_b + I_c = -I_a$:

$$\psi_a = 2\times10^{-7}\, I_a\left[\ln\frac{1}{r'} + \frac{1}{3}\ln\left(D_{12}D_{23}D_{31}\right)\right] = 2\times10^{-7}\, I_a \ln\frac{\sqrt[3]{D_{12}D_{23}D_{31}}}{r'}$$

Inductance per phase:

$$L_a = \frac{\psi_a}{I_a} = 2\times10^{-7}\ln\frac{D_{eq}}{r'}\ \text{H/m}, \qquad D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$$

$$\boxed{L = 2\times10^{-7}\ln\frac{\sqrt[3]{D_{12}D_{23}D_{31}}}{r'}\ \text{H/m per phase} = 0.2\ln\frac{D_{eq}}{r'}\ \text{mH/km}}$$

By the same steps $L_b = L_c = L_a$: transposition makes the three phases equal. D_{eq} is the **equivalent (geometric mean) spacing**; for symmetrical spacing $D_{eq} = D$ and the formula becomes $L = 2\times10^{-7}\ln(D/r')$.

3. Keyword Explanation

1. Flux linkage

The magnetic flux around a current, counted each time it surrounds the current (Wb-turns).

2. GMR (r')

Geometric mean radius: $r' = 0.7788\,r$ for a solid round wire. It folds the wire's internal flux into the formula.

3. Equivalent spacing (D_{eq})

The geometric mean of the three distances, $\sqrt[3]{D_{12}D_{23}D_{31}}$; also called the GMD (geometric mean distance).

4. Notes and Remarks

- **Typical size:** for real lines D_{eq}/r' is a few hundred to a few thousand, so L is about 1.0 to 1.4 mH/km per phase.
- **Untransposed line:** the inductances become complex numbers that differ from phase to phase, which is why the textbook result always assumes transposition.
- **Flat (horizontal) spacing** D, D, $2D$: $D_{eq} = \sqrt[3]{2}\,D = 1.26\,D$. The same D_{eq} appears for capacitance in 5(d).

1.(d) A single-phase 50 Hz power line is supported on a horizontal cross-arm. The spacing between the conductors is 3 m. A telephone line is supported symmetrically below the power line as shown in Fig. 1. Find the mutual inductance between the two circuits and the voltage induced per kilometer in the telephone line if the current in the power line is 100 A. Assume the telephone line current to be zero. Marks: 10

1. Theory

1. What is mutual inductance? (Core Idea)

When current flows in one circuit, its magnetic flux can also pass through a second circuit nearby. If the current is alternating, this flux changes all the time and induces a voltage in the second circuit, even though the two circuits are not connected. **Mutual inductance** M measures this coupling:

$$M = \frac{\psi_{21}}{I_1}, \qquad E_2 = \omega M I_1 = 2\pi f M I_1$$

where ψ_{21} is the flux linkage of circuit 2 due to current I_1 in circuit 1, and E_2 the voltage induced in circuit 2.
Power lines and telephone lines often run along the same road. A power line carries hundreds of amperes, so even a small M can induce a noticeable voltage in the telephone line: it is heard as hum and can be dangerous.

2. Flux of one wire that passes through a loop

From 1(c), a long straight wire carrying current I makes flux in rings around it. The flux in the rings with radius between D_{near} and D_{far} is

$$\psi = \frac{\mu_0 I}{2\pi} \ln \frac{D_{far}}{D_{near}} = 2 \times 10^{-7}\, I \ln \frac{D_{far}}{D_{near}} \quad \text{Wb-T/m}$$

The telephone line is a loop made of two wires, T_1 and T_2. A flux ring of the power wire links this loop only if it passes **between** T_1 and T_2, that is, if its radius is between the distance to T_1 and the distance to T_2.

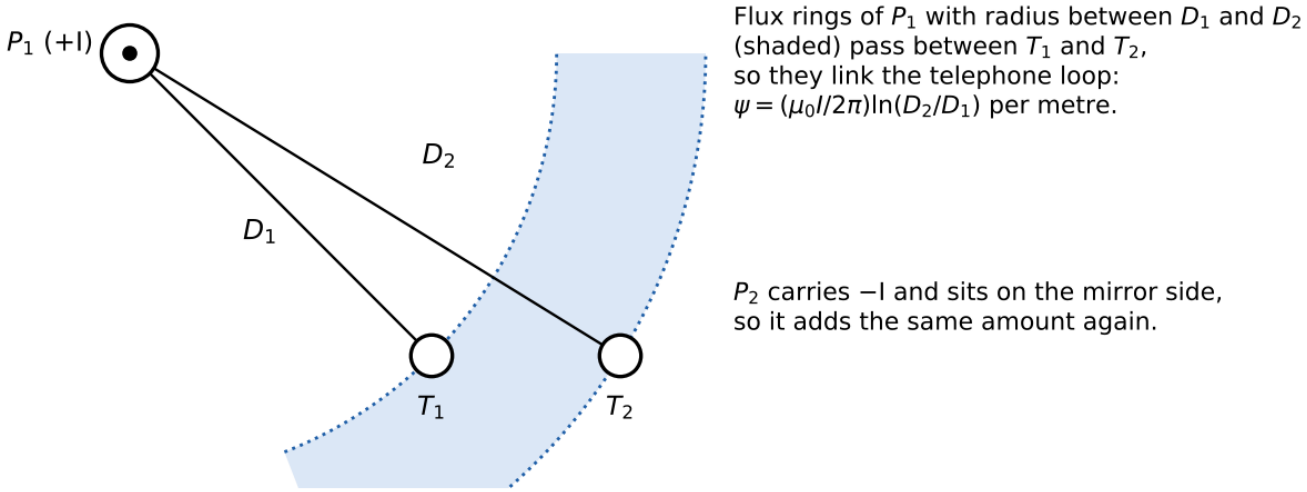


Figure 6: Flux of the power wire P1 that links the telephone loop: the shaded rings with radius between D1 and D2

Reading the figure. P_1 carries $+I$ out of the page. Its flux rings with radius smaller than D_1 pass around both telephone wires or neither, so they do not link the loop. Rings with radius larger than D_2 go around the whole loop, so they don't link it either. Only the shaded rings, between D_1 and D_2, go between T_1 and T_2 and link the loop.

3. Both power wires together

The second power wire P_2 carries the return current $-I$. By symmetry its near telephone wire is T_2 (at D_1) and its far one is T_1 (at D_2). Because both its current and its position are reversed, its flux through the loop is in the *same* direction as that of P_1, so the two add:

$$\psi = 2 \times \left(2 \times 10^{-7}\, I \ln \frac{D_2}{D_1} \right) = 4 \times 10^{-7}\, I \ln \frac{D_2}{D_1}$$

For any arrangement, the general rule is

$$M = 2 \times 10^{-7} \ln \frac{D_{P_1T_2}\, D_{P_2T_1}}{D_{P_1T_1}\, D_{P_2T_2}} \quad \text{H/m}$$

where $D_{P_1T_2}$ is the distance from P_1 to T_2, and so on. This rule is used again in 9(b).

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4. How to reduce the interference

- Increase the distance between the two lines.
- Transpose the power line or the telephone line, so that the induced voltages in different sections cancel.
- Place the telephone wires so that $D_{P_1T_2}D_{P_2T_1} = D_{P_1T_1}D_{P_2T_2}$, for example both on the centre line, one above the other; then $M = 0$.

2. Answer

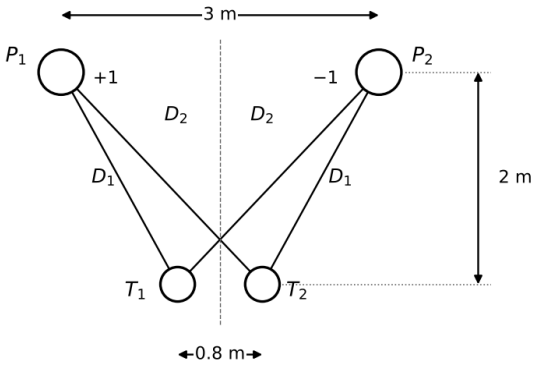


Figure 7: Repeated from the question: Fig. 1 of the paper (redrawn): figure for Q-1(d)

Given:

- Single-phase power line, $f = 50$ Hz, $I = 100$ A
- Spacing of the power conductors $P_1P_2 = 3$ m
- Telephone conductors $T_1T_2 = 0.8$ m apart, 2 m below the power line, placed symmetrically (Fig. 1)
- Telephone current $= 0$

Relationships:

$$M = 4 \times 10^{-7} \ln \frac{D_2}{D_1} \ \text{H/m}, \qquad E = 2\pi f M I$$

where $D_1 = P_1T_1 = P_2T_2$ (near distances) and $D_2 = P_1T_2 = P_2T_1$ (far distances).

Step 1: Distances D_1 and D_2

Because of the symmetry, each telephone wire is 0.4 m from the centre line and each power wire 1.5 m from it. Horizontal distances from P_1:

$$\text{to } T_1: \ 1.5 - 0.4 = 1.1 \text{ m}, \qquad \text{to } T_2: \ 1.5 + 0.4 = 1.9 \text{ m}$$

The vertical distance is 2 m, so by Pythagoras

$$D_1 = \sqrt{1.1^2 + 2^2} = \sqrt{5.21} = 2.2825 \text{ m}$$

$$D_2 = \sqrt{1.9^2 + 2^2} = \sqrt{7.61} = 2.7586 \text{ m}$$

$$\boxed{D_1 = 2.2825 \text{ m}, \quad D_2 = 2.7586 \text{ m}}$$

Step 2: Mutual inductance

$$\frac{D_2}{D_1} = \frac{2.7586}{2.2825} = 1.2086, \qquad \ln 1.2086 = 0.18944$$

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$$M = 4 \times 10^{-7} \times 0.18944 = 7.578 \times 10^{-8} \text{ H/m}$$

Per kilometre (multiply by 1000):

$$M = 7.578 \times 10^{-5} \text{ H/km} = 0.0758 \text{ mH/km}$$

$$\boxed{M = 0.0758 \text{ mH/km}}$$

Step 3: Voltage induced in the telephone line

$$E = 2\pi f M I = 2\pi \times 50 \times 7.578 \times 10^{-5} \times 100 = 2.381 \text{ V/km}$$

$$\boxed{E = 2.38 \text{ V per km}}$$

So every kilometre of parallel running adds about 2.4 V in the telephone loop. Over 10 km that is 24 V, which is why power and telephone lines are kept apart or transposed.

Final Answer: $M = 0.0758$ mH/km; induced voltage = 2.38 V/km.

3. Keyword Explanation

1. Mutual inductance

Flux linkage of one circuit per ampere of current in another circuit (H). It gives the voltage induced in the second circuit, $E = \omega M I$.

2. Electromagnetic interference

The unwanted voltage induced in communication lines by nearby power lines.

4. Notes and Remarks

- **Why not the full ln of each distance?** Only the difference of the two distances from a power wire to the two telephone wires matters, because only the flux passing between the telephone wires links the loop.
- **Electrostatic induction** also exists (from the line's voltage, through capacitance). This question asks only for the magnetic (inductive) effect.

5. Final Concise Answer

(a) Critical disruptive voltage is the minimum voltage at which the field at the conductor surface reaches the breakdown strength of air (30 kV/cm peak), so corona starts: $V_c = m_0 g_0 \delta r \ln(d/r)$ kV. Below it there is no corona; above it there is corona loss, radio interference and ozone, so the conductor size and spacing are chosen to keep the working voltage below V_c.

(b) $R_{loop} = 0.5 + 1.125 = 1.625\ \Omega$; $R_X = 1.625/3.75 = 0.4333\ \Omega$; fault at **433.3 m** from the testing end.

(c) A transposed line exchanges conductor positions so that each conductor occupies each position for one third of the length. Averaging the flux linkage over the three sections with $I_a + I_b + I_c = 0$ gives $L = 2 \times 10^{-7} \ln\left(\sqrt[3]{D_{12} D_{23} D_{31}}/r'\right)$ H/m per phase, $r' = 0.7788\, r$.

(d) $D_1 = 2.2825$ m, $D_2 = 2.7586$ m; $M = 4 \times 10^{-7} \ln(D_2/D_1) = 0.0758$ mH/km; induced voltage = 2.38 V/km.

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2. (a) What is monopolar link? Marks: 2
   (b) Write down the general construction of a 3-conductor cable with required figure. Marks: 5
   (c) A single core cable for use on 11 kV, 50 Hz system has conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find: (i) the maximum electrostatic stress in the cable, (ii) minimum electrostatic stress in the cable, (iii) capacitance of the cable per km length, and (iv) charging current. Marks: 8
   (d) Define grading of cables. Why is unequal stress distribution in a cable undesirable? Consider three dielectric layers for capacitance grading and derive the expression of potential difference between core and earthed sheath. Marks: 10

2.(a) What is monopolar link? Marks: 2

1. Theory

1. What is HVDC transmission? (Core Idea)

Power stations make **alternating current** (AC), and homes use AC. But over a very long distance (or under the sea) it can be cheaper to send the power as **direct current** (DC) at a very high voltage. This is **HVDC** (high-voltage direct current) transmission. It works in three steps:

1. At the sending end, a **rectifier** station changes AC into DC.
2. The DC is sent along the line.
3. At the receiving end, an **inverter** station changes DC back into AC.

Why DC can be cheaper for long lines is explained in 6(b).

2. How many conductors does a DC line need?

An AC line needs three conductors (three phases). A DC circuit needs a path out and a path back. The path back can be a second conductor, or the **earth** (or sea water) itself, which is a very large conductor. The three ways of arranging this are called **links**.

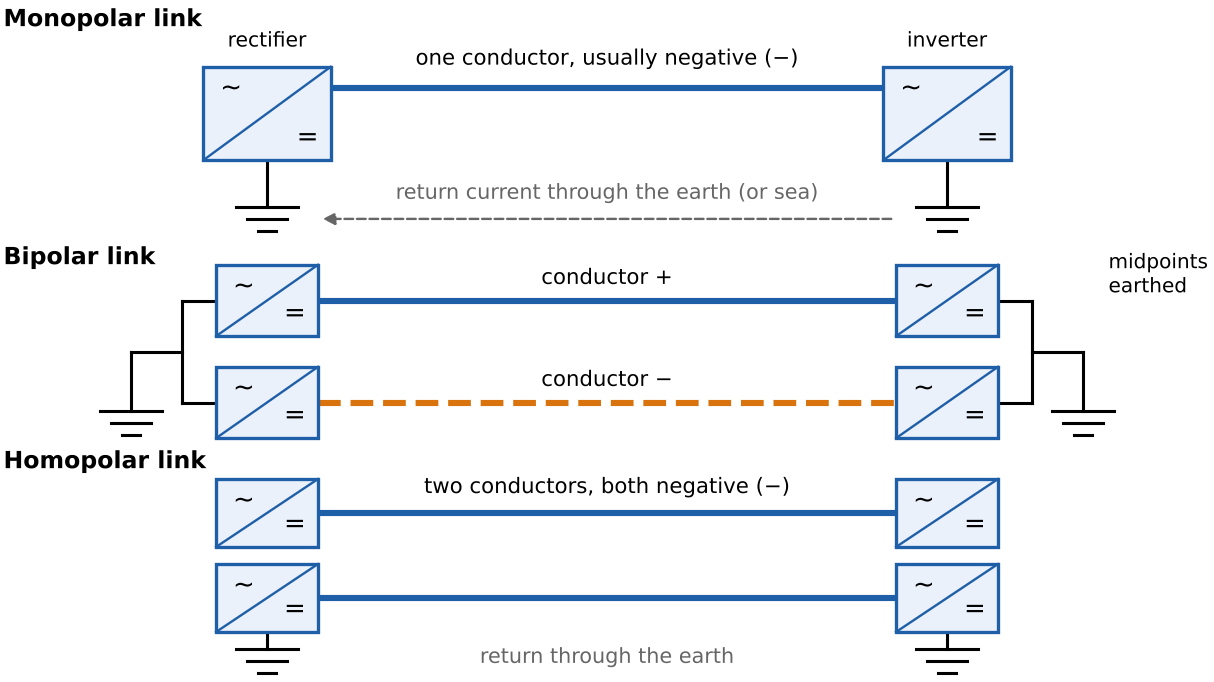


Figure 8: The three kinds of HVDC link: monopolar (one conductor, earth return), bipolar (a + and a − conductor, midpoints earthed) and homopolar (two conductors of the same polarity, earth return)

Reading the figure. Each box is a converter station (AC on one side, DC on the other). Top: one conductor, and the current comes back through the earth. Middle: one conductor at $+V$ (solid blue) and one at $-V$ (dashed orange); the return current flows in the second conductor. Bottom: two conductors at the same polarity, with the earth as the return.

- **Monopolar:** one conductor, earth return. The cheapest.
- **Bipolar:** two conductors, $+V$ and $-V$, with the midpoints of the converters earthed. The most common. If one pole fails, the other can still carry half the power with earth return.
- **Homopolar:** two or more conductors of the same polarity (usually negative), with earth return.

The conductor is usually kept **negative**, because a negative conductor produces less corona and radio interference than a positive one.

2. Answer

A **monopolar link** is an HVDC link with **only one conductor**, usually at **negative polarity**, between the rectifier station and the inverter station. The **earth** (or the sea) serves as the return path of the current. The converter stations at both ends are earthed. It is the cheapest form of HVDC link and is used for low-power transmission, especially submarine cables; a metallic return conductor can be added where earth currents are not allowed.

3. Keyword Explanation

1. Rectifier

A converter station that changes AC to DC.

2. Inverter

A converter station that changes DC back to AC.

3. Earth return

Using the earth or sea water as the return conductor of a DC circuit.

2.(b) Write down the general construction of a 3-conductor cable with required figure. Marks: 5

1. Theory

1. What is an underground cable? (Core Idea)

In cities, overhead lines are often not possible: no space for towers, danger to people, and an ugly view. Power is then carried in **underground cables**. A cable is a conductor (or several) wrapped in insulation and protective layers, laid in the ground. Because the wires are close together and surrounded by soil and water, the insulation and protection must be very good.

2. What does each layer do?

A cable is built from the inside out. Each layer solves one problem:

- **Core (conductor):** carries the current. Made of stranded copper or aluminium, so it bends easily.
- **Insulation:** keeps each core apart from the others and from the earth. Impregnated paper, varnished cambric, or (in modern cables) PVC or XLPE.
- **Metallic sheath:** usually lead or aluminium. It keeps moisture and gases out of the insulation.
- **Bedding:** fibrous material (jute or hessian tape). It protects the sheath from the armour.
- **Armouring:** galvanised steel wires or steel tape. It protects the cable from mechanical damage while it is laid and used.
- **Serving:** an outer layer of jute. It protects the armour from the weather and corrosion.

3. Why "3-core"?

Three-phase power needs three conductors. Putting all three in one cable is cheaper and smaller than laying three separate single-core cables. In a **belted** 3-core cable, each core has its own paper insulation,

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and then a common paper **belt** is wound around all three. The gaps between the round cores are filled with **jute fillers** to make the cable round.

2. Answer

A 3-conductor (3-core) cable, used for three-phase supply up to about 11 kV in its belted form, is built in layers from the centre outwards:

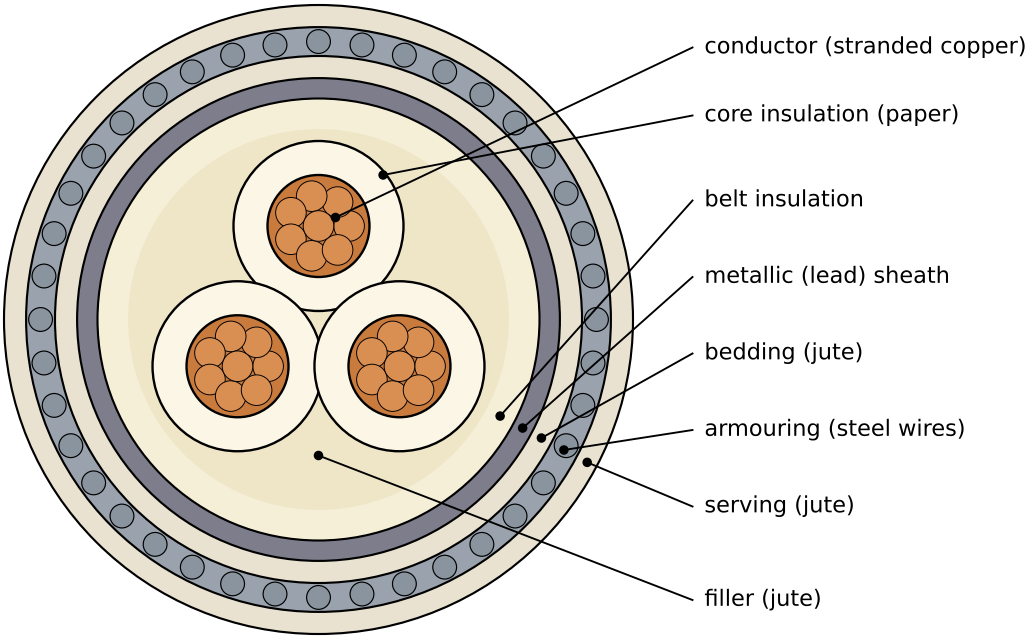


Figure 9: Cross-section of a 3-core belted cable

1. **Cores (conductors):** three stranded copper or aluminium conductors, one for each phase. Stranding makes the cable flexible.
2. **Core insulation:** each core is wrapped in impregnated paper (or PVC/XLPE), thick enough for the working voltage between cores.
3. **Fillers:** jute fillers fill the spaces between the three insulated cores, so that the bundle is round.
4. **Belt insulation:** a common layer of impregnated paper is wound around the three cores and fillers. It insulates the cores from the sheath.
5. **Metallic sheath:** a lead (or aluminium) sheath over the belt keeps moisture and gases out and protects the insulation.
6. **Bedding:** a layer of jute or hessian tape over the sheath, to protect it from the armour.
7. **Armouring:** one or two layers of galvanised steel wire or tape, to protect the cable from mechanical injury during laying and handling.
8. **Serving:** an outer layer of jute, to protect the armour from atmospheric corrosion.

3. Keyword Explanation

1. Belted cable

A 3-core cable whose cores are insulated separately and then wrapped together in a common insulating belt.

2. Armouring

A layer of steel wires or tape that gives the cable mechanical strength.

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4. Notes and Remarks

- **Above 22 kV**, belted cables are not used. The electric field in them has a part along the paper layers, where paper is weak, and tiny air gaps form. **Screened (H-type, S.L.-type)** and **pressure (oil-filled, gas)** cables are used instead.
- A **single-core** cable has the same layers around one conductor; it is studied in 2(c).

2.(c) A single core cable for use on 11 kV, 50 Hz system has conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find: (i) the maximum electrostatic stress in the cable, (ii) minimum electrostatic stress in the cable, (iii) capacitance of the cable per km length, and (iv) charging current. Marks: 8

1. Theory

1. The single-core cable as a cylindrical capacitor (Core Idea)

A single-core cable has a round conductor (diameter d) at the centre, a layer of insulation around it, and an earthed metal sheath outside (inside diameter D). The conductor and the sheath are two metal surfaces with an insulator between them, so the cable is a **capacitor** shaped like a tube.

When the core is at voltage V against the sheath, a charge $+Q$ (coulomb per metre) sits on the core and $-Q$ on the sheath. The electric field in the insulation points straight outwards, like the spokes of a wheel.

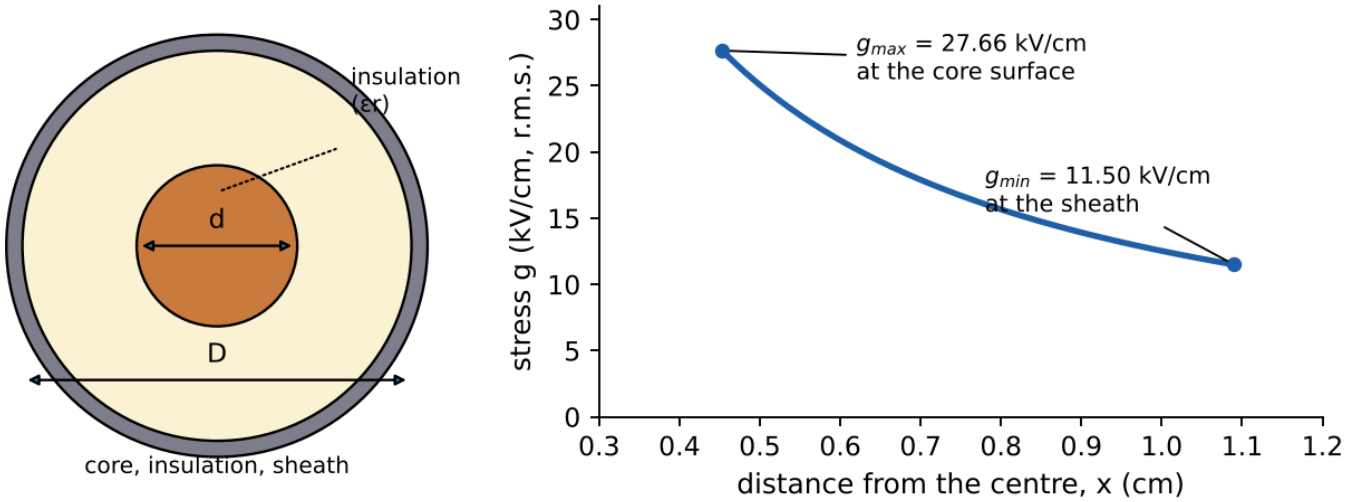


Figure 10: Left: section of a single-core cable. Right: the stress in the insulation for this question; it is highest at the core and lowest at the sheath

Reading the figure. On the left, the core (diameter d) is surrounded by insulation and the sheath (inside diameter D). On the right, the solid blue curve shows the stress across the insulation for this question: 27.66 kV/cm at the core surface, falling to 11.50 kV/cm at the sheath.

2. The electric stress (field) at radius x

By Gauss's law, the charge Q per metre spreads its field over a cylinder of radius x and surface $2\pi x$ per metre. So the field at distance x from the centre is

$$g = \frac{Q}{2\pi\varepsilon_0\varepsilon_r\, x}$$

The voltage between core and sheath is the field added up across the insulation:

$$V = \int_{d/2}^{D/2} g\, dx = \frac{Q}{2\pi\varepsilon_0\varepsilon_r} \ln\frac{D}{d}$$

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Eliminating Q gives the stress in terms of the voltage:

$$g = \frac{V}{x \ln(D/d)}$$

where

- g is the **electrostatic stress** (potential gradient) at distance x (kV/cm)
- V is the voltage between core and sheath
- d is the conductor diameter, D the inside diameter of the sheath
- $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m and ε_r the relative permittivity of the insulation

Maximum and minimum stress. The stress falls as $1/x$, so it is largest at the conductor surface ($x = d/2$) and smallest at the sheath ($x = D/2$):

$$g_{max} = \frac{2V}{d \ln(D/d)}, \qquad g_{min} = \frac{2V}{D \ln(D/d)}$$

The insulation must be strong enough for g_{max}. (This uneven stress is the reason for grading in 2(d).)

3. Capacitance and charging current

Capacitance is charge per volt, $C = Q/V$:

$$C = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)} \ \text{F/m}$$

The cable is always being charged and discharged by the AC voltage. The current that does this is the **charging current**:

$$I_c = \frac{V}{X_c} = 2\pi f C V$$

It flows even when no load is connected, and it uses up part of the cable's current rating.

2. Answer

Given:

- Voltage between core and sheath: $V = 11$ kV (r.m.s.), $f = 50$ Hz
- Conductor area $a = 0.645$ cm^2
- Inside diameter of sheath $D = 2.18$ cm
- Relative permittivity $\varepsilon_r = 3.5$
- Length $l = 1$ km $= 1000$ m

Assumption: following V.K. Mehta, the 11 kV is taken as the voltage between the core and the earthed sheath. (If 11 kV were the line voltage of a three-phase system, the core-to-sheath voltage would be $11/\sqrt{3} = 6.35$ kV; see Notes.)

Step 1: Conductor diameter

$$a = \frac{\pi d^2}{4} \quad \Rightarrow \quad d = \sqrt{\frac{4a}{\pi}} = \sqrt{\frac{4 \times 0.645}{\pi}} = 0.9062 \text{ cm}$$

$$\frac{D}{d} = \frac{2.18}{0.9062} = 2.4056, \qquad \ln\frac{D}{d} = 0.8778$$

Step 2: (i) Maximum stress

$$g_{max} = \frac{2V}{d \ln(D/d)} = \frac{2 \times 11}{0.9062 \times 0.8778} = 27.66 \text{ kV/cm (r.m.s.)}$$

$$\boxed{g_{max} = 27.66 \text{ kV/cm r.m.s.}}$$

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This is at the surface of the conductor.

Step 3: (ii) Minimum stress

$$g_{min} = \frac{2V}{D\ln(D/d)} = \frac{2 \times 11}{2.18 \times 0.8778} = 11.50 \text{ kV/cm (r.m.s.)}$$

$$\boxed{g_{min} = 11.50 \text{ kV/cm r.m.s.}}$$

This is at the inside of the sheath. Check: $g_{max}/g_{min} = D/d = 2.406$.

Step 4: (iii) Capacitance per km

$$C = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)}\, l = \frac{2\pi \times 8.854 \times 10^{-12} \times 3.5}{0.8778} \times 1000 = 2.218 \times 10^{-7} \text{ F}$$

$$\boxed{C = 0.2218 \ \mu\text{F per km}}$$

Step 5: (iv) Charging current

$$I_c = 2\pi f C V = 2\pi \times 50 \times 0.2218 \times 10^{-6} \times 11000 = 0.7665 \text{ A}$$

$$\boxed{I_c = 0.767 \text{ A per km}}$$

Final Answer:

- (i) $g_{max} = 27.66$ kV/cm r.m.s. (39.11 kV/cm peak)
- (ii) $g_{min} = 11.50$ kV/cm r.m.s. (16.26 kV/cm peak)
- (iii) $C = 0.2218$ μF per km
- (iv) $I_c = 0.767$ A per km

3. Notes and Remarks

- **Peak or r.m.s.?** Using the r.m.s. voltage gives r.m.s. stress. Insulation strength is often quoted as a peak value; multiply by $\sqrt{2}$ to compare.
- **If 11 kV is a three-phase line voltage**, then $V = 6.351$ kV, and all four answers are $\sqrt{3}$ times smaller: $g_{max} = 15.97$ kV/cm, $g_{min} = 6.64$ kV/cm, $I_c = 0.443$ A per km (C does not change).
- The same cable appears in Term Test 2 as 8(b).

2.(d) Define grading of cables. Why is unequal stress distribution in a cable undesirable? Consider three dielectric layers for capacitance grading and derive the expression of potential difference between core and earthed sheath. Marks: 10

1. Theory

The basics

A single-core cable is a round conductor inside insulation inside a metal sheath. The electric **stress** in the insulation is the voltage per unit thickness. If it exceeds the insulation's strength at any point, the insulation fails there.

1. The problem: stress is uneven (Core Idea)

In 2(c) we found that the stress in a single-core cable is $g = V/(x\ln(D/d))$. It is largest at the conductor and smallest at the sheath (2.4 times smaller there in 2(c)).

The insulation must be designed so that g_{max} is below its breakdown strength. But then the outer insulation, which feels a much smaller stress, is **under-used**: we pay for thick insulation that is mostly working at a low stress.

2. The idea of grading

Grading means arranging the insulation so that the stress is more *even* across it. Then each part of

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the insulation works near its safe limit, and a thinner cable can carry the same voltage. There are two methods:

- **Capacitance grading:** use several layers of insulation with **different permittivities**. The material with the highest ε_r goes next to the core.
- **Intersheath grading:** put thin metal sheets (intersheaths) inside one insulation and hold them at chosen voltages.

3. Why a higher permittivity near the core helps

The charge Q per metre is the same through every layer (the electric flux has nowhere else to go). In a layer of permittivity ε_r, the stress at radius x is

$$g = \frac{Q}{2\pi\varepsilon_0\varepsilon_r\, x}$$

So, at a given x, a **larger ε_r gives a smaller stress**. Near the core, where x is small and the stress would be large, we put the largest ε_r to bring the stress down. Further out we use smaller ε_r, which pushes the stress back up. The result is a stress that rises and falls in a sawtooth around a nearly constant value, instead of one big curve.

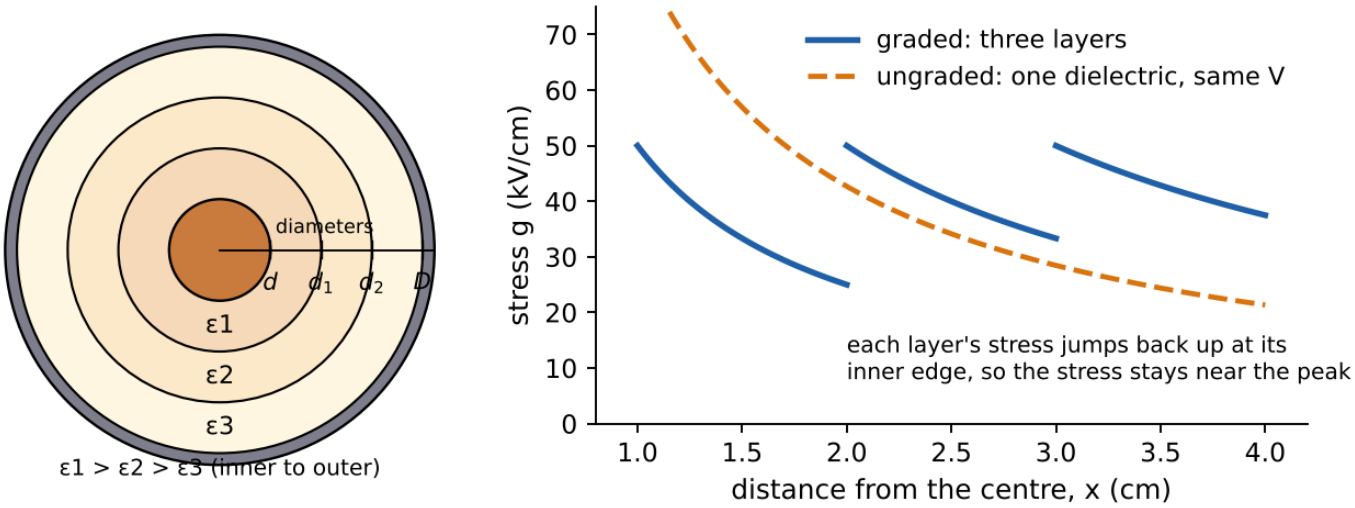


Figure 11: Left: a cable with three dielectric layers. Right: stress across the insulation; the graded cable (solid blue) stays near its peak in every layer, while the ungraded cable (dashed orange) has a high peak at the core

Reading the figure. On the left, the core (diameter d) is covered by three layers of permittivity ε_1, ε_2 and ε_3, with outer diameters d_1, d_2 and D. On the right, for the same core-to-sheath voltage (the same area under each curve), the graded cable's peak stress (solid blue) is much lower than the ungraded cable's peak (dashed orange).

2. Answer

Grading of cables: the process of making the distribution of electrostatic stress in the dielectric of a cable as uniform as possible. It is done by **capacitance grading** (layers of different permittivity) or **intersheath grading** (metallic intersheaths at intermediate voltages).

Why unequal stress is undesirable:

- The thickness of insulation must be chosen for the **maximum** stress at the core. The outer insulation then works at a much lower stress, so most of it is wasted, and the cable becomes **thick and costly**.
- Breakdown starts where the stress is highest, at the conductor surface; once a small part fails, the rest is overstressed and the whole insulation fails.
- So for the same safe maximum stress, an ungraded cable needs a larger overall diameter (or can work only at a lower voltage) than a graded one.

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Capacitance grading with three dielectrics: potential difference between core and sheath. Let the core have diameter d. It is covered by three layers of relative permittivities ε_1, ε_2 and ε_3 (with $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$), whose outer diameters are d_1, d_2 and D (figure above). Let Q be the charge per metre on the core.

Stress in each layer. The same Q passes through every layer, so at radius x:

$$g_1 = \frac{Q}{2\pi\varepsilon_0\varepsilon_1 x} \quad \left(\frac{d}{2} \le x \le \frac{d_1}{2}\right), \quad g_2 = \frac{Q}{2\pi\varepsilon_0\varepsilon_2 x} \quad \left(\frac{d_1}{2} \le x \le \frac{d_2}{2}\right), \quad g_3 = \frac{Q}{2\pi\varepsilon_0\varepsilon_3 x} \quad \left(\frac{d_2}{2} \le x \le \frac{D}{2}\right)$$

Maximum stress in each layer (at its inner surface):

$$g_{1max} = \frac{Q}{\pi\varepsilon_0\varepsilon_1 d}, \qquad g_{2max} = \frac{Q}{\pi\varepsilon_0\varepsilon_2 d_1}, \qquad g_{3max} = \frac{Q}{\pi\varepsilon_0\varepsilon_3 d_2}$$

Voltage across each layer (stress added up across the layer):

$$V_1 = \int_{d/2}^{d_1/2} g_1\, dx = \frac{Q}{2\pi\varepsilon_0\varepsilon_1} \ln\frac{d_1}{d} = \frac{g_{1max}\, d}{2} \ln\frac{d_1}{d}$$

$$V_2 = \frac{Q}{2\pi\varepsilon_0\varepsilon_2} \ln\frac{d_2}{d_1} = \frac{g_{2max}\, d_1}{2} \ln\frac{d_2}{d_1}, \qquad V_3 = \frac{Q}{2\pi\varepsilon_0\varepsilon_3} \ln\frac{D}{d_2} = \frac{g_{3max}\, d_2}{2} \ln\frac{D}{d_2}$$

Potential difference between core and earthed sheath $V = V_1 + V_2 + V_3$:

$$\boxed{V = \frac{g_{1max}\, d}{2} \ln\frac{d_1}{d} + \frac{g_{2max}\, d_1}{2} \ln\frac{d_2}{d_1} + \frac{g_{3max}\, d_2}{2} \ln\frac{D}{d_2}}$$

or, in terms of the charge, $V = \dfrac{Q}{2\pi\varepsilon_0}\left[\dfrac{1}{\varepsilon_1}\ln\dfrac{d_1}{d} + \dfrac{1}{\varepsilon_2}\ln\dfrac{d_2}{d_1} + \dfrac{1}{\varepsilon_3}\ln\dfrac{D}{d_2}\right]$.

Same maximum stress in all layers. If the layers are chosen so that $g_{1max} = g_{2max} = g_{3max} = g_{max}$, then from the maximum-stress expressions

$$\varepsilon_1 d = \varepsilon_2 d_1 = \varepsilon_3 d_2$$

and the voltage becomes

$$V = \frac{g_{max}}{2}\left[d \ln\frac{d_1}{d} + d_1 \ln\frac{d_2}{d_1} + d_2 \ln\frac{D}{d_2}\right]$$

This is larger than the ungraded value $V = \dfrac{g_{max}\, d}{2} \ln\dfrac{D}{d}$ for the same g_{max} and D, so the graded cable can work at a higher voltage (or be thinner for the same voltage).

3. Keyword Explanation

1. Dielectric (relative permittivity ε_r)

How easily an insulating material lets the electric field pass; a higher ε_r gives a lower field for the same charge.

2. Electric stress

The potential gradient inside the insulation (kV/cm). Breakdown occurs if it exceeds the material's strength.

4. Notes and Remarks

- **Why grading is rare in practice:** it is hard to find materials with the right range of permittivities, and they can change with temperature. Modern cables use better insulation (XLPE) instead.
- **A numerical example** of capacitance grading with two layers is 4(d).

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5. Final Concise Answer

(a) A monopolar link is an HVDC link with one conductor, usually negative, between the converter stations; the earth or sea is the return path.

(b) From the centre: three stranded cores, each with paper (or PVC/XLPE) insulation; jute fillers; a common paper belt; a lead sheath; jute bedding; steel-wire or tape armour; jute serving.

(c) $d = 0.906$ cm; (i) $g_{max} = 27.66$ kV/cm, (ii) $g_{min} = 11.50$ kV/cm (both r.m.s.), (iii) $C = 0.2218\ \mu$F/km, (iv) $I_c = 0.767$ A per km.

(d) Grading makes the stress in the dielectric more uniform; uneven stress wastes outer insulation and makes the cable thick and costly. With three layers: $V = \frac{g_{1max}d}{2}\ln\frac{d_1}{d} + \frac{g_{2max}d_1}{2}\ln\frac{d_2}{d_1} + \frac{g_{3max}d_2}{2}\ln\frac{D}{d_2}$; for equal maximum stresses, $\varepsilon_1 d = \varepsilon_2 d_1 = \varepsilon_3 d_2$.

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3. (a) How electrical breakdown happen in a pin-type insulator? Marks: 2
   (b) What kind of line supports are used for distribution purposes at low voltages (upto 11 kV)? Marks: 5
   (c) In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) String efficiency. Marks: 8
   (d) Derive the mathematical expression of String efficiency with proper diagram. What will happen if:
      (i) $K = 0.2$ ($K = \frac{shunt\ capacitance}{self\ capacitance}$)
      (ii) $K = 0.8$
      (iii) the number of disc increases in the string. Marks: 10

3.(a) How electrical breakdown happen in a pin-type insulator? Marks: 2

1. Theory

1. Why do overhead lines need insulators? (Core Idea)

The conductors of an overhead line are bare and at a high voltage, but they must be fixed to towers or poles that are connected to the earth. If a conductor touched the support directly, current would flow into the earth: a short circuit. **Insulators** hold the conductor to the support while keeping the current out of it. They are made of glazed **porcelain** (or glass, or polymer), which is a very good insulator.

2. The pin-type insulator

A **pin-type insulator** is screwed onto a steel **pin**, which is bolted to the cross-arm of the pole. The conductor lies in a groove on top of the insulator and is tied to it with wire. The porcelain has one or more umbrella-shaped **sheds** (petticoats). The sheds make the path along the surface, from the conductor down to the pin, much longer, and keep the lower surfaces dry in rain. Pin insulators are used up to about 33 kV.

3. Two ways the insulation can fail

The voltage between the conductor (top) and the pin (bottom) tries to push a current between them. There are two possible paths:

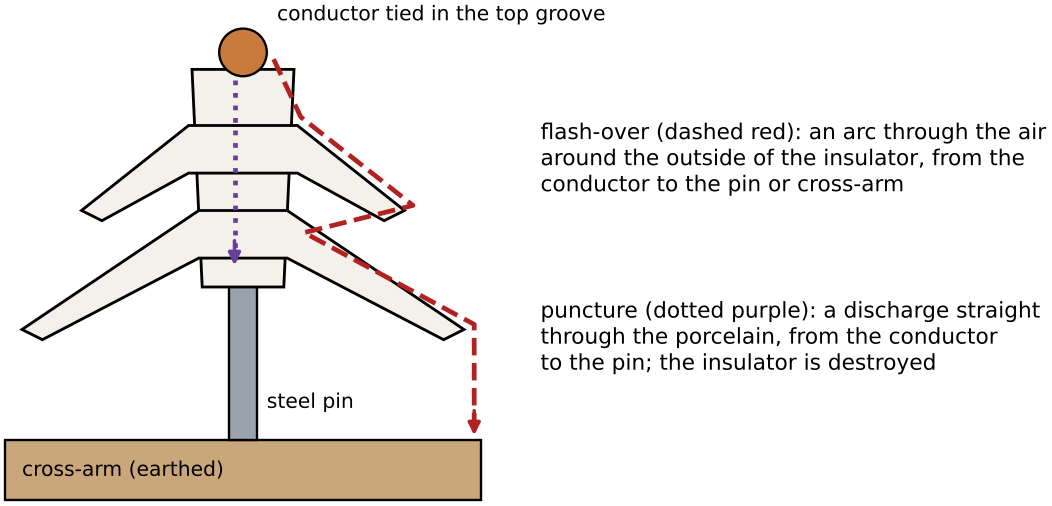


Figure 12: Pin-type insulator: the flash-over path through the air outside (dashed red) and the puncture path through the porcelain (dotted purple)

Reading the figure. The conductor sits in the top groove; the steel pin holds the insulator on the earthed cross-arm. The dashed red path goes around the outside of the sheds through the air: this is **flash-over**. The dotted purple path goes straight through the porcelain from the top to the pin: this is **puncture**.

- **Flash-over:** the air around the insulator breaks down (as in 1(a)) and an arc jumps from the conductor, over the sheds, to the pin or cross-arm. The insulator usually survives, though the heat of the arc can crack it.
- **Puncture:** the porcelain itself breaks down, and the discharge goes straight through the body of the insulator. The insulator is permanently destroyed.

4. Safety factor

Because puncture destroys the insulator, it is designed so that flash-over always happens first:

$$\text{Safety factor} = \frac{\text{puncture strength}}{\text{flash-over voltage}}$$

It is about 10 for pin insulators.

2. Answer

Electrical breakdown of a pin-type insulator can occur in two ways:

1. **Flash-over:** an arc is formed between the line conductor and the insulator pin (earth) through the **surrounding air**, along the outside of the sheds. The insulator generally keeps working afterwards, although the heat may damage it.
2. **Puncture:** the discharge passes from the conductor to the pin **through the body of the porcelain**. The insulator is permanently damaged and must be replaced.

To make flash-over happen before puncture, the porcelain is made thick enough that the safety factor (puncture strength / flash-over voltage) is high, about 10.

3. Keyword Explanation

1. Shed (petticoat)

The umbrella-like skirt of an insulator; it lengthens the surface path and keeps part of the surface dry.

2. Flash-over voltage

The voltage at which an arc forms through the air around the insulator.

3.(b) What kind of line supports are used for distribution purposes at low voltages (upto 11 kV)? Marks: 5

1. Theory

1. What is a line support? (Core Idea)

A **line support** is the pole or tower that holds the conductors (with their insulators) at a safe height above the ground. It must carry the weight of the conductors and the force of the wind and ice, for many years.

2. What makes a good support?

A good line support should be:

- **strong** mechanically, to withstand the weight of the conductors and wind loads;
- **light**, so it is easy to transport and erect;
- **cheap**, and cheap to maintain;
- **long-lasting**;
- **easy to reach** for the line workers;
- good-looking (in towns), with a small ground area.

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3. The main kinds of support

There are four main kinds: wooden poles, steel poles, RCC (reinforced cement concrete) poles and steel towers. Steel towers are big lattice structures used for high-voltage transmission; the other three are poles, used for distribution.

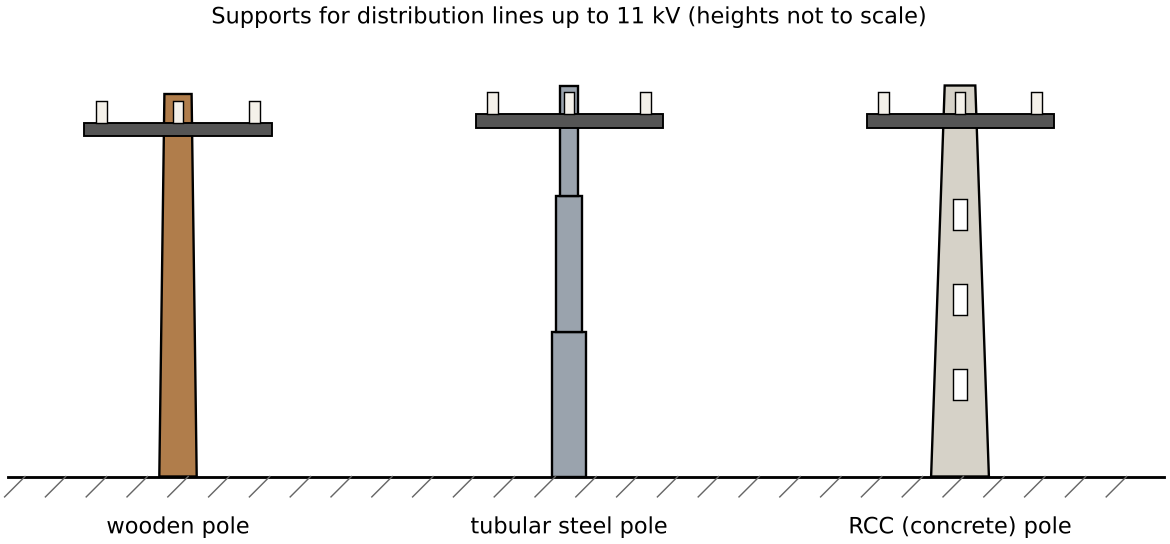


Figure 13: The three kinds of pole used for distribution lines up to 11 kV

Reading the figure. Left: a wooden pole. Middle: a tubular steel pole made of sections of decreasing diameter. Right: a tapered RCC pole with holes that make it lighter and give footholds for climbing. Each carries a cross-arm with pin insulators.

2. Answer

For distribution at low voltages (up to 11 kV), **poles** are used as line supports; steel towers are needed only for high-voltage transmission. The poles used are:

1. **Wooden poles:** made of seasoned wood (sal or chir), impregnated with preservative.
   - Cheap, easy to handle, and a good insulator, so fewer insulators are needed.
   - Used for short spans (about 40 to 50 m) and in rural areas.
   - They rot at ground level and have a short life (20 to 25 years); they are weaker than the other types. Double poles (A-type or H-type) are used for heavier loads.
2. **Steel poles:** rail poles, tubular poles (sections of decreasing diameter) or rolled steel joists.
   - Stronger, longer life, and allow longer spans (about 50 to 80 m).
   - Used for distribution in towns and cities; they need painting to prevent corrosion and must be earthed.
3. **RCC (reinforced cement concrete) poles:**
   - Strong, long life, need little maintenance, look good, and allow longer spans (about 80 to 200 m).
   - Heavy, so they are costly to transport; often cast with holes to reduce the weight. Widely used for 11 kV and 400 V distribution.

3. Keyword Explanation

1. Span

The horizontal distance between two neighbouring supports.

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2. A-type and H-type poles

Two poles joined together (in the shape of an A or an H) to carry larger loads or longer spans.

4. Notes and Remarks

- **Steel towers** (lattice towers) are used for long-distance lines of 66 kV and above, with spans of 300 m or more.
- **Mark of this part:** on the scanned paper the mark of 3(b) is covered by a handwritten circle; 5 is used here because the parts of Question 3 must add up to 25 (2 + 5 + 8 + 10).

3.(c) In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) String efficiency. Marks: 8

1. Theory

1. Suspension insulators and strings (Core Idea)

Above about 33 kV a single pin insulator would have to be very large. Instead, the conductor **hangs** below the cross-arm from a chain of identical porcelain **discs** linked by metal caps and pins. This chain is a **string**, and each disc is a **unit**. To insulate a higher voltage, more discs are added.

2. Two kinds of capacitance in a string

Each disc is a small capacitor: the metal cap and the metal pin are two conductors with porcelain between them. Its capacitance is the **self capacitance** C.

Each metal link between two discs is also close to the earthed tower. So there is a small capacitance from each link to the tower: the **shunt capacitance** C_1. We write it as a fraction of the self capacitance:

$$K = \frac{C_1}{C} = \frac{\text{shunt capacitance}}{\text{self capacitance}}$$

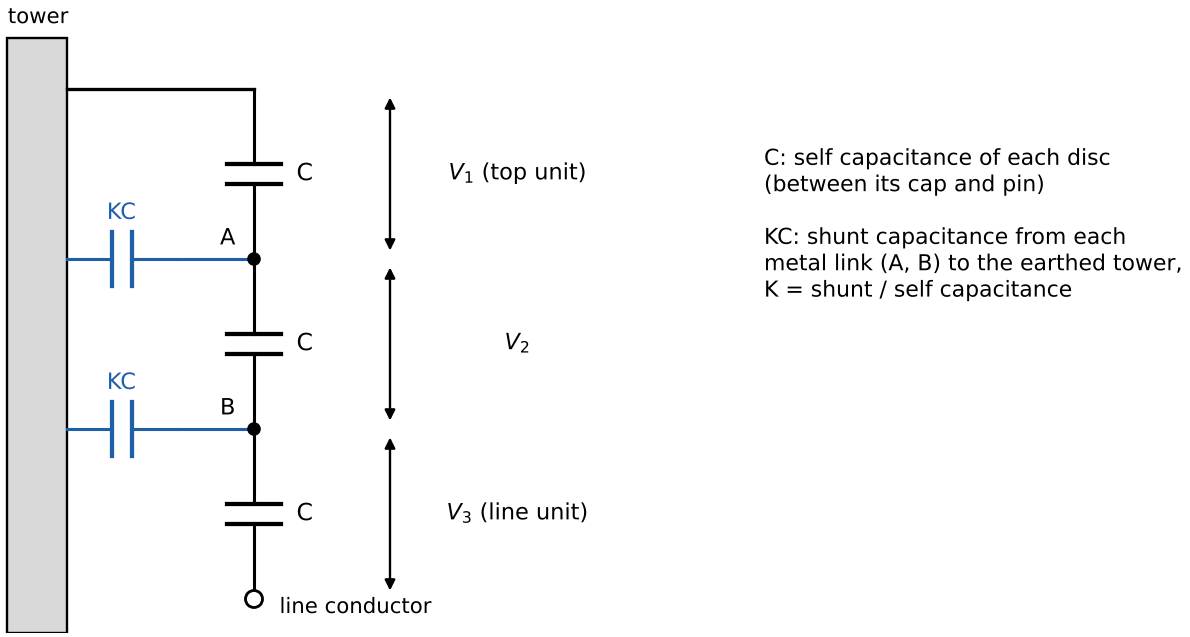


Figure 14: Equivalent circuit of a 3-disc string: self capacitance C for each disc, shunt capacitance KC from each link to the tower

Reading the figure. The tower (earth) is on the left. The three discs are the capacitors C in series, from the tower (top unit) down to the line conductor (line unit). From each metal link, A and B, a shunt capacitor KC leads to the tower.

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3. Why the voltage is not shared equally

If there were no shunt capacitance ($K = 0$), the same current would flow through all three discs and each would take one third of the voltage. But at each link, part of the current leaks to the tower through KC. So the disc **nearest the line conductor** carries the most current (its own current plus all the leaking currents above it) and so has the **largest voltage**. The top disc, nearest the tower, has the smallest voltage.

4. The equations (Kirchhoff's current law at each link)

Let V_1, V_2, V_3 be the voltages across the top, middle and line units, and ω the angular frequency. A capacitor C with voltage V carries current ωCV.

At link A: the current through disc 2 = the current through disc 1 + the current through the shunt capacitance at A. The voltage of A against the tower is V_1, so

$$\omega CV_2 = \omega CV_1 + \omega KCV_1 \quad \Rightarrow \quad V_2 = V_1(1+K)$$

At link B: the voltage of B against the tower is $V_1 + V_2$, so

$$\omega CV_3 = \omega CV_2 + \omega KC(V_1+V_2) \quad \Rightarrow \quad V_3 = V_2 + K(V_1+V_2)$$

Total: the voltage of the line conductor against earth (the **phase voltage**) is

$$V = V_1 + V_2 + V_3$$

5. String efficiency

$$\eta_{string} = \frac{\text{voltage across the string}}{n \times \text{voltage across the disc nearest the conductor}} = \frac{V}{n\,V_n}$$

It is 100 % when all discs share equally, and it falls as the sharing becomes more unequal. Its derivation is in 3(d).

2. Answer

Given:

- Line voltage $V_L = 33$ kV, so the voltage across the string (phase voltage) is $V = 33/\sqrt{3} = 19.053$ kV
- Number of units $n = 3$
- $K = 11\% = 0.11$

Relationships (figure above, V_1 = top unit, V_3 = line unit):

$$V_2 = V_1(1+K), \qquad V_3 = V_2 + K(V_1+V_2), \qquad V = V_1 + V_2 + V_3$$

Step 1: Express V_2 and V_3 through V_1

$$V_2 = 1.11\,V_1$$

$$V_3 = 1.11\,V_1 + 0.11(V_1 + 1.11\,V_1) = 1.11\,V_1 + 0.2321\,V_1 = 1.3421\,V_1$$

Step 2: Find V_1 from the phase voltage

$$V = V_1(1 + 1.11 + 1.3421) = 3.4521\,V_1$$

$$V_1 = \frac{19.053}{3.4521} = 5.519 \text{ kV}$$

Step 3: (i) Voltage on each unit

$$V_1 = 5.519 \text{ kV}, \qquad V_2 = 1.11 \times 5.519 = 6.126 \text{ kV}, \qquad V_3 = 1.3421 \times 5.519 = 7.407 \text{ kV}$$

$$\boxed{V_1 = 5.52 \text{ kV (top)}, \quad V_2 = 6.13 \text{ kV}, \quad V_3 = 7.41 \text{ kV (line unit)}}$$

Check: $5.519 + 6.126 + 7.407 = 19.05$ kV. The unit nearest the line takes the most voltage.

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Step 4: (ii) String efficiency

$$\eta = \frac{V}{nV_3} = \frac{19.053}{3 \times 7.407} = 0.8574$$

$$\boxed{\eta = 85.74\ \%}$$

Final Answer: $V_1 = 5.52$ kV, $V_2 = 6.13$ kV, $V_3 = 7.41$ kV; string efficiency 85.74 %.

3. Notes and Remarks

- **Why the phase voltage?** The string is between the line conductor and the earthed tower, so it carries the line-to-earth (phase) voltage, $V_L/\sqrt{3}$.
- The same method, run backwards (from measured unit voltages to K), is 5(c); with five units it is 8(a).

3.(d) Derive the mathematical expression of String efficiency with proper diagram. What will happen if: (i) $K = 0.2$ ($K = \frac{shunt\ capacitance}{self\ capacitance}$) (ii) $K = 0.8$ (iii) the number of disc increases in the string. Marks: 10

1. Theory

The basic ideas (self capacitance C, shunt capacitance KC, and why the unit nearest the line takes the most voltage) are explained in the Theory of 3(c). Here we derive the result in general form and see what changes it.

1. Why string efficiency matters (Core Idea)

The string fails when its **worst** disc (the one nearest the line) is overstressed. If the voltage were shared equally, each disc would take V/n. With unequal sharing, the line unit takes more than V/n, so the other discs are not used to their full strength. String efficiency measures how well the discs are used: 100 % means perfectly equal sharing.

2. Answer

String efficiency is the ratio of the voltage across the whole string to n times the voltage across the unit nearest the line conductor:

$$\eta_{string} = \frac{V}{n \times V_n} \times 100\ \%$$

Derivation (string of three units). Let C be the self capacitance of each unit and KC the shunt capacitance from each metal link to the tower (diagram in 3(c)). Let V_1, V_2, V_3 be the voltages across the top, middle and line units, and V the voltage across the string.

At junction A (between units 1 and 2), by Kirchhoff's current law, $I_2 = I_1 + i_1$:

$$\omega C V_2 = \omega C V_1 + \omega K C V_1 \quad \Rightarrow \quad V_2 = (1+K)V_1 \qquad (1)$$

At junction B (between units 2 and 3), $I_3 = I_2 + i_2$, and B is at voltage $V_1 + V_2$ to the tower:

$$\omega C V_3 = \omega C V_2 + \omega K C (V_1 + V_2)$$

$$V_3 = V_2 + K(V_1 + V_2) = (1+K)V_1 + K(2+K)V_1 = (1 + 3K + K^2)V_1 \qquad (2)$$

Voltage across the string:

$$V = V_1 + V_2 + V_3 = V_1\left[1 + (1+K) + (1+3K+K^2)\right] = V_1(3 + 4K + K^2) \qquad (3)$$

String efficiency (the line unit has the largest voltage, V_3):

$$\boxed{\eta = \frac{V}{3V_3} = \frac{3 + 4K + K^2}{3(1 + 3K + K^2)}}$$

For any number of units the same steps give the rule $V_{k+1} = V_k + K(V_1 + V_2 + \cdots + V_k)$, and $\eta = V/(nV_n)$.

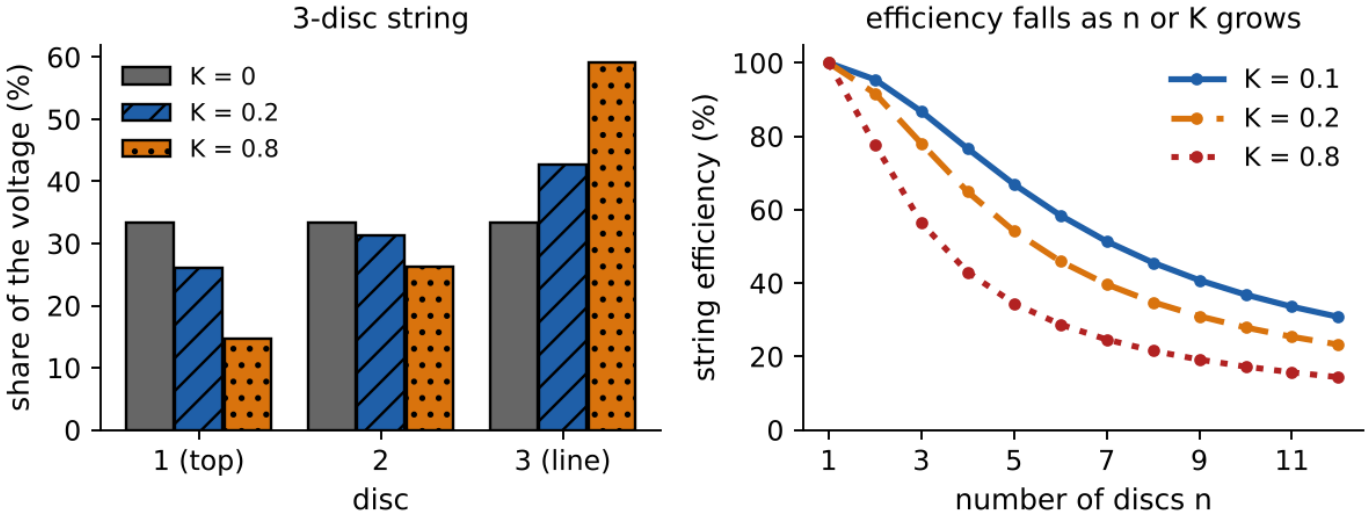


Figure 15: Left: the share of the voltage on each disc of a 3-disc string for K = 0, 0.2 and 0.8. Right: string efficiency against the number of discs for K = 0.1 (solid blue), 0.2 (dashed orange) and 0.8 (dotted red)

What happens if:

(i) $K = 0.2$ (three units):

$$V_2 = 1.2\,V_1, \qquad V_3 = (1 + 0.6 + 0.04)V_1 = 1.64\,V_1, \qquad V = 3.84\,V_1$$

$$\eta = \frac{3.84}{3 \times 1.64} = 78.05\ \%$$

The line unit takes $1.64/3.84 = 42.7$ % of the voltage instead of 33.3 %. The sharing is clearly unequal.

(ii) $K = 0.8$**:**

$$V_2 = 1.8\,V_1, \qquad V_3 = (1 + 2.4 + 0.64)V_1 = 4.04\,V_1, \qquad V = 6.84\,V_1$$

$$\eta = \frac{6.84}{3 \times 4.04} = 56.44\ \%$$

The line unit now takes 59 % of the whole voltage, and the top unit only 15 %. A large shunt capacitance (conductor too close to the tower, short cross-arm) makes the distribution very poor, and the line unit may break down.

(iii) The number of discs increases: each extra disc adds another leakage path to the tower, so the voltage piles up more and more on the units near the line. The efficiency **decreases** as n grows. For $K = 0.1$: $n = 3$ gives 86.8 %, $n = 5$ gives 66.9 %, $n = 8$ gives 45.6 % and $n = 12$ gives 30.8 %. So adding discs raises the string's voltage rating by less than in proportion: the extra discs near the tower carry little voltage.

Conclusion: the efficiency is highest when K is small and the string is short. It is improved by:

- **longer cross-arms**, which move the string away from the tower and reduce K;
- **grading the units**, using discs of larger capacitance near the line;
- a **guard ring** (static shielding), a metal ring around the bottom unit connected to the line, which supplies the leakage currents.

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3. Keyword Explanation

1. Self capacitance (C)

The capacitance of one disc, between its metal cap and pin.

2. Shunt capacitance (KC)

The capacitance between a metal link of the string and the earthed tower.

3. Guard ring

A metal ring around the line-end unit, connected to the conductor, that equalises the voltage sharing.

4. Notes and Remarks

- **Check with $K = 0$:** $\eta = 3/3 = 100$ %, as expected for equal sharing.
- **Rain makes it worse:** a wet string has leakage paths over its surface, which can change the sharing further.

5. Final Concise Answer

(a) By flash-over (an arc through the surrounding air from the conductor to the pin; the insulator survives) or puncture (a discharge through the porcelain; the insulator is destroyed). Safety factor = puncture strength / flash-over voltage ≈ 10, so flash-over occurs first.

(b) Poles: wooden poles (cheap, short spans of 40–50 m, rural lines), steel poles (rail or tubular, 50–80 m spans, towns) and RCC poles (strong, long life, 80–200 m spans); A- or H-type double poles for heavy loads.

(c) $V = 19.05$ kV; $V_1 = 5.52$ kV, $V_2 = 6.13$ kV, $V_3 = 7.41$ kV; $\eta = 85.74$ %.

(d) $V_2 = (1+K)V_1$, $V_3 = (1+3K+K^2)V_1$, $\eta = (3+4K+K^2)/[3(1+3K+K^2)]$. $K = 0.2$: $\eta = 78.05$ %; $K = 0.8$: $\eta = 56.44$ %; more discs: efficiency falls (for $K = 0.1$, from 86.8 % with 3 discs to 30.8 % with 12).

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4. (a) What is Stringing charts? Marks: 2
   (b) The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2.5 cm and the resistivity of insulation is $4.5 \times 10^{14}\ \Omega - cm$, find the insulation thickness. Marks: 5
   (c) Derive the expression of inductance of composite conductor lines. Marks: 8
   (d) A 66 kV single-core lead sheathed cable is graded by using two dielectrics of relative permittivity 5 and 3 respectively; thickness of each being 1 cm. The core diameter is 2 cm. Determine the maximum stress in the two dielectrics. Marks: 10

4.(a) What is Stringing charts? Marks: 2

1. Theory

1. What is sag? (Core Idea)

A conductor hung between two supports cannot be perfectly straight: its own weight pulls it down into a curve. The vertical distance between the level of the supports and the lowest point of the conductor is the **sag**.

We could pull the conductor tighter to reduce the sag, but that raises its **tension**, and too much tension can break it. Too little tension gives a large sag, and the conductor may come too close to the ground. The sag must be a safe compromise.

2. How sag and tension are related

For a span l between supports at the same level, a conductor of weight w per metre pulled with tension T hangs (very nearly) as a parabola with sag

$$S = \frac{w\, l^2}{8\, T}$$

where S is the sag (m), w the weight per metre (N/m), l the span (m) and T the tension (N). For the same span, **more tension means less sag**.

3. Temperature changes everything

A metal conductor gets **longer when it is hot** and shorter when it is cold.

- On a hot day the conductor lengthens, so it hangs lower: **sag rises and tension falls**.
- On a cold day it shortens, so it is pulled tight: **sag falls and tension rises**. Ice and wind add to the load, so the worst tension comes in cold, icy, windy weather.

The line must stay safe in both extremes: enough ground clearance at the highest temperature, and tension below the safe limit at the lowest temperature with ice and wind.

4. Stringing the line

When a line is first erected (**stringing**), the workers pull each conductor to a chosen sag. But the sag that is right depends on the temperature on that day. So they need a ready-made graph that tells them the correct sag and tension for the temperature at which they are working.

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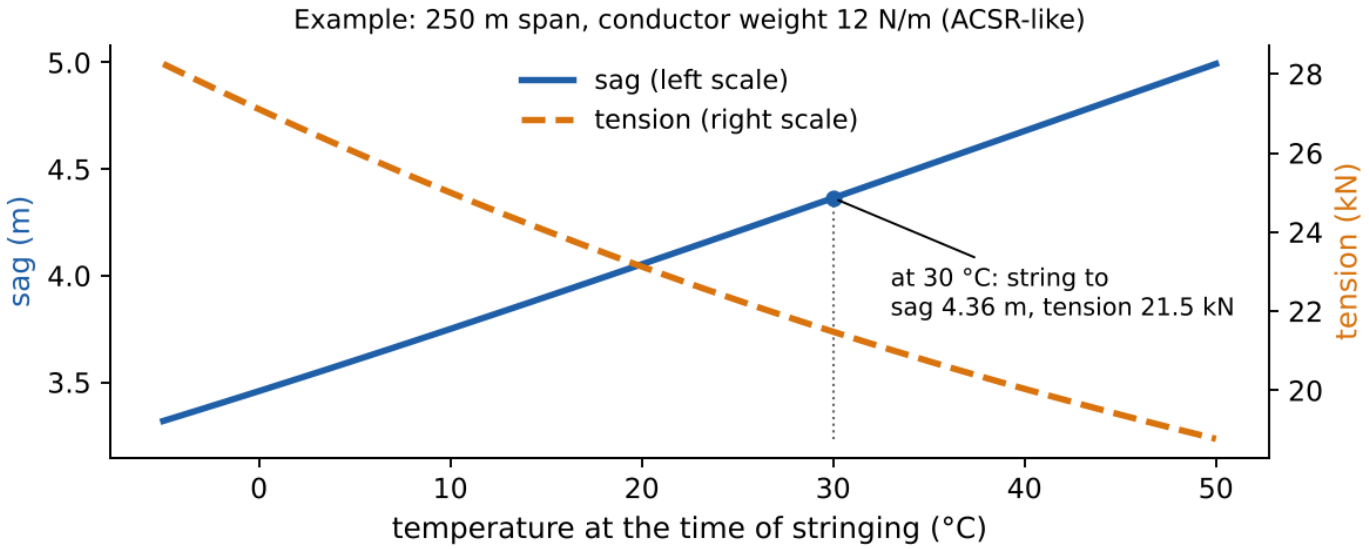


Figure 16: A stringing chart: for one span, sag (solid blue, left scale) rises and tension (dashed orange, right scale) falls as the temperature rises

Reading the figure. This example is a 250 m span of a conductor weighing 12 N/m. At 10 °C the correct tension is 25 kN (sag 3.75 m); if the line is strung on a 30 °C day, it must be pulled to only 21.5 kN, with a sag of 4.36 m. These curves are calculated so that the line is safe at both temperature extremes.

2. Answer

A **stringing chart** is a set of curves, drawn for a given conductor and span, that shows the **sag** and the **tension** of the conductor against the **temperature**. It is used while erecting (stringing) an overhead line: from the temperature at the time of erection, the lineman reads the sag (or tension) to which the conductor must be pulled, so that the conductor keeps safe ground clearance at the highest temperature and safe tension at the lowest temperature (with ice and wind). As the temperature rises, the sag increases and the tension decreases.

3. Keyword Explanation

1. Sag

The vertical distance between the point of support and the lowest point of the conductor.

2. Span

The horizontal distance between two neighbouring supports.

4.(b) The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2.5 cm and the resistivity of insulation is $4.5 \times 10^{14}\ \Omega - cm$, find the insulation thickness. Marks: 5

1. Theory

1. What is insulation resistance? (Core Idea)

No insulator is perfect. A tiny **leakage current** always flows from the core of a cable, through the insulation, to the earthed sheath. The opposition to this current is the **insulation resistance** of the cable. It should be very high (hundreds of megohms). A low value means damaged or damp insulation.

2. Which way does the leakage current flow?

The leakage current flows **radially**, straight outwards from the core to the sheath, not along the cable. So its path length is the **thickness** of the insulation, and the area it flows through is the area of a cylinder around the core.

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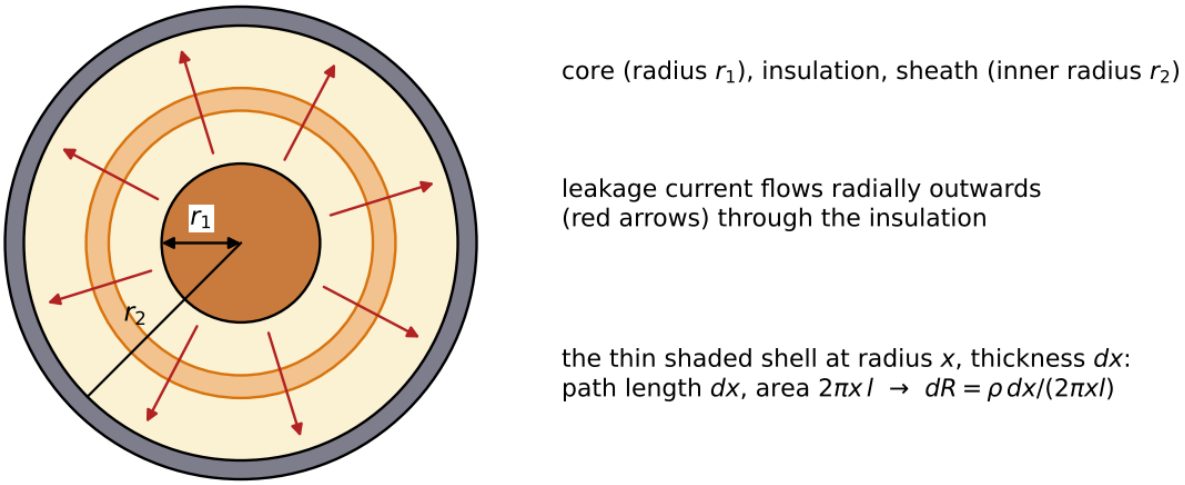


Figure 17: Leakage current flows radially from the core (radius r1) to the sheath (radius r2); a thin shell of radius x and thickness dx is a small resistance in the path

Reading the figure. The red arrows show the leakage current flowing outwards. The shaded thin shell at radius x has thickness dx (its path length) and area $2\pi x\,l$ for a cable of length l.

3. The formula

The resistance of any piece of material is $R = \rho \times \dfrac{\text{length}}{\text{area}}$. For the thin shell:

$$dR = \frac{\rho\,dx}{2\pi x\,l}$$

The shells are in series from the core ($x = r_1$) to the sheath ($x = r_2$), so we add them up:

$$R = \int_{r_1}^{r_2} \frac{\rho\,dx}{2\pi x\,l} = \frac{\rho}{2\pi l} \ln \frac{r_2}{r_1}$$

where

- R is the insulation resistance (Ω)
- ρ is the resistivity of the insulation (Ω-cm)
- l is the length of the cable (cm, to match ρ)
- r_1 is the core radius and r_2 the inner radius of the sheath

Two things to notice: R falls as the cable gets longer (more leakage paths in parallel), which is why it is quoted "per km"; and R depends on the **ratio** r_2/r_1, not on the thickness alone.

2. Answer

Given:

- Insulation resistance $R = 495$ MΩ $= 495 \times 10^6$ Ω for $l = 1$ km $= 10^5$ cm
- Core diameter $= 2.5$ cm, so $r_1 = 1.25$ cm
- Resistivity $\rho = 4.5 \times 10^{14}$ Ω-cm

Relationship:

$$R = \frac{\rho}{2\pi l} \ln \frac{r_2}{r_1}$$

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Step 1: Find $\ln(r_2/r_1)$

$$\ln \frac{r_2}{r_1} = \frac{2\pi l R}{\rho} = \frac{2\pi \times 10^5 \times 495 \times 10^6}{4.5 \times 10^{14}} = 0.6912$$

Step 2: Find the sheath radius

$$\frac{r_2}{r_1} = e^{0.6912} = 1.996, \qquad r_2 = 1.996 \times 1.25 = 2.495 \text{ cm}$$

Step 3: Insulation thickness

$$t = r_2 - r_1 = 2.495 - 1.25 = 1.245 \text{ cm}$$

$$\boxed{t = 1.245 \text{ cm} \approx 12.5 \text{ mm}}$$

The ratio r_2/r_1 is almost exactly 2, so the insulation is about as thick as the core radius.
Final Answer: the insulation thickness is **1.245 cm**.

3. Notes and Remarks

- **Units:** ρ is in Ω-cm, so l must be in cm (1 km $= 10^5$ cm). Mixing metres and centimetres is the usual mistake here.

4.(c) Derive the expression of inductance of composite conductor lines. Marks: 8

1. Theory

1. What is a composite conductor? (Core Idea)

Most conductors are not one solid wire. They are made of many thin wires (**strands**) twisted together, so the conductor is flexible. A conductor made of several parallel strands, all carrying the same direction of current, is a **composite conductor**. Bundled conductors (7(a)) are also composite conductors.

To find its inductance we treat each strand as a small round wire and use the flux-linkage result of 1(c):

- a round wire of radius r carrying current I has a flux linkage up to a distance D of $2 \times 10^{-7}\, I \ln(D/r')$, with $r' = 0.7788\, r$;
- when the currents of a group add up to zero, the flux linkage of one wire is $2\times10^{-7} \sum I_k \ln(1/D_k)$, where D_k is its distance to wire k (and r' for itself).

2. Two new words: GMD and GMR

- **GMD (geometric mean distance), D_m:** take the distance from every strand of one conductor to every strand of the other conductor, multiply them all, and take the root of the same order as the number of distances.
- **GMR (self GMD), D_s:** do the same with the distances between every pair of strands **of the same conductor**, counting each strand's distance to itself as its own r'.

The GMD says how far the two conductors are from each other on average; the GMR says how "spread out" one conductor is. A conductor that is more spread out has a larger GMR and a smaller inductance.

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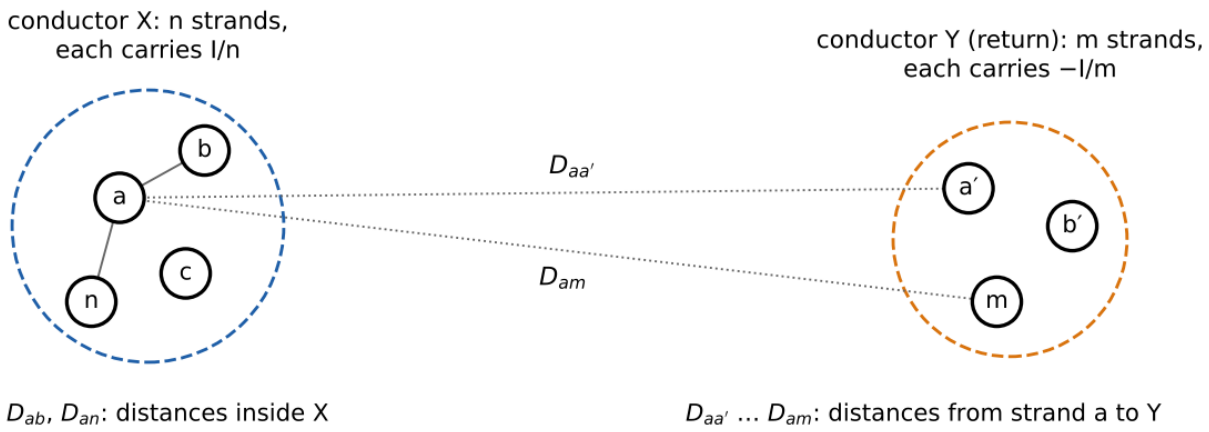


Figure 18: Conductor X with n strands, each carrying I/n, and the return conductor Y with m strands, each carrying −I/m

Reading the figure. Conductor X (left, n strands) carries the current I out, split equally between its strands; conductor Y (right, m strands) carries it back. D_{ab} and D_{an} are distances between strands of X; $D_{aa'}$ to D_{am} are distances from strand a to the strands of Y.

2. Answer

Consider a single-phase line made of two composite conductors (figure above). Conductor X has n identical parallel strands, each carrying I/n; the return conductor Y has m strands, each carrying $-I/m$. The currents add up to zero.

Where the flux linkage comes from. A long wire carrying I_k links the flux $2\times10^{-7}I_k\ln(1/D_k)$ Wb-T/m with a wire a distance D_k away. The fluxes of all the wires add, and the wire's own current counts too, with D_k replaced by its GMR r' (which includes the flux inside the wire). The total is

$$\psi = 2\times10^{-7}\sum_k I_k \ln\frac{1}{D_k}$$

The sum is finite because the currents add up to zero.

Flux linkage of strand a of X (its own current and the other $n-1$ strands of X, then the m strands of Y):

$$\psi_a = 2\times10^{-7}\left[\frac{I}{n}\left(\ln\frac{1}{r'_a}+\ln\frac{1}{D_{ab}}+\cdots+\ln\frac{1}{D_{an}}\right)-\frac{I}{m}\left(\ln\frac{1}{D_{aa'}}+\ln\frac{1}{D_{ab'}}+\cdots+\ln\frac{1}{D_{am}}\right)\right]$$

Combine the logarithms. Three rules are used: $\ln A+\ln B=\ln(AB)$, $\frac{1}{n}\ln A=\ln\sqrt[n]{A}$, and $\ln\frac{1}{A}-\ln\frac{1}{B}=\ln\frac{B}{A}$. The first rule joins each bracket into one logarithm:

$$\psi_a = 2\times10^{-7}\,I\left[\frac{1}{n}\ln\frac{1}{r'_a D_{ab}\cdots D_{an}}-\frac{1}{m}\ln\frac{1}{D_{aa'}D_{ab'}\cdots D_{am}}\right]$$

The second rule moves $1/n$ and $1/m$ inside as roots:

$$\psi_a = 2\times10^{-7}\,I\left[\ln\frac{1}{\sqrt[n]{r'_a D_{ab}\cdots D_{an}}}-\ln\frac{1}{\sqrt[m]{D_{aa'}D_{ab'}\cdots D_{am}}}\right]$$

The third rule joins the two:

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$$\psi_a = 2\times10^{-7}\, I \ln \frac{\sqrt[m]{D_{aa'}D_{ab'}\cdots D_{am}}}{\sqrt[n]{r'_a D_{ab}\cdots D_{an}}} \quad \text{Wb-T/m}$$

Inductance of strand a. Inductance is flux linkage per ampere, and this strand carries only I/n. Dividing by I/n cancels I and brings in the factor n:

$$L_a = \frac{\psi_a}{I/n} = 2n\times10^{-7} \ln \frac{\sqrt[m]{D_{aa'}D_{ab'}\cdots D_{am}}}{\sqrt[n]{r'_a D_{ab}\cdots D_{an}}} \quad \text{H/m}$$

Inductance of strand b, in the same way:

$$L_b = 2n\times10^{-7} \ln \frac{\sqrt[m]{D_{ba'}D_{bb'}\cdots D_{bm}}}{\sqrt[n]{D_{ba} r'_b \cdots D_{bn}}} \quad \text{H/m}$$

and so on for $L_c, \ldots, L_n$.

Inductance of conductor X. The average inductance of one strand is $L_{av} = (L_a + L_b + \cdots + L_n)/n$. The n strands are in parallel, all with (nearly) this inductance, and n equal inductors in parallel give one n-th of the value, so

$$L_X = \frac{L_{av}}{n} = \frac{L_a + L_b + \cdots + L_n}{n^2}$$

Put in the strand inductances. Each $L_a, L_b, \ldots$ has the factor $2n\times10^{-7}$, and there are n of them. Call the logarithm in L_a simply $\ln(A_a/B_a)$, where A_a is its m-th root and B_a its n-th root, and likewise for the others:

$$L_X = \frac{2n\times10^{-7}}{n^2}\left[\ln\frac{A_a}{B_a} + \ln\frac{A_b}{B_b} + \cdots + \ln\frac{A_n}{B_n}\right] = 2\times10^{-7}\cdot\frac{1}{n}\ln\frac{A_aA_b\cdots A_n}{B_aB_b\cdots B_n}$$

The factor $1/n$ becomes an n-th root of the whole fraction:

$$L_X = 2\times10^{-7}\ln\frac{\sqrt[n]{A_aA_b\cdots A_n}}{\sqrt[n]{B_aB_b\cdots B_n}}$$

Each A is already an m-th root, so the top is an mn-th root of all the distances from X to Y. Each B is already an n-th root, so the bottom is an n^2-th root of all the distances within X:

$$L_X = 2\times10^{-7}\ln\frac{\sqrt[mn]{(D_{aa'}D_{ab'}\cdots D_{am})(D_{ba'}D_{bb'}\cdots D_{bm})\cdots(D_{na'}\cdots D_{nm})}}{\sqrt[n^2]{(r'_aD_{ab}\cdots D_{an})(D_{ba}r'_b\cdots D_{bn})\cdots(D_{na}D_{nb}\cdots r'_n)}}$$

Giving the top the name D_m and the bottom the name D_{sX}:

$$\boxed{L_X = 2\times10^{-7}\ln\frac{D_m}{D_{sX}} \quad \text{H/m}}$$

where

$$D_m = \sqrt[mn]{(D_{aa'}D_{ab'}\cdots D_{am})(D_{ba'}D_{bb'}\cdots D_{bm})\cdots(D_{na'}\cdots D_{nm})}$$

is the **mutual GMD** between X and Y (mn distances), and

$$D_{sX} = \sqrt[n^2]{(D_{aa}D_{ab}\cdots D_{an})(D_{ba}D_{bb}\cdots D_{bn})\cdots(D_{na}\cdots D_{nn})}, \qquad D_{aa} = r'_a,\ D_{bb} = r'_b,\ \ldots$$

is the **self GMD (GMR)** of X (n^2 distances).

Inductance of Y. The same steps with the roles of X and Y swapped give $L_Y = 2\times10^{-7}\ln(D_m/D_{sY})$, with the same D_m.

Loop inductance of the line. The go and return conductors are in series, so their inductances add, and $\ln\frac{D_m}{D_{sX}} + \ln\frac{D_m}{D_{sY}} = \ln\frac{D_m^2}{D_{sX}D_{sY}}$:

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$$L = L_X + L_Y = 2 \times 10^{-7} \ln \frac{D_m^2}{D_{sX} D_{sY}} \text{ H/m}$$

Check: for two solid round wires ($n = m = 1$) a distance d apart, $D_m = d$ and $D_s = r'$, so $L = 4 \times 10^{-7} \ln(d/r')$, the familiar single-phase result.

3. Keyword Explanation

1. GMD (mutual geometric mean distance)

The mn-th root of the product of all distances between the strands of one conductor and the strands of the other.

2. GMR (self geometric mean distance)

The n^2-th root of the product of all distances between the strands of one conductor, each strand's distance to itself taken as $r' = 0.7788\, r$.

4. Notes and Remarks

- **Why GMD and GMR are so useful:** the inductance formula for any arrangement keeps the same simple form, $2 \times 10^{-7} \ln(\text{GMD}/\text{GMR})$. Only the two distances change.
- **Stranded conductors:** the GMR of a 7-strand conductor is $0.726\, R$ and of a 19-strand conductor $0.758\, R$, where R is the overall radius; tables give these values.

4.(d) A 66 kV single-core lead sheathed cable is graded by using two dielectrics of relative permittivity 5 and 3 respectively; thickness of each being 1 cm. The core diameter is 2 cm. Determine the maximum stress in the two dielectrics. Marks: 10

1. Theory

The basics

A cable's insulation carries an electric stress, the voltage per unit thickness. The stress is highest next to the conductor. **Grading** evens this out by using insulation of a higher permittivity near the conductor.
This is the two-layer case of the capacitance grading derived in 2(d). The key facts from there:

1. The same charge passes through every layer (Core Idea)

The charge Q per metre on the core sends the same electric flux through each layer. In a layer of relative permittivity ε_r, the stress at radius x is

$$g = \frac{Q}{2\pi\varepsilon_0\varepsilon_r\, x}$$

so each layer has its maximum stress at its **inner** surface:

$$g_{1max} = \frac{Q}{\pi\varepsilon_0\varepsilon_1 d}, \qquad g_{2max} = \frac{Q}{\pi\varepsilon_0\varepsilon_2 d_1}$$

where d is the core diameter and d_1 the outer diameter of the inner layer.

2. Two equations for two unknowns

Dividing the two maximum stresses gives the link between them:

$$\varepsilon_1 d\, g_{1max} = \varepsilon_2 d_1\, g_{2max}$$

The voltage between core and sheath is the sum of the voltages across the two layers:

$$V = \frac{g_{1max}\, d}{2} \ln \frac{d_1}{d} + \frac{g_{2max}\, d_1}{2} \ln \frac{D}{d_1}$$

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where D is the inside diameter of the sheath. These two equations give g_{1max} and g_{2max}.

2. Answer

Given:

- Voltage between core and sheath: $V = 66$ kV (r.m.s.; see the assumption below)
- Core diameter $d = 2$ cm
- Inner dielectric: $\varepsilon_1 = 5$, thickness 1 cm, so $d_1 = 2 + 2 \times 1 = 4$ cm
- Outer dielectric: $\varepsilon_2 = 3$, thickness 1 cm, so $D = 4 + 2 \times 1 = 6$ cm

Assumption: following V.K. Mehta, the 66 kV is taken as the r.m.s. voltage between the core and the earthed sheath, so the stresses come out as r.m.s. values (peak values are given at the end).

Step 1: Relation between the two maximum stresses

$$\varepsilon_1 d\, g_{1max} = \varepsilon_2 d_1\, g_{2max} \quad \Rightarrow \quad g_{2max} = \frac{\varepsilon_1 d}{\varepsilon_2 d_1}\, g_{1max} = \frac{5 \times 2}{3 \times 4}\, g_{1max} = 0.8333\, g_{1max}$$

Step 2: Voltage equation

$$V = \frac{g_{1max}\, d}{2} \ln \frac{d_1}{d} + \frac{g_{2max}\, d_1}{2} \ln \frac{D}{d_1}$$

$$66 = g_{1max} \times 1 \times \ln 2 + 0.8333\, g_{1max} \times 2 \times \ln 1.5$$

$$66 = g_{1max}\,(0.6931 + 0.6758) = 1.3689\, g_{1max}$$

Step 3: Maximum stress in each dielectric

$$g_{1max} = \frac{66}{1.3689} = 48.21 \text{ kV/cm}$$

$$g_{2max} = 0.8333 \times 48.21 = 40.18 \text{ kV/cm}$$

$$\boxed{g_{1max} = 48.21 \text{ kV/cm (inner, at the core)}, \qquad g_{2max} = 40.18 \text{ kV/cm (outer, at } d_1)}$$

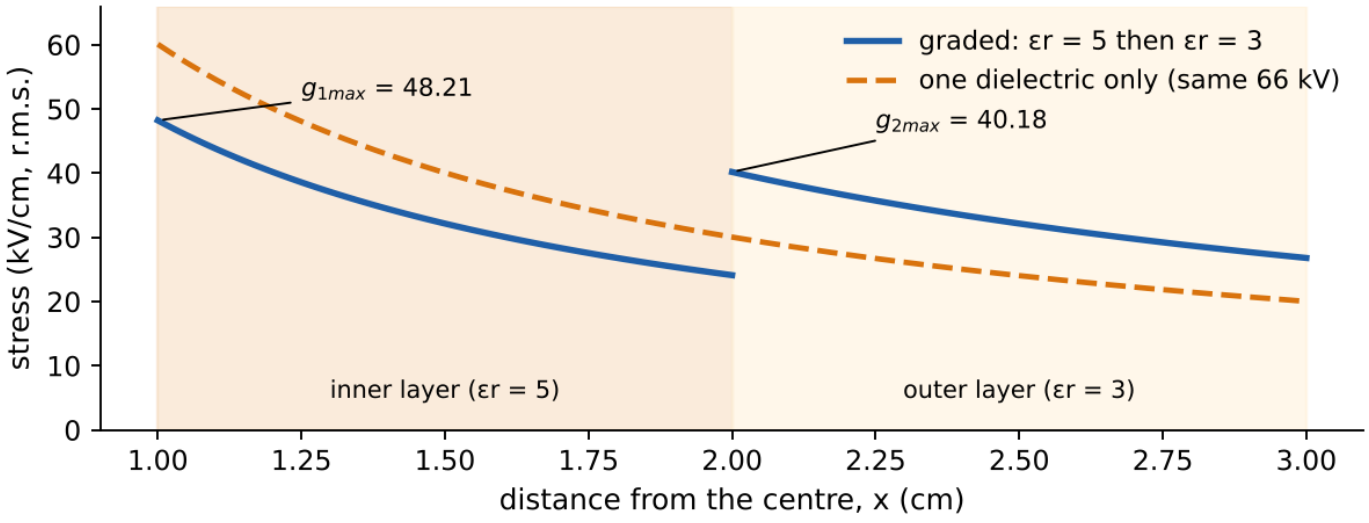


Figure 19: Stress across the graded cable (solid blue) and across the same cable with one dielectric only (dashed orange)

Reading the figure. In the inner layer (εr $= 5$) the stress falls from 48.21 kV/cm at the core to 24.1 kV/cm at d_1; there it jumps up to 40.18 kV/cm in the outer layer (εr $= 3$) and falls to 26.8 kV/cm at the sheath. With one dielectric only (dashed orange), the peak at the core would be 60.08 kV/cm.

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Step 4: Check the voltage split

$$V_1 = 48.21 \times 1 \times \ln 2 = 33.42 \text{ kV}, \qquad V_2 = 40.18 \times 2 \times \ln 1.5 = 32.58 \text{ kV}$$

$V_1 + V_2 = 66.0$ kV, as required. The two layers share the voltage almost equally.

Final Answer: maximum stress in the inner dielectric (εr $= 5$) = **48.21 kV/cm**; in the outer dielectric (εr $= 3$) = **40.18 kV/cm** (r.m.s. values).

3. Notes and Remarks

- **Peak values:** if 66 kV is treated as the r.m.s. value and the peak stresses are wanted, multiply by $\sqrt{2}$: $g_{1max} = 68.18$ kV/cm and $g_{2max} = 56.82$ kV/cm.
- **What grading gained:** with a single dielectric and the same overall diameter, $g_{max} = \dfrac{66}{1 \times \ln 3} = 60.08$ kV/cm. Grading lowers the peak by 20 %.
- **Higher permittivity inside:** if the layers were swapped ($\varepsilon = 3$ inside), the inner stress would rise above the ungraded value; the material with the larger ε_r must always be next to the core.

4. Final Concise Answer

(a) A stringing chart is a set of curves of sag and tension against temperature for a given conductor and span; it tells the lineman the correct sag (or tension) for the temperature at the time of erection, so the line stays safe at both temperature extremes.

(b) $\ln(r_2/r_1) = 2\pi lR/\rho = 0.6912$; $r_2 = 2.495$ cm; insulation thickness **1.245 cm**.

(c) $L_X = 2 \times 10^{-7} \ln(D_m/D_{sX})$ H/m, with D_m the mutual GMD (mn-th root of all mn distances between the two conductors) and D_{sX} the GMR (n^2-th root of all distances within X, $D_{aa} = r'$); loop inductance $L = L_X + L_Y$.

(d) $g_{2max} = 0.8333\, g_{1max}$; $66 = 1.3689\, g_{1max}$; $g_{1max} = 48.21$ kV/cm, $g_{2max} = 40.18$ kV/cm (r.m.s.; peak 68.18 and 56.82 kV/cm).

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5. (a) How Earth affect on transmission line capacitance? Marks: 2

(b) A three phase 50 Hz transmission line has flat horizontal spacing with 3.5 m between adjacent conductors. The conductors are No. 2/0 hard/drawn seven strand copper (outside conductor diameter= 1.05 cm). The voltage of the line is 110 kV. Find the capacitance to neutral and the charging current per kilometre of line. Marks: 5

(c) A 3-phase transmission line is being supported by three-disc insulators. The potentials across top unit (near to the tower) and middle unit are 8 kV and 11 kV, respectively.

(i) Draw the equivalent circuit
(ii) Calculate the ratio of capacitance between pin and earth to the self-capacitance of each unit
(iii) Calculate the line voltage and
(iv) Calculate string efficiency. Marks: 8

(d) Derive the expression of capacitance of a three phase line with unsymmetrical spacing. Marks: 10

5.(a) How Earth affect on transmission line capacitance? Marks: 2

1. Theory

1. Why does a line have capacitance? (Core Idea)

A **capacitor** is two conductors separated by an insulator. The conductors of an overhead line, with air between them, are exactly that. When a voltage is applied, positive charge collects on one wire and negative charge on the other, and electric field lines run from one to the other. The **capacitance** is the charge stored per volt, $C = q/V$. It draws a **charging current** from the supply even when no load is connected.
For a single-phase line of two wires of radius r, spacing d, far from everything else:

$$C = \frac{\pi\varepsilon_0}{\ln(d/r)} \quad \text{F/m (between the wires)}$$

2. The earth is a conductor too

The ground under the line is (slightly damp) soil, which conducts. It is at **zero potential**. So some of the field lines from the conductors end on the earth instead of on the other wire. The earth changes the shape of the field, and so it changes the capacitance.

3. The method of images

A neat trick replaces the earth by an imaginary charge. A conducting plane at zero potential has the same effect as an **image charge** of the opposite sign, placed at the same depth below the surface as the real charge is above it (like a mirror image).

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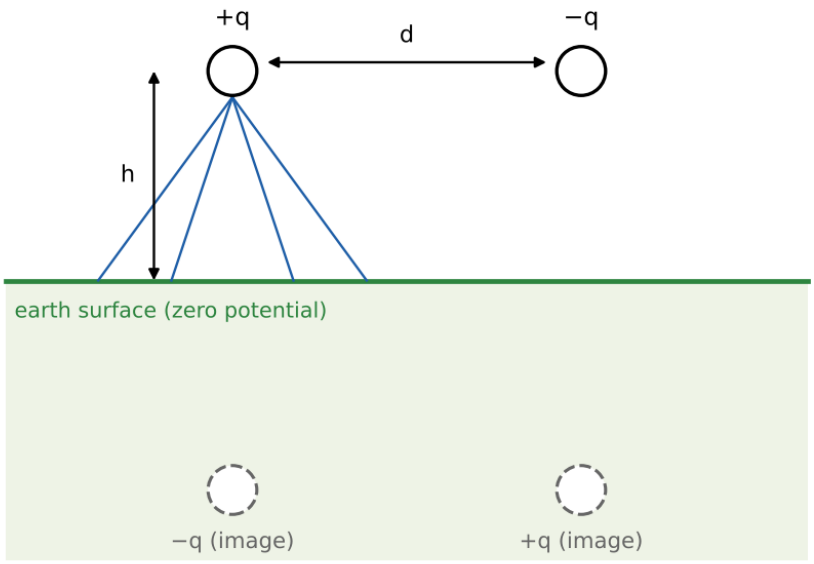


Figure 20: Two-wire line at height h above the earth. Field lines end on the earth; the earth acts like image charges of opposite sign at depth h below the surface

Reading the figure. The real conductors carry $+q$ and $-q$ at height h. The earth behaves as if a $-q$ and a $+q$ (dashed grey) sat at depth h below it. The capacitance can then be found from four charges in free space, with no earth.
With the images included, the capacitance of the single-phase line becomes

$$C = \frac{\pi\varepsilon_0}{\ln\left(\dfrac{d}{r}\cdot\dfrac{2h}{\sqrt{4h^2+d^2}}\right)} \ \text{F/m}$$

Since $2h/\sqrt{4h^2+d^2} < 1$, the logarithm is smaller than $\ln(d/r)$, so the capacitance is **larger** than without the earth.

2. Answer

The earth is a conductor at zero potential, so it changes the electric field of the line. Its effect is found by the **method of images**: the earth is replaced by **image conductors** carrying equal and opposite charges at the same depth below the ground as the real conductors are above it.

- The images bring opposite charges effectively closer to each conductor, so **the line capacitance increases** slightly. For a single-phase line, $C = \pi\varepsilon_0/\ln\left[\dfrac{d}{r}\cdot\dfrac{2h}{\sqrt{4h^2+d^2}}\right]$ instead of $\pi\varepsilon_0/\ln(d/r)$.
- The effect is small when the conductors are high above the ground compared with their spacing ($h \gg d$), so it is usually **neglected** in calculations; it matters for low lines and for unbalanced (earth-fault) conditions.

3. Keyword Explanation

1. Image charge

An imaginary charge of opposite sign below a conducting plane, which produces the same field above the plane as the plane itself.

5.(b) A three phase 50 Hz transmission line has flat horizontal spacing with 3.5 m between adjacent conductors. The conductors are No. 2/0 hard/drawn seven strand copper (outside conductor diameter= 1.05 cm). The voltage of the line is 110 kV. Find the capacitance to neutral and the charging current per kilometre of line. Marks: 5

1. Theory

1. Capacitance to neutral (Core Idea)

In a three-phase line, each conductor forms a capacitor with the other two. To treat the three phases separately, we use the **capacitance to neutral** C_n: the capacitance between one conductor and the (imaginary) neutral point. Each phase then has a charging current

$$I_c = \frac{V_{ph}}{X_c} = 2\pi f\, C_n\, V_{ph}$$

where $V_{ph} = V_L/\sqrt{3}$ is the phase (line-to-neutral) voltage.

2. The formula with unsymmetrical spacing

For a transposed three-phase line (derived in 5(d)):

$$C_n = \frac{2\pi\varepsilon_0}{\ln(D_{eq}/r)} \text{ F/m}, \qquad D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$$

where r is the **outside radius** of the conductor. (For inductance we used $r' = 0.7788\,r$, because current flows inside the wire. Charge sits on the **surface**, so capacitance uses the real radius r.)

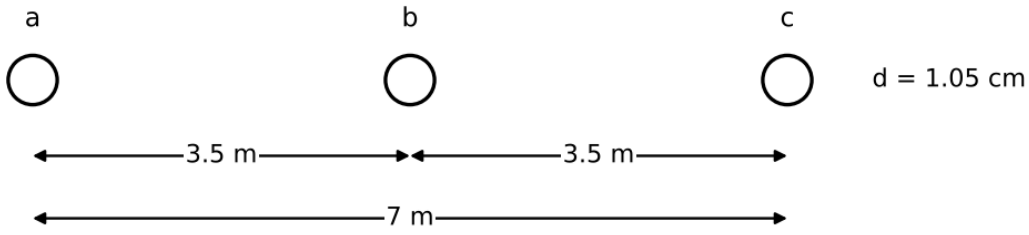


Figure 21: The three conductors in a horizontal row: 3.5 m between neighbours, 7 m between the outer two

2. Answer

Given:

- Flat spacing: $D_{12} = D_{23} = 3.5$ m, $D_{31} = 7$ m
- Conductor outside diameter 1.05 cm, so $r = 0.525$ cm $= 0.00525$ m
- Line voltage $V_L = 110$ kV, $f = 50$ Hz
- $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

Relationships:

$$C_n = \frac{2\pi\varepsilon_0}{\ln(D_{eq}/r)}, \qquad I_c = 2\pi f\, C_n\, V_{ph}$$

Step 1: Equivalent spacing

$$D_{eq} = \sqrt[3]{3.5 \times 3.5 \times 7} = \sqrt[3]{85.75} = 4.410 \text{ m}$$

Step 2: Capacitance to neutral

$$\frac{D_{eq}}{r} = \frac{4.410}{0.00525} = 840.0, \qquad \ln 840.0 = 6.733$$

$$C_n = \frac{2\pi \times 8.854 \times 10^{-12}}{6.733} = 8.262 \times 10^{-12} \text{ F/m}$$

Per kilometre:

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$$C_n = 8.262 \times 10^{-9} \text{ F/km} = 0.00826 \ \mu\text{F/km}$$

$$\boxed{C_n = 0.00826 \ \mu\text{F per km (to neutral)}}$$

Step 3: Charging current

$$V_{ph} = \frac{110}{\sqrt{3}} = 63.51 \text{ kV}$$

$$I_c = 2\pi \times 50 \times 8.262 \times 10^{-9} \times 63\,510 = 0.1648 \text{ A}$$

$$\boxed{I_c = 0.165 \text{ A per km}}$$

Final Answer: $C_n = 0.00826 \ \mu$F/km (8.26 nF/km) to neutral; charging current 0.165 A per km.

3. Notes and Remarks

- **"No. 2/0, seven strand"** only names the conductor size; for capacitance only the outside diameter (1.05 cm) is needed.
- **Earth neglected:** the conductor height is not given, so the effect of the earth (5(a)) is ignored, as usual.

5.(c) A 3-phase transmission line is being supported by three-disc insulators. The potentials across top unit (near to the tower) and middle unit are 8 kV and 11 kV, respectively. (i) Draw the equivalent circuit (ii) Calculate the ratio of capacitance between pin and earth to the self-capacitance of each unit (iii) Calculate the line voltage and (iv) Calculate string efficiency. Marks: 8

1. Theory

The basics

An overhead line hangs from a string of porcelain discs. Each disc is a capacitor, with self capacitance C. The metal links between discs also form a small **shunt capacitance** KC to the earthed tower. So the discs do not carry equal currents, and the disc nearest the line takes the largest voltage. **String efficiency** measures how uneven this is: the string's voltage over n times the largest disc's voltage.

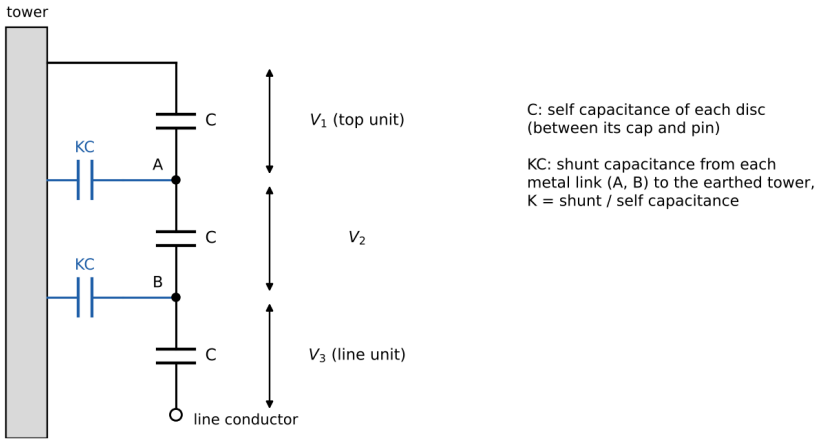


Figure 22: A string of insulators with its self and shunt capacitances

Reading the figure. Each shunt capacitance draws current from the links, so the discs near the line carry more current and take more voltage.

This uses the string theory of 3(c) and 3(d). In short:

1. The two equations of a three-unit string (Core Idea)

With self capacitance C per unit and shunt capacitance KC from each link to the tower, Kirchhoff's current law at the two links gives

$$V_2 = V_1(1+K), \qquad V_3 = V_2 + K(V_1+V_2)$$

where V_1 is the top unit (next to the tower) and V_3 the unit next to the line. The phase voltage is $V = V_1 + V_2 + V_3$, and the string efficiency is $\eta = V/(3V_3)$.

2. Working backwards

In 3(c) we knew K and found the voltages. Here we know two of the voltages, so the first equation gives K directly, and the rest follows.

2. Answer

Given: $V_1 = 8$ kV (top unit, near the tower), $V_2 = 11$ kV (middle unit), three units.

Step 1: (i) Equivalent circuit

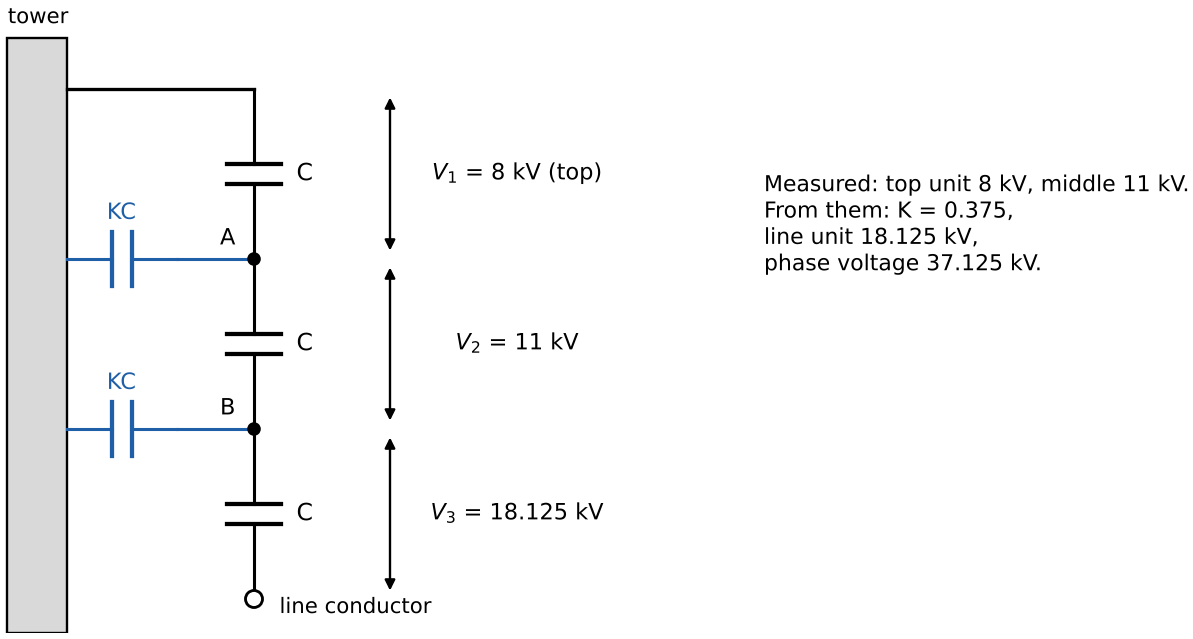


Figure 23: Equivalent circuit of the three-disc string with the measured voltages: self capacitance C per unit, shunt capacitance KC from links A and B to the tower

Each unit is a capacitor C (between its cap and pin); each metal link (A, B) has a shunt capacitance KC to the earthed tower.

Step 2: (ii) Ratio K

At link A:

$$V_2 = V_1(1+K) \quad \Rightarrow \quad K = \frac{V_2}{V_1} - 1 = \frac{11}{8} - 1 = 0.375$$

$$\boxed{K = 0.375}$$

The shunt capacitance is 37.5 % of the self capacitance.

Step 3: Voltage on the line unit

At link B:

$$V_3 = V_2 + K(V_1+V_2) = 11 + 0.375 \times (8+11) = 11 + 7.125 = 18.125 \text{ kV}$$

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Step 4: (iii) Line voltage

$$V_{ph} = V_1 + V_2 + V_3 = 8 + 11 + 18.125 = 37.125 \text{ kV}$$

$$V_L = \sqrt{3} \times 37.125 = 64.30 \text{ kV}$$

$$\boxed{V_L = 64.3 \text{ kV}}$$

The string carries the phase voltage, so the line voltage is $\sqrt{3}$ times larger.

Step 5: (iv) String efficiency

$$\eta = \frac{V_{ph}}{3V_3} = \frac{37.125}{3 \times 18.125} = 0.6828$$

$$\boxed{\eta = 68.28\ \%}$$

Final Answer: (ii) $K = 0.375$; (iii) line voltage 64.3 kV (phase voltage 37.125 kV, line unit 18.125 kV); (iv) $\eta = 68.28\ \%$.

3. Notes and Remarks

- **Why so low?** $K = 0.375$ is a large shunt capacitance (compare $K = 0.11$ in 3(c), which gave 85.7 %). The line unit takes almost half of the phase voltage.

5.(d) Derive the expression of capacitance of a three phase line with unsymmetrical spacing. Marks: 10

1. Theory

1. The voltage near a charged wire (Core Idea)

A long straight wire carrying a charge q coulomb per metre sends its electric field straight outwards. By Gauss's law, the field at distance x is

$$g = \frac{q}{2\pi\varepsilon_0\, x}$$

The voltage difference between two points P_1 and P_2 at distances D_1 and D_2 from the wire is the field added up from P_1 to P_2:

$$V_{12} = \int_{D_1}^{D_2} \frac{q}{2\pi\varepsilon_0 x}\, dx = \frac{q}{2\pi\varepsilon_0} \ln \frac{D_2}{D_1}$$

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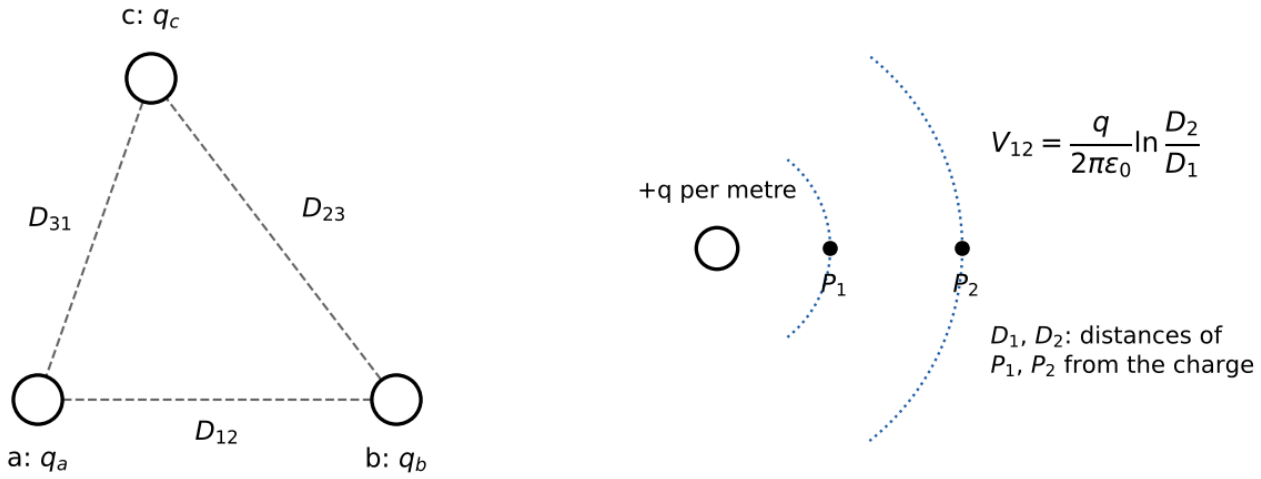


Figure 24: Left: three conductors with charges qa, qb, qc at unequal distances. Right: the voltage between two points near a line charge

Reading the figure. On the left are the three line conductors a, b, c at the corners of an unequal triangle. On the right, the point P_1 (nearer the charge) is at a higher voltage than P_2 by $\frac{q}{2\pi\varepsilon_0} \ln \frac{D_2}{D_1}$.

2. Voltage between two conductors of a group

When several charged wires are present, their effects simply add (**superposition**). The voltage between conductors a and b is the sum of the contributions of q_a, q_b and q_c. For example, q_a contributes $\frac{q_a}{2\pi\varepsilon_0} \ln \frac{D_{ab}}{r}$: from the surface of a (distance r) to b (distance D_{ab}).

3. Symmetrical spacing first

If all three distances are equal to D, the result (shown in the Answer, with $D_{eq} = D$) is

$$C_n = \frac{2\pi\varepsilon_0}{\ln(D/r)} \quad \text{F/m to neutral}$$

With **unsymmetrical** spacing, each phase would have a different capacitance, so the line is **transposed**: each conductor takes each position for one third of the length (see the transposition figure in 1(c)).

2. Answer

Set-up. Conductors a, b, c of radius r are at three positions 1, 2, 3 with distances D_{12}, D_{23}, D_{31}, and the line is transposed. The charges per metre are q_a, q_b, q_c, with $q_a + q_b + q_c = 0$ for balanced conditions; we assume each conductor keeps the same charge in all three sections.

Voltage V_{ab} in each section. Using $V = \frac{q}{2\pi\varepsilon_0} \ln \frac{D_2}{D_1}$ for each charge:

Section 1 (a at 1, b at 2, c at 3):

$$V_{ab1} = \frac{1}{2\pi\varepsilon_0} \left[q_a \ln \frac{D_{12}}{r} + q_b \ln \frac{r}{D_{12}} + q_c \ln \frac{D_{23}}{D_{31}} \right]$$

Section 2 (a at 2, b at 3, c at 1):

$$V_{ab2} = \frac{1}{2\pi\varepsilon_0} \left[q_a \ln \frac{D_{23}}{r} + q_b \ln \frac{r}{D_{23}} + q_c \ln \frac{D_{31}}{D_{12}} \right]$$

Section 3 (a at 3, b at 1, c at 2):

$$V_{ab3} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln\frac{D_{31}}{r} + q_b \ln\frac{r}{D_{31}} + q_c \ln\frac{D_{12}}{D_{23}}\right]$$

Average (the three sections are of equal length):

$$V_{ab} = \frac{V_{ab1}+V_{ab2}+V_{ab3}}{3} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln\frac{D_{eq}}{r} + q_b \ln\frac{r}{D_{eq}}\right], \qquad D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$$

The q_c terms cancel, because $\dfrac{D_{23}}{D_{31}}\cdot\dfrac{D_{31}}{D_{12}}\cdot\dfrac{D_{12}}{D_{23}} = 1$.

In the same way:

$$V_{ac} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln\frac{D_{eq}}{r} + q_c \ln\frac{r}{D_{eq}}\right]$$

Add the two and use $q_b + q_c = -q_a$:

$$V_{ab} + V_{ac} = \frac{1}{2\pi\varepsilon_0}\left[2q_a \ln\frac{D_{eq}}{r} - (q_b+q_c)\ln\frac{D_{eq}}{r}\right] = \frac{3q_a}{2\pi\varepsilon_0}\ln\frac{D_{eq}}{r}$$

Relate to the phase voltage. For balanced three-phase voltages, $V_{ab} = \sqrt{3}\,V_{an}\angle 30°$ and $V_{ac} = \sqrt{3}\,V_{an}\angle -30°$, so their sum is $V_{ab} + V_{ac} = 3V_{an}$. Therefore

$$3V_{an} = \frac{3q_a}{2\pi\varepsilon_0}\ln\frac{D_{eq}}{r} \quad\Rightarrow\quad V_{an} = \frac{q_a}{2\pi\varepsilon_0}\ln\frac{D_{eq}}{r}$$

Capacitance to neutral:

$$\boxed{C_n = \frac{q_a}{V_{an}} = \frac{2\pi\varepsilon_0}{\ln\left(\sqrt[3]{D_{12}D_{23}D_{31}}/r\right)} \text{ F/m per phase}}$$

or, with $\varepsilon_0 = 8.854\times10^{-12}$ F/m, $C_n = \dfrac{0.0556}{\ln(D_{eq}/r)}$ μF/km. For symmetrical spacing $D_{eq} = D$, and the result becomes $2\pi\varepsilon_0/\ln(D/r)$.

3. Keyword Explanation

1. Capacitance to neutral

The capacitance between one conductor and the neutral point of a three-phase system; the charging current per phase is $\omega C_n V_{ph}$.

2. Equivalent spacing D_{eq}

$\sqrt[3]{D_{12}D_{23}D_{31}}$, the same as in the inductance formula of 1(c).

4. Notes and Remarks

- **r, not r':** for capacitance the charge sits on the conductor surface, so the outside radius r is used; for inductance the GMR $r' = 0.7788\,r$ is used.
- **Inductance and capacitance together:** $L = 2\times10^{-7}\ln(D_{eq}/r')$ and $C_n = 2\pi\varepsilon_0/\ln(D_{eq}/r)$ have the same shape. A larger spacing raises L and lowers C.
- **The same derivation** is asked in Term Test 1 as 7(b).

5. Final Concise Answer

(a) The earth is a zero-potential conductor; by the method of images it acts as opposite charges at the same depth below the ground, which **increases** the line capacitance slightly (single-phase: $C = \pi\varepsilon_0/\ln[(d/r)\cdot 2h/\sqrt{4h^2+d^2}]$). The effect is small for high lines and is usually neglected.

(b) $D_{eq} = 4.410$ m, $r = 0.525$ cm; $C_n = 0.00826$ μF/km; $I_c = 0.165$ A per km.
(c) (i) Three capacitors C in series with KC from links to the tower; (ii) $K = 0.375$; (iii) $V_3 = 18.125$ kV, $V_{ph} = 37.125$ kV, $V_L = 64.3$ kV; (iv) $\eta = 68.28$ %.
(d) Averaging V_{ab} and V_{ac} over the three transposition sections and using $q_a + q_b + q_c = 0$ and $V_{ab} + V_{ac} = 3V_{an}$ gives $C_n = 2\pi\varepsilon_0 / \ln(D_{eq}/r)$ F/m, $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$.

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6. (a) Find the most economical size of a single-core cable working on a 132 kV, 3- phase system, if a dielectric stress of 60 kV/cm can be allowed. Marks: 2

(b) Define HVDC. Discuss the Economics of Power transmission. Marks: 5

(c) Explain what happens for sudden increase in mechanical input to generator using $Pe - \delta$ diagram. Marks: 8

(d) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus ($|V| = 1.0\ pu$), with the generator terminal voltage ($|V_t| = 1.0\ pu$). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (i) System healthy, (ii) One line shorted (3 phase) in the middle and (iii) One line open. Plot all the three power angle curves. Marks: 10

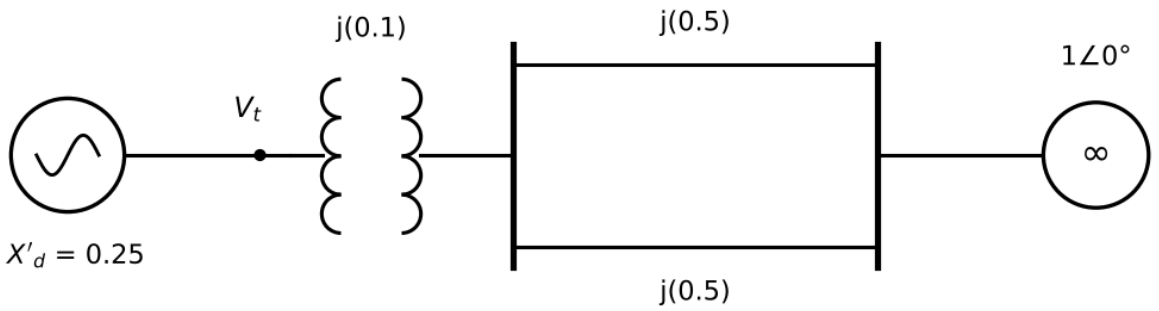


Figure 25: Fig. 2 of the paper (redrawn): figure for Q-6(d)

6.(a) Find the most economical size of a single-core cable working on a 132 kV, 3- phase system, if a dielectric stress of 60 kV/cm can be allowed. Marks: 2

1. Theory

1. What does "most economical size" mean? (Core Idea)

The cost of a cable depends mostly on its **overall diameter**: a fatter cable needs more insulation, more lead and more armour. So, for a given working voltage and a given safe stress, we want the **smallest sheath diameter** D.
From 2(c), the stress is highest at the core surface:

$$g_{max} = \frac{2V}{d \ln(D/d)}$$

where V is the voltage between core and sheath, d the core diameter and D the sheath diameter.

2. Two opposite effects of the core size

Fix V and g_{max}, and ask what D must be for each core size d:

$$\ln \frac{D}{d} = \frac{2V}{g_{max}\, d} \quad \Rightarrow \quad D = d\, e^{2V/(g_{max} d)}$$

- If the core is **very thin**, the field at its sharp surface is very high, so a lot of insulation (a large D/d) is needed.
- If the core is **very thick**, the core itself makes D large.

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In between there is a core size that gives the smallest D.

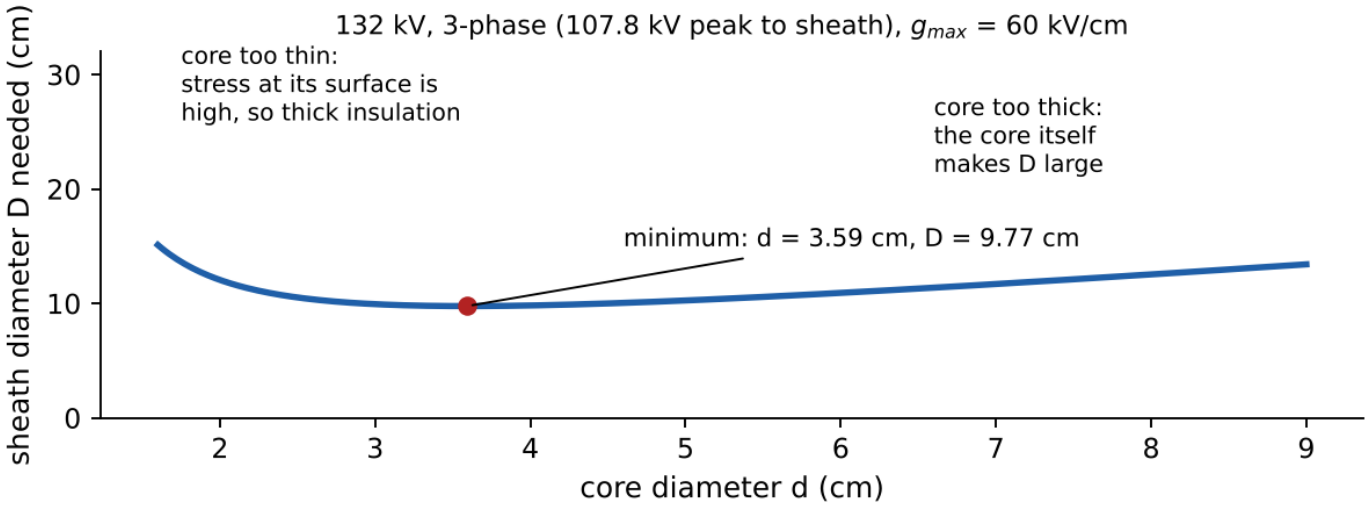


Figure 26: Sheath diameter needed against core diameter, for this question's 132 kV cable and 60 kV/cm; the minimum is at d = 3.59 cm

3. Finding the minimum

Set the slope of $D(d)$ to zero:

$$\frac{dD}{dd} = e^{2V/(g_{max}d)}\left(1 - \frac{2V}{g_{max}\,d}\right) = 0 \quad \Rightarrow \quad d = \frac{2V}{g_{max}}$$

At this point $\ln(D/d) = 1$, so

$$D = e\,d = 2.718\,d$$

4. Which voltage?

- A single-core cable in a three-phase system has the **phase voltage** between its core and its earthed sheath: $V_L/\sqrt{3}$.
- The allowed dielectric stress is a **peak** value (breakdown depends on the peak), so the peak voltage $\sqrt{2}\,V_{ph}$ is used.

2. Answer

Given: $V_L = 132$ kV (3-phase), $g_{max} = 60$ kV/cm (peak).

Voltage between core and sheath (peak):

$$V = \frac{132}{\sqrt{3}} \times \sqrt{2} = 107.78 \text{ kV (peak)}$$

Most economical core diameter ($\ln(D/d) = 1$):

$$d = \frac{2V}{g_{max}} = \frac{2 \times 107.78}{60} = 3.593 \text{ cm}$$

Sheath diameter:

$$D = e\,d = 2.718 \times 3.593 = 9.766 \text{ cm}$$

$$\boxed{d = 3.59 \text{ cm}, \qquad D = 9.77 \text{ cm}}$$

The insulation is $(9.766 - 3.593)/2 = 3.09$ cm thick.

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3. Notes and Remarks

- **Current rating:** the core found this way is often larger than needed for the current. Then a hollow core or a different material can be used to reach the required diameter cheaply.

6.(b) Define HVDC. Discuss the Economics of Power transmission. Marks: 5

1. Theory

1. HVDC in one line (Core Idea)

HVDC sends power as direct current at a very high voltage: a rectifier at the sending end turns AC into DC, and an inverter at the receiving end turns it back into AC. The ways of arranging the conductors (monopolar, bipolar, homopolar links) are in 2(a).

2. What decides the cost of a transmission line?

Every line has two kinds of cost:

- **Capital cost**, paid once: conductors, towers, insulators, transformers, switchgear. Each year it costs **interest and depreciation** on that money.
- **Running cost**, paid every year: mainly the **energy lost** as heat in the conductors (I^2R).

Making one cheaper usually makes the other dearer. Economics means finding the best balance.

3. Conductor size: Kelvin's law

A **thicker conductor** costs more (more metal), so the annual capital charge rises with the cross-section a. But its resistance is lower, so less energy is lost and the annual loss cost falls as $1/a$.

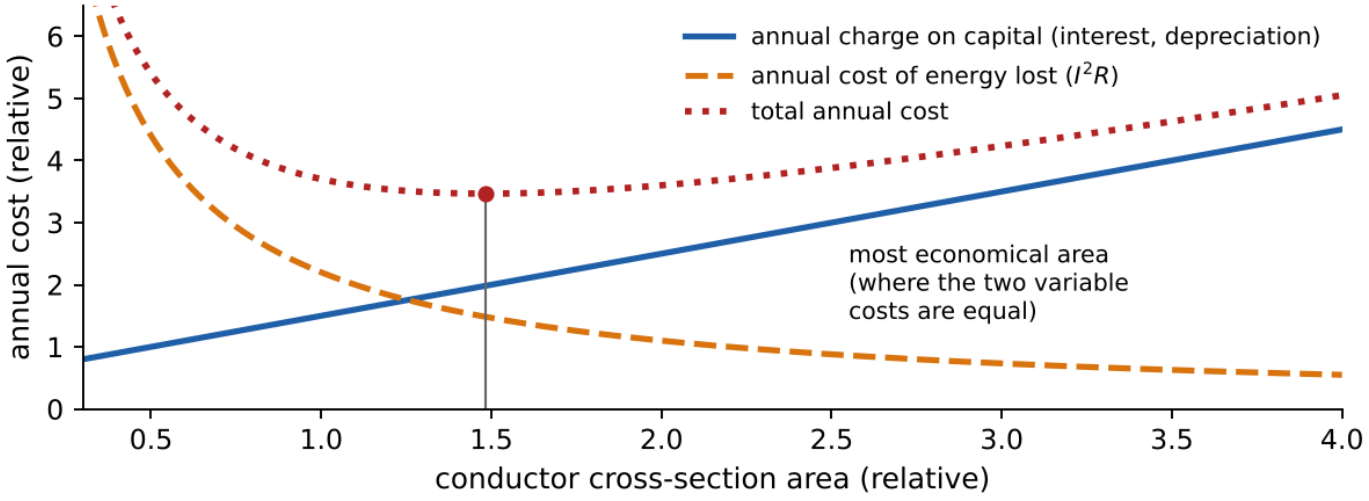


Figure 27: Kelvin's law: the annual capital charge (solid blue) rises and the annual cost of losses (dashed orange) falls with conductor area; their total (dotted red) is smallest where the two variable costs are equal

Writing the total annual cost as $P_1 + P_2 a + \dfrac{P_3}{a}$ and setting its slope to zero gives $P_2 a = P_3/a$: **the most economical area is the one at which the annual cost of energy lost equals the annual interest and depreciation on the part of the capital cost that depends on the area.** This is **Kelvin's law**.

4. Transmission voltage

For the same power, a **higher voltage** means a smaller current ($P = \sqrt{3} V I \cos\phi$). Then:

- the conductors can be thinner (less copper or aluminium) and the I^2R loss is smaller;
- but insulators, towers (clearances), transformers and switchgear cost more.

So there is an economical voltage for each line; it rises with the power and the distance.

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5. AC or DC?

An HVDC line needs only one or two conductors, smaller towers and no charging current, and has no stability limit, so its **line cost per km is lower**. But its **converter stations are expensive**. So DC is cheaper only beyond a certain distance.

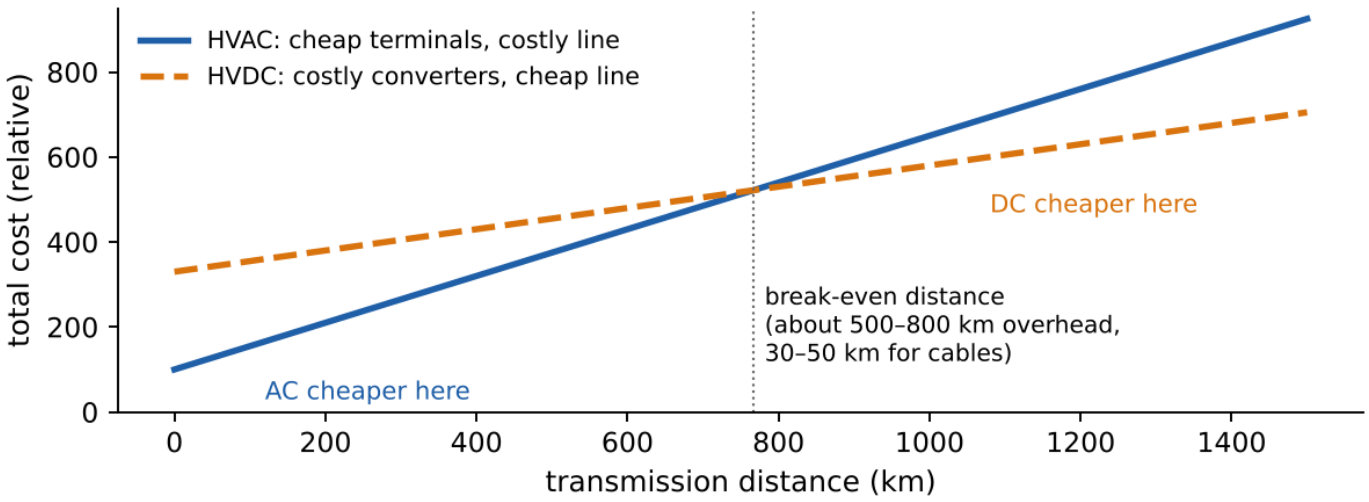


Figure 28: Total cost against distance: HVAC (solid blue) starts low but rises fast; HVDC (dashed orange) starts high but rises slowly; they cross at the break-even distance

2. Answer

HVDC: high-voltage direct-current transmission is the transmission of electric power by direct current at a very high voltage (hundreds of kV). AC from the generating station is converted to DC by a rectifier station, sent over the DC line, and converted back to AC by an inverter station at the receiving end.

Economics of power transmission. The aim is the minimum total annual cost (annual charges on the capital + annual cost of energy lost) for the power to be sent. It involves three choices:

1. **Economic size of conductor (Kelvin's law).** A larger conductor raises the capital cost (interest and depreciation, $\propto a$) but lowers the I^2R energy loss ($\propto 1/a$). The total is least when *the annual cost of the energy wasted equals the annual interest and depreciation on the part of the capital cost proportional to the conductor area.*
2. **Economic transmission voltage.** A higher voltage reduces the current, so it saves conductor material and losses, but it increases the cost of insulators, towers, transformers and switchgear. The voltage at which the total cost is least is chosen; it increases with the power and the length of the line.
3. **HVAC versus HVDC.** An HVDC line needs fewer conductors, lighter towers and a narrower right of way, has no charging current, no skin effect and no stability limit, so its cost per km and its losses are lower. But the converter stations are costly. Beyond the **break-even distance** (about 500–800 km for overhead lines, 30–50 km for underground or submarine cables) HVDC is cheaper; below it, HVAC is cheaper.

3. Keyword Explanation

1. Kelvin's law

The rule that gives the most economical conductor size: annual energy-loss cost = annual capital charge on the area-dependent part.

2. Break-even distance

The line length at which HVAC and HVDC transmission cost the same.

4. Notes and Remarks

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- **Limits of Kelvin's law:** the loss is hard to estimate (the load varies), interest and depreciation are hard to fix, and the size it gives may not satisfy other needs (voltage drop, mechanical strength, corona). It is a guide, not the final answer.

6.(c) Explain what happens for sudden increase in mechanical input to generator using $Pe - \delta$ diagram. Marks: 8

1. Theory

1. The generator and the power-angle equation (Core Idea)

A synchronous generator is turned by a turbine. The turbine gives **mechanical power** P_m to the rotor; the generator sends **electrical power** P_e to the network. When the generator (internal emf E) is connected through a reactance X to a very large system of fixed voltage V (an **infinite bus**), the electrical power is

$$P_e = \frac{E\,V}{X}\sin\delta = P_{max}\sin\delta$$

where δ is the **rotor (power) angle**: how far the generator's emf is ahead of the bus voltage. This is the **power-angle curve**, a sine wave with peak $P_{max} = EV/X$ at $\delta = 90°$.

2. What makes the rotor speed up or slow down?

The rotor is heavy and spinning. Like any heavy wheel:

- if $P_m > P_e$, the extra power **accelerates** the rotor, so δ increases;
- if $P_m < P_e$, the rotor **decelerates**, so δ decreases;
- if $P_m = P_e$, the speed stays at synchronous speed and δ stays fixed.

This is the **swing equation**: $M\dfrac{d^2\delta}{dt^2} = P_m - P_e$, where M is the inertia constant.

3. The equal area criterion

The energy that speeds the rotor up is the area between the P_m line and the P_e curve while $P_m > P_e$. The rotor returns to synchronous speed only when the same amount of energy has been taken back, while $P_e > P_m$. So the rotor's maximum swing is where the two areas become **equal**. This is the **equal area criterion**.

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2. Answer

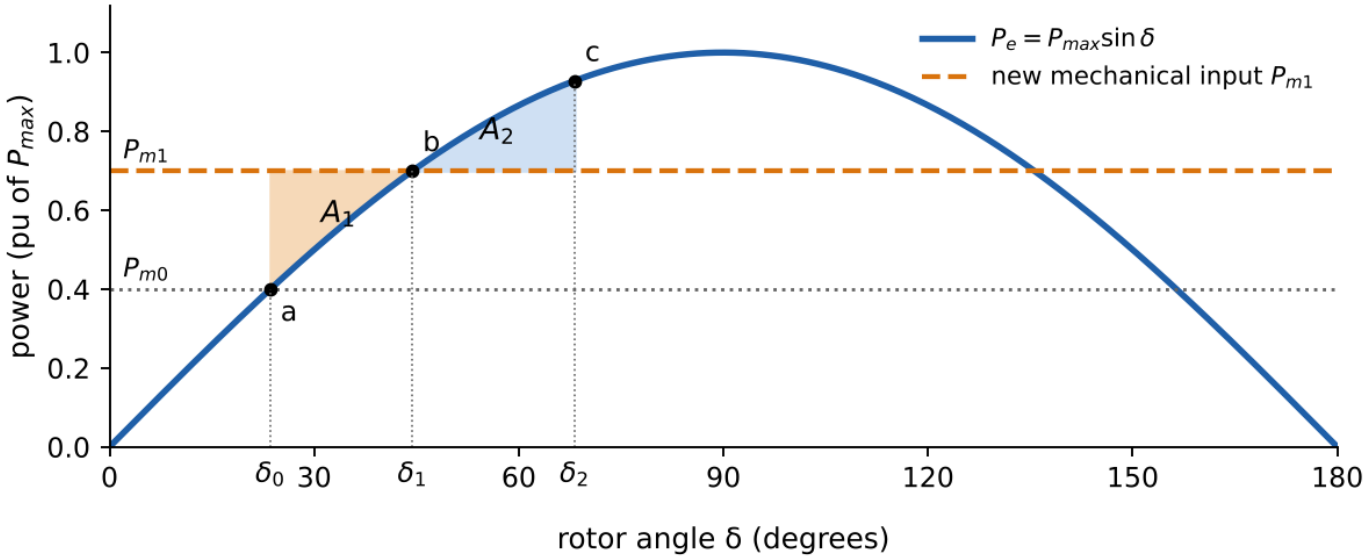


Figure 29: Sudden increase in mechanical input from Pm0 to Pm1: the operating point moves from a to b, overshoots to c, and the areas A1 and A2 are equal

Let the generator run steadily at point a on the power-angle curve $P_e = P_{max} \sin\delta$, with mechanical input $P_{m0} = P_e$ and rotor angle δ_0. Now the mechanical input is suddenly raised to P_{m1}.

1. **Acceleration ($\delta_0 \to \delta_1$).** At δ_0 the electrical output is still $P_{m0} < P_{m1}$. The extra input accelerates the rotor above synchronous speed, so δ increases and P_e rises along the curve. The rotor gains kinetic energy equal to the area A_1 (between P_{m1} and the curve, from δ_0 to δ_1).
2. **Overshoot ($\delta_1 \to \delta_2$).** At point b (δ_1) $P_e = P_{m1}$, so the acceleration is zero, but the rotor is now running **faster** than synchronous speed, so δ keeps increasing. Beyond δ_1, $P_e > P_{m1}$, so the rotor decelerates and gives back energy (area A_2).
3. **Maximum swing.** The rotor reaches synchronous speed again at point c (δ_2), where $A_2 = A_1$ (**equal area criterion**). δ is then largest.
4. **Oscillation and settling.** At δ_2, $P_e > P_{m1}$, so the rotor slows below synchronous speed and δ swings back past δ_1 towards δ_0. Without damping it would swing between δ_0 and δ_2; with damping (losses, damper windings) the swings die out and the generator settles at the **new operating point b**, angle δ_1, delivering P_{m1}.

Stability limit. The decelerating area cannot extend beyond $\delta_{max} = 180° - \delta_1$, where the curve falls below P_{m1} again. If the increase is so large that A_1 is bigger than the largest possible A_2, the rotor passes δ_{max}, keeps accelerating and **loses synchronism**. The largest sudden increase that keeps the system stable is the one for which A_2 (up to $180° - \delta_1$) just equals A_1. Even a slow (steady-state) increase can never exceed P_{max}.

3. Keyword Explanation

1. Rotor (power) angle δ

The angle by which the generator's internal emf leads the infinite-bus voltage.

2. Infinite bus

A very large system whose voltage and frequency do not change, whatever one generator does.

3. Equal area criterion

A machine stays stable after a disturbance if the decelerating area can become equal to the accelerating area before δ reaches its maximum.

4. Notes and Remarks

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- **Transient stability limit:** because of the overshoot, the largest **sudden** step of input is smaller than P_{max}, the steady-state limit.
- The same curves, for a real system, are drawn in 6(d).

6.(d) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus ($|V| = 1.0\ pu$), with the generator terminal voltage ($|V_t| = 1.0\ pu$). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (i) System healthy, (ii) One line shorted (3 phase) in the middle and (iii) One line open. Plot all the three power angle curves. Marks: 10

1. Theory

1. Per unit, infinite bus and transient reactance (Core Idea)

- **Per unit (pu):** every quantity is given as a fraction of a chosen base value. Here $|V| = 1.0$ pu simply means rated voltage.
- **Infinite bus:** the large system on the right of Fig. 2, at a fixed $1\angle 0°$ pu.
- **Transient reactance X'_d:** in the first second or so after a disturbance, a generator behaves like a fixed voltage E' (the **emf behind transient reactance**) in series with X'_d. So for stability studies the generator is drawn as E' behind $X'_d = 0.25$ pu.

2. Power transfer between two voltages

From 6(c), two voltages E' and V joined by a reactance X exchange

$$P = \frac{|E'||V|}{X}\sin\delta, \qquad P_{max} = \frac{|E'||V|}{X}$$

So P_{max} depends on the **transfer reactance** X between them. Anything that raises X (opening a line, a fault) lowers P_{max}.

3. Reducing a network with a fault

A fault connects a point of the network to earth (the zero-voltage reference). The network then has three nodes (E′, the infinite bus and the earth) and is no longer one simple series path. A **star–delta conversion** turns it into one reactance between E' and the bus. For a star with arms X_a, X_b, X_c (to nodes a, b, c), the delta arm between a and b is

$$X_{ab} = \frac{X_aX_b + X_bX_c + X_cX_a}{X_c}$$

2. Answer

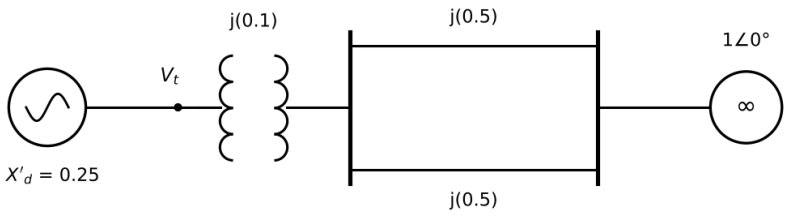


Figure 30: Repeated from the question: Fig. 2 of the paper (redrawn): figure for Q-6(d)

Given (Fig. 2): $X'_d = 0.25$, transformer $X_t = 0.1$, two parallel lines of 0.5 each (all pu); $P = 1.0$ pu; $|V_t| = 1.0$; infinite bus $V = 1\angle 0°$.

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Step 1: Angle of the terminal voltage

Between V_t and the bus: $X = X_t + 0.5 \parallel 0.5 = 0.1 + 0.25 = 0.35$ pu.

$$P = \frac{|V_t||V|}{0.35}\sin\theta \quad \Rightarrow \quad \sin\theta = 1.0 \times 0.35 = 0.35, \qquad \theta = 20.49^\circ$$

$$V_t = 1\angle 20.49^\circ = 0.9367 + j0.3500 \text{ pu}$$

Step 2: Current

$$I = \frac{V_t - V}{j0.35} = \frac{-0.0633 + j0.35}{j0.35} = 1.0000 + j0.1807 = 1.0162\angle 10.24^\circ \text{ pu}$$

Step 3: EMF behind transient reactance

$$E' = V_t + jX'_dI = (0.9367 + j0.35) + j0.25(1.0 + j0.1807)$$

$$E' = (0.9367 - 0.0452) + j(0.35 + 0.25) = 0.8916 + j0.6000 = 1.0747\angle 33.94^\circ \text{ pu}$$

$$\boxed{E' = 1.0747\angle 33.94^\circ \text{ pu}}$$

So the initial rotor angle is $\delta_0 = 33.94^\circ$.

Step 4: (i) System healthy

$$X_I = X'_d + X_t + \frac{0.5}{2} = 0.25 + 0.1 + 0.25 = 0.60 \text{ pu}$$

$$P_{max,I} = \frac{1.0747 \times 1.0}{0.60} = 1.791 \text{ pu}$$

$$\boxed{P_{max} = 1.791 \text{ pu (healthy)}}$$

Check: $1.791 \sin 33.94^\circ = 1.0$ pu, the power being delivered.

Step 5: (ii) Three-phase short at the middle of one line

The faulted line is split into two halves of $j0.25$, with the fault point F at earth potential.

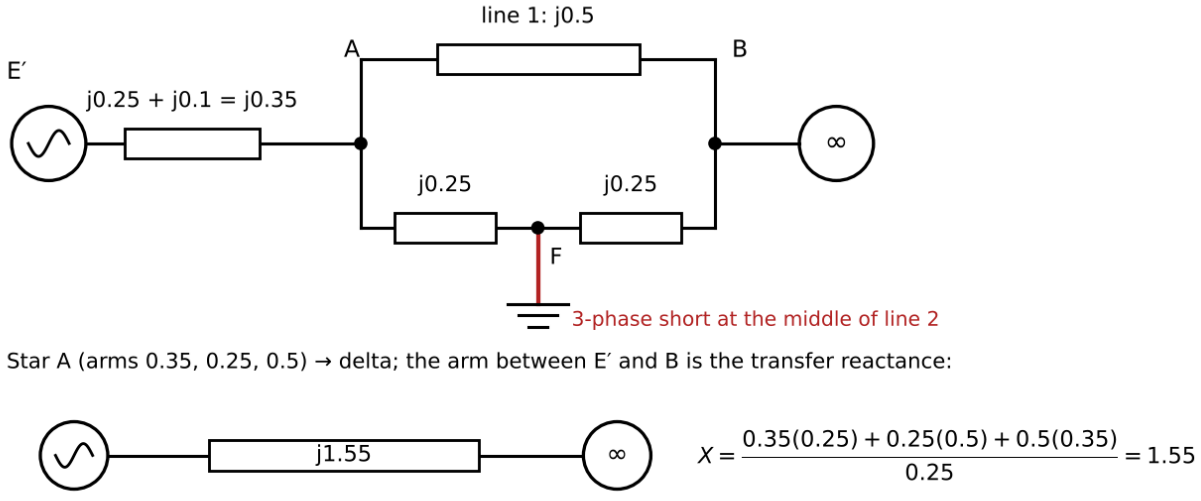


Figure 31: Network with the fault at the middle of line 2, and its reduction to one transfer reactance of j1.55

At node A (the sending end of the lines) there is a star of three reactances:

- to E': $X_a = 0.25 + 0.1 = 0.35$

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- to the fault (earth): $X_c = 0.25$ (half of line 2)
- to the infinite bus B: $X_b = 0.5$ (line 1)

The other half of line 2 (B to F, $j0.25$) hangs from the infinite bus to earth, so it does not affect the power from E'. Converting the star to a delta, the arm between E' and B is

$$X_{II} = \frac{X_a X_b + X_b X_c + X_c X_a}{X_c} = \frac{0.35 \times 0.5 + 0.5 \times 0.25 + 0.25 \times 0.35}{0.25} = \frac{0.3875}{0.25} = 1.55 \text{ pu}$$

$$P_{max,II} = \frac{1.0747 \times 1.0}{1.55} = 0.693 \text{ pu}$$

$$\boxed{P_{max} = 0.693 \text{ pu (fault)}}$$

This is **less than the 1.0 pu input**: during the fault the generator cannot deliver its mechanical input, so it accelerates. It stays in step only if the fault is cleared quickly enough (equal area criterion, 6(c)).

Step 6: (iii) One line open

$$X_{III} = 0.25 + 0.1 + 0.5 = 0.85 \text{ pu}$$

$$P_{max,III} = \frac{1.0747 \times 1.0}{0.85} = 1.264 \text{ pu}$$

$$\boxed{P_{max} = 1.264 \text{ pu (one line open)}}$$

With one line out, the 1.0 pu is carried at a new angle $\delta = \sin^{-1}(1/1.264) = 52.27°$.

Step 7: Power-angle curves

$$P_I = 1.791 \sin\delta, \qquad P_{II} = 0.693 \sin\delta, \qquad P_{III} = 1.264 \sin\delta \quad \text{pu}$$

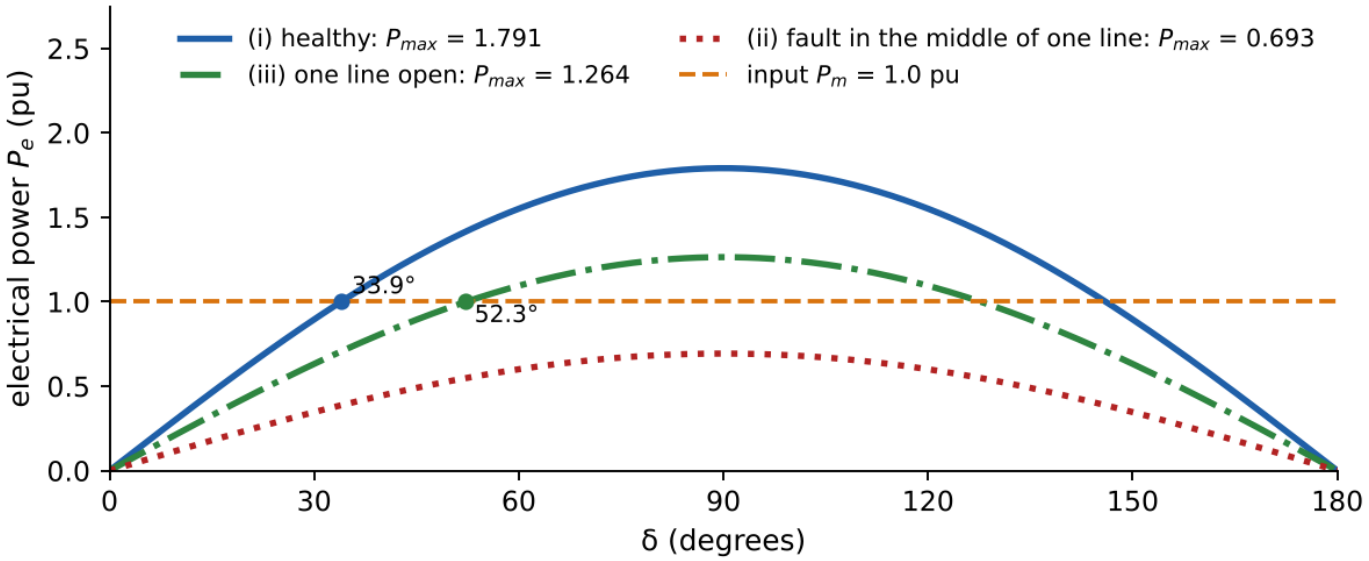


Figure 32: Power-angle curves: healthy (solid blue), one line open (dash-dot green) and fault in the middle of one line (dotted red), with the 1.0 pu input (dashed orange)

Reading the figure. All three curves are sine waves with different heights. The healthy curve meets the 1.0 pu input at 33.9°, the one-line-open curve at 52.3°, and the fault curve never reaches 1.0 pu.
Final Answer: $E' = 1.0747\angle 33.94°$ pu; $P_{max} = 1.791$ pu (healthy), 0.693 pu (3-phase short at the middle of one line), 1.264 pu (one line open).

3. Notes and Remarks

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- **Why the fault hurts so much:** the short at F carries much of the generator's current to earth, so the effective coupling to the infinite bus is weak: the transfer reactance rises from 0.60 to 1.55 pu.
- **Fault at the sending end of a line** (at A) would make $X_c = 0$ and the transfer reactance infinite: no power at all would reach the bus.

4. Final Concise Answer

(a) $V = 132\sqrt{2}/\sqrt{3} = 107.78$ kV peak; $d = 2V/g_{max} = 3.59$ cm; $D = e\,d = 9.77$ cm.

(b) HVDC transmits power as high-voltage DC between a rectifier and an inverter station. Economics: Kelvin's law for conductor size (annual loss cost = annual capital charge on the area-dependent part), an economic voltage balancing conductor savings against insulation and equipment cost, and HVDC cheaper than HVAC beyond the break-even distance ($\approx$ 500–800 km overhead, 30–50 km cable).

(c) A sudden rise from P_{m0} to P_{m1} accelerates the rotor ($P_m > P_e$); δ passes the new equilibrium δ_1, overshoots to δ_2 where the decelerating area equals the accelerating area, then oscillates and settles at δ_1. If A_1 exceeds the area available up to $180° - \delta_1$, synchronism is lost.

(d) $V_t = 1\angle 20.49°$; $E' = 1.0747\angle 33.94°$ pu; $P_{max} = 1.791$ pu (healthy, $X = 0.60$), 0.693 pu (mid-line fault, $X = 1.55$), 1.264 pu (one line open, $X = 0.85$); curves $P = P_{max} \sin\delta$.

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7. (a) A 500 kV line has a bundling arrangement of 2 conductors per phase as shown in the Figure 1. Compute the inductive reactance per phase of this line at 50 Hz. Each conductor carries 50% of the phase current. Marks: 5

(b) Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing.

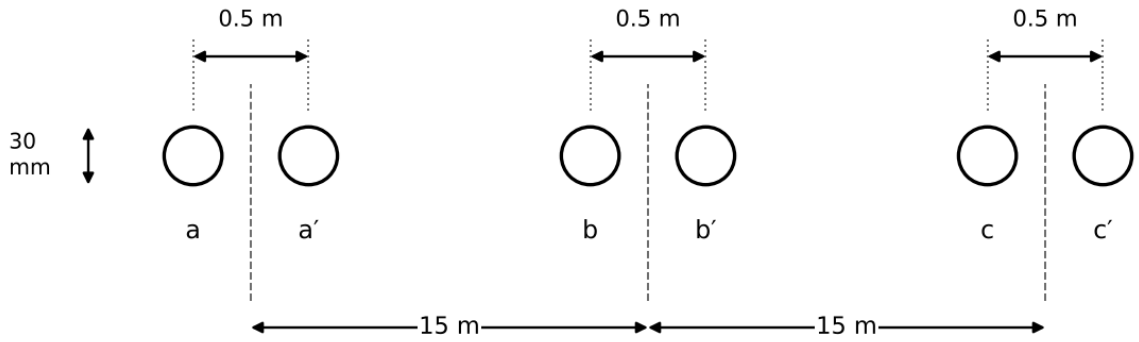


Figure 33: Figure 1 of Term Test 1 (redrawn)

7.(a) A 500 kV line has a bundling arrangement of 2 conductors per phase as shown in the Figure 1. Compute the inductive reactance per phase of this line at 50 Hz. Each conductor carries 50% of the phase current. Marks: 5

1. Theory

1. What is a bundled conductor? (Core Idea)

At very high voltages (220 kV and above), each phase is often made of **two or more sub-conductors** held a short distance apart (30 to 50 cm) by metal spacers, instead of one thick conductor. This is a **bundled conductor**. In this question each phase has two sub-conductors 0.5 m apart, and they share the phase current equally (50 % each).

2. Why bundle?

- **Less corona:** the charge is shared between the sub-conductors, so the field at each surface is lower; the bundle behaves like one conductor of a much larger radius, which raises the critical disruptive voltage (1(a)).
- **Less inductance:** the bundle's GMR is much larger than that of one conductor, so the reactance falls and the line can carry more power.
- Less radio interference and lower losses.

3. The GMR of a two-conductor bundle

A bundle is a composite conductor (4(c)) with $n = 2$ strands carrying equal currents. Its GMR is the 4th root ($n^2 = 4$) of all the distances inside the bundle: each sub-conductor to itself (r') and to the other (d):

$$D_s = \sqrt[4]{r' \cdot d \cdot d \cdot r'} = \sqrt{r'\, d}$$

where $r' = 0.7788\, r$ is the GMR of one sub-conductor and d the sub-conductor spacing.

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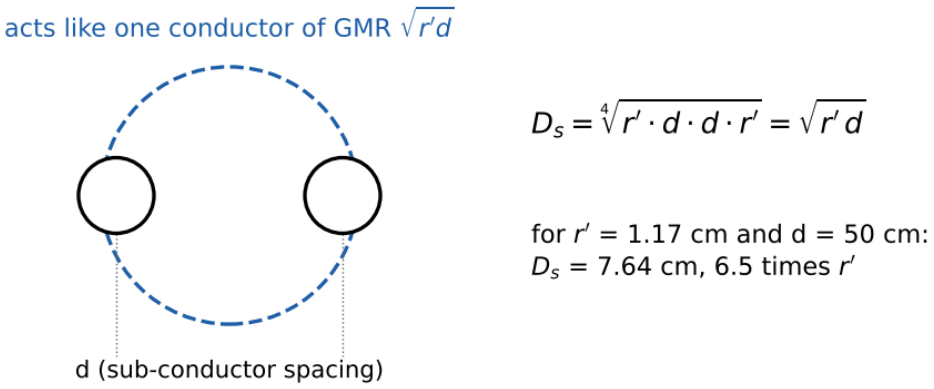


Figure 34: A two-conductor bundle acts like one conductor of GMR the square root of r' times d

4. Inductance and reactance per phase

The phases are far apart compared with d, so the distances between phases can be measured between the **centres** of the bundles. With the line transposed (1(c)):

$$L = 2 \times 10^{-7} \ln \frac{D_{eq}}{D_s} \ \text{H/m}, \qquad X_L = 2\pi f L$$

with $D_{eq} = \sqrt[3]{D_{ab} D_{bc} D_{ca}}$.

2. Answer

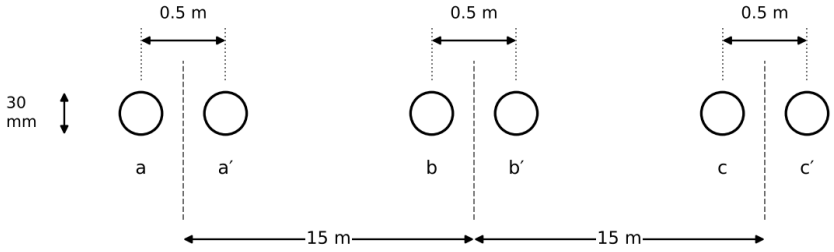


Figure 35: Repeated from the question: Figure 1 of Term Test 1 (redrawn)

Given (Figure 1):

- Conductor diameter 30 mm, so $r = 15$ mm $= 0.015$ m
- Sub-conductor spacing $d = 0.5$ m (2 per phase, equal current sharing)
- Distances between bundle centres: $D_{ab} = D_{bc} = 15$ m, $D_{ca} = 30$ m
- $f = 50$ Hz; the line is assumed transposed

Step 1: GMR of one sub-conductor

$$r' = 0.7788 \times 0.015 = 0.011682 \text{ m}$$

Step 2: GMR of the bundle

$$D_s = \sqrt{r'\, d} = \sqrt{0.011682 \times 0.5} = 0.07643 \text{ m}$$

Step 3: Equivalent spacing

$$D_{eq} = \sqrt[3]{15 \times 15 \times 30} = \sqrt[3]{6750} = 18.899 \text{ m}$$

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Step 4: Inductance per phase

$$\frac{D_{eq}}{D_s} = \frac{18.899}{0.07643} = 247.3, \qquad \ln 247.3 = 5.511$$

$$L = 2 \times 10^{-7} \times 5.511 = 1.1021 \times 10^{-6}\ \text{H/m} = 1.102\ \text{mH/km}$$

Step 5: Inductive reactance per phase

$$X_L = 2\pi \times 50 \times 1.1021 \times 10^{-3} = 0.3462\ \Omega/\text{km}$$

$$\boxed{X_L = 0.346\ \Omega \text{ per km per phase}}$$

Final Answer: $L = 1.102$ mH/km and $X_L = 0.346\ \Omega$/km per phase.

3. Notes and Remarks

- **Gain from bundling:** with a single 30 mm conductor per phase at the same spacing, $X_L = 2\pi f \times 2 \times 10^{-7} \ln(18.899/0.011682) = 0.464\ \Omega$/km. The bundle reduces the reactance by about 25 %.
- **Exact distances:** using all four sub-conductor distances between two bundles (for example $\sqrt[4]{15 \times 15.5 \times 14.5 \times 15} = 14.996$ m instead of 15 m) gives $D_{eq} = 18.895$ m and the same $X_L = 0.346\ \Omega$/km.

7.(b) Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing.

1. Theory

1. What is capacitance of a line? (Core Idea)

Two conductors separated by air form a capacitor. Charge on them sets up a voltage, and the capacitance is the charge per volt, $C = q/V$. For a three-phase line we want the capacitance of one phase to the neutral.

2. What is the voltage near a charged wire?

A long wire with charge q per metre has an electric field $q/(2\pi\varepsilon_0 x)$ at distance x. Integrating that field from D_1 to D_2 gives the voltage between the two distances:

$$V = \frac{q}{2\pi\varepsilon_0} \ln \frac{D_2}{D_1}$$

3. How do several wires add?

Voltages add, so V_{ab} is the sum of the voltage each of the three charges makes between a and b. This is **superposition**.

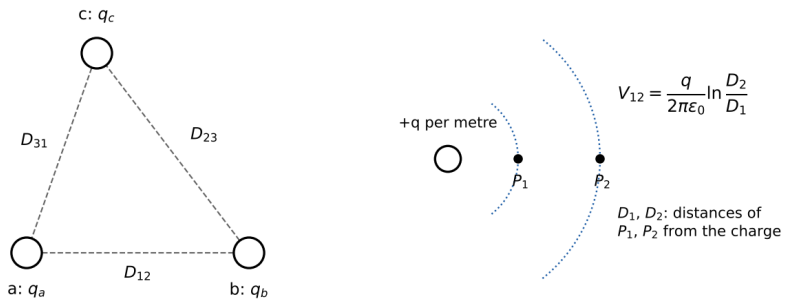


Figure 36: Three charged conductors with unsymmetrical spacing

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Reading the figure. Each conductor is at a different distance from the others, so without transposition each phase would have a different capacitance.

4. Why transpose?

In a **transposed** line each phase takes each position for a third of the length. Averaging over the three sections replaces the three distances by one equivalent spacing, $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$. All three phases then have the same capacitance.

5. How does the neutral come in?

For balanced phases $V_{ab} + V_{ac} = 3V_{an}$, and $q_b + q_c = -q_a$. These two facts remove q_b and q_c. The full derivation is also in 5(d).

2. Answer

Set-up. Conductors a, b, c of radius r sit at three positions 1, 2, 3, with distances D_{12}, D_{23}, D_{31} between the positions. The line is transposed: each conductor takes each position for one third of the length. The charges per metre are q_a, q_b, q_c, and for balanced conditions

$$q_a + q_b + q_c = 0$$

We assume each conductor keeps the same charge in all three sections (the usual textbook assumption).

Step 1: Voltage due to one charge. A long wire with charge q per metre has a field $E = q/(2\pi\varepsilon_0 x)$ at distance x. The voltage between a point at distance D_1 and a point at distance D_2 is the integral of this field:

$$V = \int_{D_1}^{D_2} \frac{q}{2\pi\varepsilon_0 x}\, dx = \frac{q}{2\pi\varepsilon_0} \ln \frac{D_2}{D_1}$$

Step 2: V_{ab} in each section. By superposition, V_{ab} is the sum of the voltages that q_a, q_b and q_c each make between a and b. For each charge, D_1 is its distance from a and D_2 its distance from b.

Section 1 (a at 1, b at 2, c at 3). For q_a: $D_1 = r$, $D_2 = D_{12}$. For q_b: $D_1 = D_{12}$, $D_2 = r$. For q_c: $D_1 = D_{31}$, $D_2 = D_{23}$. So

$$V_{ab1} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln \frac{D_{12}}{r} + q_b \ln \frac{r}{D_{12}} + q_c \ln \frac{D_{23}}{D_{31}}\right]$$

Section 2 (a at 2, b at 3, c at 1):

$$V_{ab2} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln \frac{D_{23}}{r} + q_b \ln \frac{r}{D_{23}} + q_c \ln \frac{D_{31}}{D_{12}}\right]$$

Section 3 (a at 3, b at 1, c at 2):

$$V_{ab3} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln \frac{D_{31}}{r} + q_b \ln \frac{r}{D_{31}} + q_c \ln \frac{D_{12}}{D_{23}}\right]$$

Step 3: Average over the three sections. The sections are of equal length, so $V_{ab} = (V_{ab1} + V_{ab2} + V_{ab3})/3$. Add the terms of each charge, using $\ln A + \ln B + \ln C = \ln(ABC)$. Write $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$, so that $D_{12}D_{23}D_{31} = D_{eq}^3$.

The q_a terms:

$$\ln \frac{D_{12}}{r} + \ln \frac{D_{23}}{r} + \ln \frac{D_{31}}{r} = \ln \frac{D_{eq}^3}{r^3} = 3 \ln \frac{D_{eq}}{r}$$

The q_b terms, in the same way:

$$\ln \frac{r}{D_{12}} + \ln \frac{r}{D_{23}} + \ln \frac{r}{D_{31}} = 3 \ln \frac{r}{D_{eq}}$$

The q_c terms cancel:

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$$\ln\frac{D_{23}}{D_{31}}+\ln\frac{D_{31}}{D_{12}}+\ln\frac{D_{12}}{D_{23}}=\ln 1=0$$

Dividing by 3:

$$V_{ab}=\frac{1}{2\pi\varepsilon_0}\left[q_a\ln\frac{D_{eq}}{r}+q_b\ln\frac{r}{D_{eq}}\right] \qquad (1)$$

Step 4: V_{ac} in the same way. The same steps between a and c now cancel the q_b terms:

$$V_{ac}=\frac{1}{2\pi\varepsilon_0}\left[q_a\ln\frac{D_{eq}}{r}+q_c\ln\frac{r}{D_{eq}}\right] \qquad (2)$$

Step 5: Add (1) and (2). Since $\ln(r/D_{eq})=-\ln(D_{eq}/r)$:

$$V_{ab}+V_{ac}=\frac{1}{2\pi\varepsilon_0}\left[2q_a\ln\frac{D_{eq}}{r}-(q_b+q_c)\ln\frac{D_{eq}}{r}\right]$$

The balance condition gives $q_b+q_c=-q_a$, so

$$V_{ab}+V_{ac}=\frac{3q_a}{2\pi\varepsilon_0}\ln\frac{D_{eq}}{r} \qquad (3)$$

Step 6: Bring in the phase voltage. For balanced three-phase voltages, $V_{ab}=\sqrt{3}\,V_{an}\angle 30^\circ$ and $V_{ac}=\sqrt{3}\,V_{an}\angle -30^\circ$. Their sum is

$$V_{ab}+V_{ac}=\sqrt{3}\,V_{an}\,(\angle 30^\circ+\angle -30^\circ)=\sqrt{3}\,V_{an}\times 2\cos 30^\circ=3V_{an}$$

Putting this into (3):

$$3V_{an}=\frac{3q_a}{2\pi\varepsilon_0}\ln\frac{D_{eq}}{r}\quad\Rightarrow\quad V_{an}=\frac{q_a}{2\pi\varepsilon_0}\ln\frac{D_{eq}}{r}$$

Step 7: Capacitance to neutral. Capacitance is charge per volt:

$$\boxed{C_n=\frac{q_a}{V_{an}}=\frac{2\pi\varepsilon_0}{\ln\left(\sqrt[3]{D_{12}D_{23}D_{31}}/r\right)}\ \text{F/m to neutral}}$$

With $\varepsilon_0=8.854\times10^{-12}$ F/m, this is $C_n=\dfrac{0.0556}{\ln(D_{eq}/r)}$ μF/km. For symmetrical spacing $D_{eq}=D$, and the result becomes $2\pi\varepsilon_0/\ln(D/r)$.

3. Notes and Remarks

- The term-test paper prints no mark for this question; with 10 marks in total and 5 for question 1, it is worth 5.

4. Final Concise Answer

(a) $r'=0.011682$ m; $D_s=\sqrt{r'd}=0.07643$ m; $D_{eq}=18.899$ m; $L=1.102$ mH/km; $X_L=0.346$ Ω/km per phase.
(b) $C_n=2\pi\varepsilon_0/\ln(D_{eq}/r)$ F/m, $D_{eq}=\sqrt[3]{D_{12}D_{23}D_{31}}$ (derivation in 5(d)).

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8. (a) A string of 5 insulators is connected across a 100 kV line. If the capacitance of each disc to earth is 0.1 of the capacitance of the insulator, calculate (i) the distribution of voltage on the insulator discs and (ii) the string efficiency. Marks: 5

(b) A single core cable for use on 11 kV, 50 Hz system has a conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find (i) the maximum electrostatic stress in the cable (ii) the minimum electrostatic stress in the cable (iii) the capacitance of the cable per km length (iv) the charging current. Marks: 5

8.(a) A string of 5 insulators is connected across a 100 kV line. If the capacitance of each disc to earth is 0.1 of the capacitance of the insulator, calculate (i) the distribution of voltage on the insulator discs and (ii) the string efficiency. Marks: 5

1. Theory

The basics

An overhead line hangs from a string of porcelain discs. Each disc is a capacitor, with self capacitance C, and each metal link between discs has a small shunt capacitance KC to the earthed tower. The shunt currents add up as we move towards the line, so the disc nearest the line carries the most current and takes the most voltage. **String efficiency** is the string's voltage over n times that largest disc's voltage.

The string theory is in 3(c) (equivalent circuit and why the line unit takes the most voltage) and 3(d) (efficiency). The one new step is going from three units to five.

1. The rule for any number of units (Core Idea)

Each unit has self capacitance C, and each metal link has a shunt capacitance KC to the tower. Number the units from the tower (1) to the line (5). At the link below unit k, Kirchhoff's current law says: the current in unit $k+1$ = the current in unit k + the current leaking to the tower. That link is at voltage $V_1 + \cdots + V_k$ above the tower, so

$$V_{k+1} = V_k + K\,(V_1 + V_2 + \cdots + V_k)$$

Each unit's voltage is the previous one plus K times the total voltage of all the units above it. Starting from V_1, we get every voltage as a multiple of V_1.

2. Which voltage is across the string?

The string hangs from the earthed tower and holds the line conductor, so it carries the **phase voltage** $V_L/\sqrt{3}$ (V.K. Mehta reads "a 100 kV line" this way).

2. Answer

Given: $n = 5$ units, $K = 0.1$, line voltage 100 kV, so the voltage across the string is $V = 100/\sqrt{3} = 57.735$ kV.

Step 1: Each unit voltage in terms of V_1

$$V_2 = V_1 + 0.1V_1 = 1.1\,V_1$$

$$V_3 = 1.1V_1 + 0.1(1 + 1.1)V_1 = 1.31\,V_1$$

$$V_4 = 1.31V_1 + 0.1(1 + 1.1 + 1.31)V_1 = 1.651\,V_1$$

$$V_5 = 1.651V_1 + 0.1(1 + 1.1 + 1.31 + 1.651)V_1 = 2.1571\,V_1$$

Step 2: Find V_1

$$V = V_1(1 + 1.1 + 1.31 + 1.651 + 2.1571) = 7.2181\,V_1 = 57.735 \text{ kV}$$

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$$V_1 = \frac{57.735}{7.2181} = 7.999 \text{ kV}$$

Step 3: (i) Voltage distribution

$$V_1 = 8.00, \quad V_2 = 8.80, \quad V_3 = 10.48, \quad V_4 = 13.21, \quad V_5 = 17.25 \text{ kV}$$

$$\boxed{8.00,\ 8.80,\ 10.48,\ 13.21,\ 17.25 \text{ kV (top to line unit)}}$$

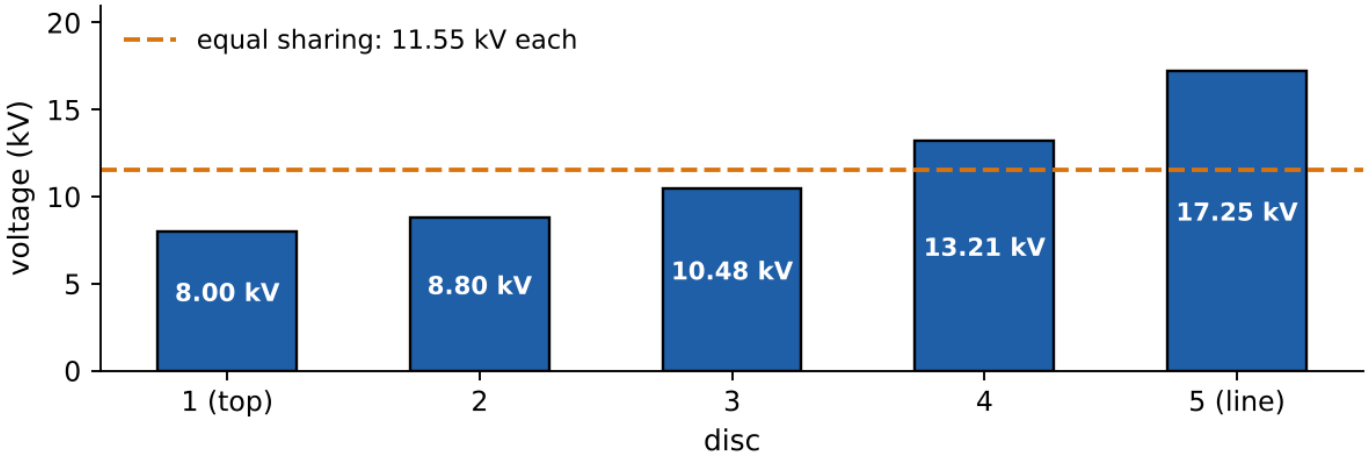


Figure 37: Voltage on each disc of the five-disc string; the dashed orange line is what each disc would take with equal sharing

The line unit carries 17.25 kV, about 30 % of the whole string voltage, while equal sharing would give 11.55 kV each.

Step 4: (ii) String efficiency

$$\eta = \frac{V}{nV_5} = \frac{57.735}{5 \times 17.254} = 0.6692$$

$$\boxed{\eta = 66.92\ \%}$$

Final Answer: 8.00, 8.80, 10.48, 13.21 and 17.25 kV from the top unit to the line unit; string efficiency 66.92 %.

3. Notes and Remarks

- **If 100 kV were across the string itself**, all voltages scale by $\sqrt{3}$: 13.85, 15.24, 18.15, 22.87 and 29.89 kV. The efficiency (66.92 %) does not change, because it depends only on K and n.
- **Compare with 3(d):** for the same $K = 0.1$, three units give 86.8 % and five units give 66.9 %: more discs, lower efficiency.

8.(b) A single core cable for use on 11 kV, 50 Hz system has a conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find (i) the maximum electrostatic stress in the cable (ii) the minimum electrostatic stress in the cable (iii) the capacitance of the cable per km length (iv) the charging current. Marks: 5

1. Theory

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1. What is a single-core cable? (Core Idea)

It is a round conductor of diameter d, inside a layer of insulation, inside a metal sheath of inner diameter D. The conductor and the sheath, with the insulation between them, form a cylindrical capacitor.

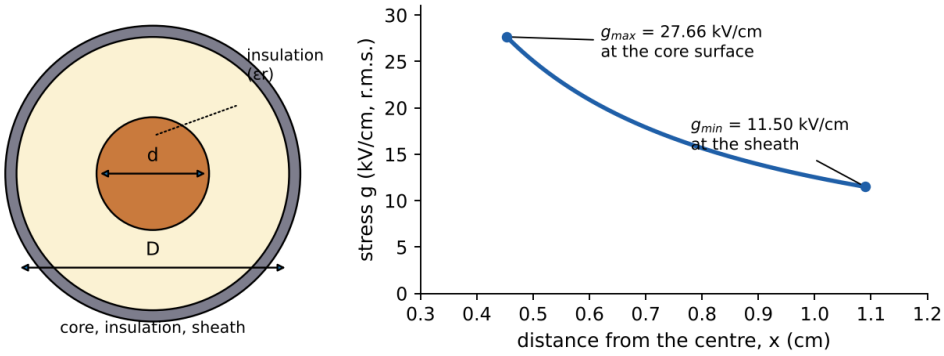


Figure 38: Stress in the insulation of a single-core cable

Reading the figure. The stress is largest at the conductor and falls with distance towards the sheath.

2. What is the stress at a point?

Let the conductor hold charge Q per metre. By Gauss's law, the stress at radius x is

$$g = \frac{Q}{2\pi\varepsilon_0\varepsilon_r x}$$

Integrating g from $d/2$ to $D/2$ gives the voltage, $V = \frac{Q}{2\pi\varepsilon_0\varepsilon_r}\ln\frac{D}{d}$. Removing Q gives

$$g = \frac{V}{x\ln(D/d)}$$

Putting $x = d/2$ and $x = D/2$ gives $g_{max} = 2V/(d\ln(D/d))$ and $g_{min} = 2V/(D\ln(D/d))$.

3. What are the capacitance and charging current?

From $C = Q/V$,

$$C = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)} \ \text{F/m}$$

An ac voltage charges and discharges this capacitance each cycle, so even an unloaded cable draws a charging current, $I_c = 2\pi f C V$, where V is the voltage from the core to the sheath.
The same cable is also worked in 2(c).

2. Answer

With $V = 11$ kV, $d = \sqrt{4\times 0.645/\pi} = 0.9062$ cm, $D = 2.18$ cm, $\ln(D/d) = 0.8778$:

- (i) $g_{max} = \dfrac{2V}{d\ln(D/d)} = \dfrac{2\times 11}{0.9062\times 0.8778} = 27.66$ kV/cm (r.m.s.)
- (ii) $g_{min} = \dfrac{2V}{D\ln(D/d)} = \dfrac{2\times 11}{2.18\times 0.8778} = 11.50$ kV/cm (r.m.s.)
- (iii) $C = \dfrac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)} = \dfrac{2\pi\times 8.854\times 10^{-12}\times 3.5}{0.8778}\times 1000 = 0.2218$ μF per km
- (iv) $I_c = 2\pi f C V = 2\pi\times 50\times 0.2218\times 10^{-6}\times 11000 = 0.767$ A per km

3. Final Concise Answer

(a) $V = 57.735$ kV; $V_k = 8.00, 8.80, 10.48, 13.21, 17.25$ kV (top to line); $\eta = 66.92$ %.

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(b) $g_{max} = 27.66$ kV/cm, $g_{min} = 11.50$ kV/cm, $C = 0.2218$ μF/km, $I_c = 0.767$ A/km (as 2(c)).

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9. (a) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. Marks: 5

(b) A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. Marks: 5

9.(a) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. Marks: 5

1. Theory

1. Why do we need a fault test? (Core Idea)

An underground cable is buried, so a damaged point cannot be seen. If its insulation fails at one point, the core touches earth there. We find the distance from one end by measuring resistance, since a uniform conductor's resistance is proportional to its length.

2. What is a Wheatstone bridge?

A bridge has four resistors in a loop, a supply across one diagonal and a galvanometer across the other. When the galvanometer reads zero, the bridge is **balanced**, and the ratio of two arms equals the ratio of the other two.

3. How does the Murray loop use it?

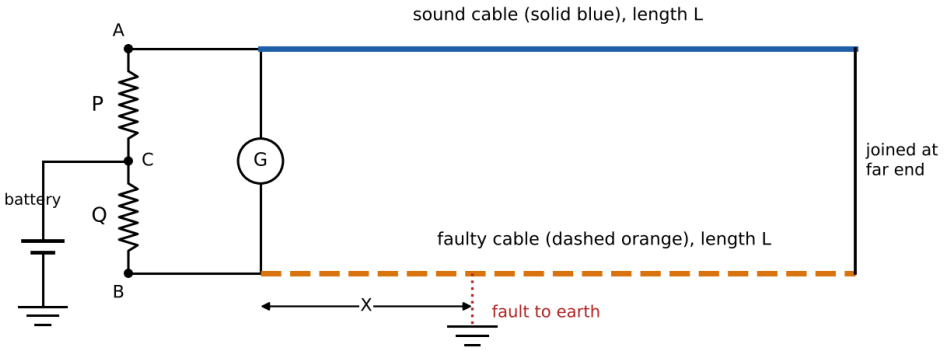


Figure 39: Murray loop test

Reading the figure. The faulty cable is joined at the far end to a sound cable, making a loop of resistance R_{loop}. The earth fault divides the loop into two parts: R_X from the testing end to the fault, and the rest. These two parts are one side of the bridge, and the adjustable resistors P and Q are the other. At balance

$$\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X}$$

So $R_X = R_{loop}\, Q/(P+Q)$, and dividing by the resistance per metre gives the distance. The test is also solved in 1(b).

2. Answer

- Faulty cable: $0.001 \times 500 = 0.5\ \Omega$; sound cable: $0.00225 \times 500 = 1.125\ \Omega$; loop $R_{loop} = 1.625\ \Omega$.
- At balance $\dfrac{P}{Q} = \dfrac{R_{loop} - R_X}{R_X} = 2.75$, so $R_X = \dfrac{1.625}{3.75} = 0.4333\ \Omega$.
- Distance $= \dfrac{0.4333}{0.001} = 433.3$ m.

Fault at 433.3 m from the testing end

9.(b) A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. Marks: 5

1. Theory

The basics

Current in one circuit makes magnetic flux, and some of that flux passes through a nearby circuit. The **mutual inductance** M is the flux linkage in the second circuit per ampere in the first. A long straight wire carrying I makes a flux between distances D_1 and D_2 of $\frac{\mu_0 I}{2\pi} \ln \frac{D_2}{D_1}$ per metre. When the flux changes in time, it induces a voltage in the second circuit, and this is how power lines disturb telephone lines.

1. The rule from 1(d) (Core Idea)

From 1(d): the flux of a power wire links the telephone loop only through the rings that pass **between** the two telephone wires. For a power line P_1P_2 (currents $+I$ and $-I$) and a telephone line T_1T_2, this gives

$$M = 2 \times 10^{-7} \ln \frac{D_{P_1T_2}\, D_{P_2T_1}}{D_{P_1T_1}\, D_{P_2T_2}} \ \text{H/m}$$

and the induced voltage $E = 2\pi f M I$.

2. Why this arrangement gives a small M

Here all four wires are in one horizontal row, and the telephone line is far away (20 m) compared with its own width (0.6 m). The flux of P_1 and the flux of P_2 through the telephone loop are almost equal and opposite, so they nearly cancel. M comes out very small.

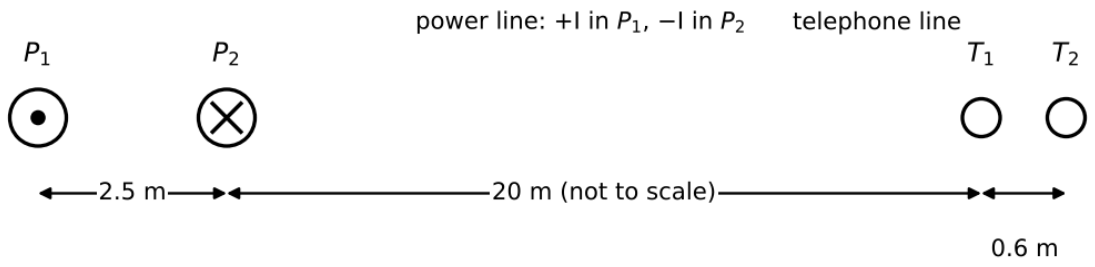


Figure 40: Power line P1, P2 and telephone line T1, T2 in one horizontal plane

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2. Answer

Given: power conductors 2.5 m apart, telephone conductors 0.6 m apart, 20 m between the nearest conductors (P_2 to T_1), all in one plane; $I = 150$ A, $f = 50$ Hz.

Step 1: Distances

Putting P_1 at 0, the positions are $P_2 = 2.5$ m, $T_1 = 22.5$ m and $T_2 = 23.1$ m:

$$D_{P_1T_1} = 22.5 \text{ m}, \quad D_{P_1T_2} = 23.1 \text{ m}, \quad D_{P_2T_1} = 20 \text{ m}, \quad D_{P_2T_2} = 20.6 \text{ m}$$

Step 2: Mutual inductance

$$\frac{D_{P_1T_2}\, D_{P_2T_1}}{D_{P_1T_1}\, D_{P_2T_2}} = \frac{23.1 \times 20}{22.5 \times 20.6} = \frac{462}{463.5} = 0.996764$$

$$M = 2 \times 10^{-7} \ln(0.996764) = 2 \times 10^{-7} \times (-0.0032415) = -6.483 \times 10^{-10} \text{ H/m}$$

The minus sign only gives the direction of the induced voltage. Per kilometre:

$$|M| = 6.483 \times 10^{-7} \text{ H/km} = 0.648 \ \mu\text{H/km}$$

$$\boxed{M = 0.648 \ \mu\text{H per km}}$$

Step 3: Induced voltage

$$E = 2\pi f |M| I = 2\pi \times 50 \times 6.483 \times 10^{-7} \times 150 = 0.0306 \text{ V/km}$$

$$\boxed{E = 30.6 \text{ mV per km}}$$

Final Answer: $M = 0.648$ μH/km; induced voltage 30.6 mV per km.

3. Notes and Remarks

- **Compare with 1(d):** there the telephone line was 2 m directly below and $M = 75.8$ μH/km. Moving the telephone line 20 m to the side reduces M more than a hundred times.

4. Final Concise Answer

(a) $R_{loop} = 1.625$ Ω, $R_X = 0.4333$ Ω; fault at 433.3 m from the testing end (as 1(b)).

(b) $D_{P_1T_1} = 22.5$, $D_{P_1T_2} = 23.1$, $D_{P_2T_1} = 20$, $D_{P_2T_2} = 20.6$ m; $M = 2 \times 10^{-7} \ln(462/463.5) \Rightarrow |M| = 0.648$ μH/km; $E = 30.6$ mV/km.

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FORMULA SHEET

EEE337

Power System II

Formula Sheet

Session 2021–22 paper

Every formula behind the Final Examination 2024 and Term Tests 1–3

TOPICS

1 Inductance of lines — 16 formulas

2 Capacitance of lines — 6 formulas

3 Corona — 5 formulas

4 Insulators — 8 formulas

5 Sag — 1 formula

6 Cables — 13 formulas

7 Economics of transmission — 2 formulas

8 Stability — 5 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

Before you use the sheet

How to read a row

Inductance	$L = \frac{\psi}{I}$	ψ = flux linkage (Wb-T/m) I = current (A)	1(c)
the name you search for	the formula, exactly as derived	every new symbol with its unit	link: opens the derivation

In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Inductance of lines 4

2 Capacitance of lines 5

3 Corona 6

4 Insulators 6

5 Sag 7

6 Cables 7

7 Economics of transmission 8

8 Stability 8

9 Key formulas 9

10 Symbol index 11

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Conventions (V.K. Mehta; Grainger & Stevenson for stability): $\mu_0 = 4\pi \times 10^{-7}$ H/m and $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m; ln is the natural log; line voltages are line-to-line unless marked V_{ph}; r is the conductor radius and D a centre-to-centre spacing (same unit); in a cable d is the core diameter and D the inside diameter of the sheath; K is the ratio of shunt to self capacitance of a string unit. References point to the solutions of the 2021–22 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Inductance of lines:

Title	Equation	Notation	Ref.
Inductance	$L = \dfrac{\psi}{I}$	ψ = flux linkage (Wb-T/m) I = current (A)	1(c)
Internal flux linkage	$\psi_{int} = \dfrac{\mu_0 I}{8\pi}$	uniform current in a solid round conductor	1(c)
External flux linkage	$\psi_{12} = \dfrac{\mu_0 I}{2\pi} \ln \dfrac{D_2}{D_1}$	D_1, D_2 = distances of two points from the conductor (m)	1(c)
★ **GMR of a solid conductor**	$r' = r\, e^{-1/4} = 0.7788\, r$	r' = geometric mean radius (m)	1(c)
Symmetrical 3-phase line	$L = 2 \times 10^{-7} \ln \dfrac{D}{r'}$ H/m	D = equal spacing of the phases (m)	1(c)
★ **Unsymmetrical transposed line**	$L = 2 \times 10^{-7} \ln \dfrac{D_{eq}}{r'} = 0.2 \ln \dfrac{D_{eq}}{r'}$ mH/km	per phase	1(c)
★ **Equivalent spacing**	$D_{eq} = \sqrt[3]{D_{12} D_{23} D_{31}}$	D_{12}, D_{23}, D_{31} = the three phase spacings (m)	1(c)
★ **Composite conductor**	$L_X = 2 \times 10^{-7} \ln \dfrac{D_m}{D_{sX}}$ H/m	D_m = mutual GMD between conductors X and Y (m) D_{sX} = self GMD of X (m)	4(c)
Mutual GMD	$D_m = \sqrt[mn]{(D_{aa'} D_{ab'} \cdots D_{am}) \cdots (D_{na'} \cdots D_{nm})}$	n, m = strands in X and Y	4(c)
Self GMD	$D_{sX} = \sqrt[n^2]{(D_{aa} D_{ab} \cdots D_{an}) \cdots (D_{na} \cdots D_{nn})}$	$D_{aa} = r'_a$, $D_{bb} = r'_b$, …	4(c)
Composite line loop	$L = 2 \times 10^{-7} \ln \dfrac{D_m^2}{D_{sX} D_{sY}}$ H/m	both sides of a single-phase line	4(c)
Two-conductor bundle	$D_s = \sqrt{r'\, d}$	d = spacing of the sub-conductors (m)	7(a)
★ **Inductive reactance**	$X_L = 2\pi f L$	f = frequency (Hz)	7(a)
★ **Mutual inductance**	$M = 2 \times 10^{-7} \ln \dfrac{D_{P_1T_2}\, D_{P_2T_1}}{D_{P_1T_1}\, D_{P_2T_2}}$ H/m	$D_{P_iT_j}$ = distance from power wire i to telephone wire j (m)	1(d)
Symmetrical arrangement	$M = 4 \times 10^{-7} \ln \dfrac{D_2}{D_1}$ H/m	D_1, D_2 = near and far distances (m)	1(d)

Title	Equation	Notation	Ref.
Induced voltage	$E = 2\pi f M I$	E = voltage in the telephone line (V/m)	1(d)

Table 1: Inductance of lines

2. Capacitance of lines:

Title	Equation	Notation	Ref.
Field of a line charge	$g = \dfrac{q}{2\pi\varepsilon_0\, x}$	q = charge (C/m) x = distance from the conductor (m)	5(d)
Potential difference	$V_{12} = \dfrac{q}{2\pi\varepsilon_0} \ln \dfrac{D_2}{D_1}$	between two points at D_1 and D_2	5(d)
Single-phase line	$C = \dfrac{\pi\varepsilon_0}{\ln(d/r)}$ F/m	d = spacing of the two wires (m)	5(a)
Effect of the earth	$C = \dfrac{\pi\varepsilon_0}{\ln\left(\dfrac{d}{r} \cdot \dfrac{2h}{\sqrt{4h^2 + d^2}}\right)}$	h = height above the earth (m)	5(a)
★ **Capacitance to neutral**	$C_n = \dfrac{2\pi\varepsilon_0}{\ln(D_{eq}/r)}$ F/m	transposed 3-phase line, per phase	5(d)
★ **Charging current**	$I_c = 2\pi f\, C_n\, V_{ph}$	V_{ph} = phase voltage (V)	5(b)

Table 2: Capacitance of lines

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3. Corona:

Title	Equation	Notation	Ref.
Field near the conductor	$g = \dfrac{V}{x \ln(d/r)}$	V = phase-to-neutral voltage (kV) d = spacing (cm)	1(a)
Surface field	$g_{max} = \dfrac{V}{r \ln(d/r)}$	largest at the conductor surface	1(a)
Strength of air	$g_0 = 30$ kV/cm peak $= 21.2$ kV/cm r.m.s.	g_0 = breakdown strength of air	1(a)
★ **Critical disruptive voltage**	$V_c = m_0\, g_0\, \delta\, r \ln \dfrac{d}{r}$ kV	m_0 = irregularity factor (1 smooth, 0.80–0.87 stranded) r = conductor radius (cm)	1(a)
Air density factor	$\delta = \dfrac{3.92\, b}{273 + t}$	b = pressure (cm of Hg) t = temperature (°C)	1(a)

Table 3: Corona

4. Insulators:

Title	Equation	Notation	Ref.
Safety factor	$\text{SF} = \dfrac{\text{puncture strength}}{\text{flash-over voltage}}$	about 10 for pin insulators	3(a)
★ **Capacitance ratio**	$K = \dfrac{C_1}{C}$	C_1 = shunt capacitance to earth (F) C = self capacitance of a unit (F)	3(c)
★ **Unit voltages**	$V_2 = V_1(1 + K),\ V_3 = V_2 + K(V_1 + V_2)$	V_1 = voltage of the top (cross-arm) unit (V)	3(c)
General unit	$V_{k+1} = V_k + K\,(V_1 + V_2 + \cdots + V_k)$	node equation at the joint below unit k	8(a)
String voltage	$V = V_1 + V_2 + \cdots + V_n$	V = phase voltage across the string (V)	3(c)
★ **String efficiency**	$\eta = \dfrac{V}{n\, V_n} \times 100\ \%$	n = number of units V_n = voltage of the unit nearest the line (V)	3(d)
Three-unit string	$\eta = \dfrac{3 + 4K + K^2}{3(1 + 3K + K^2)}$	falls as K or n grows	3(d)
K from measurements	$K = \dfrac{V_2}{V_1} - 1$	two measured unit voltages	5(c)

Table 4: Insulators

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5. Sag:

Title	Equation	Notation	Ref.
★ **Sag, level supports**	$S = \frac{w\,l^2}{8\,T}$	S = sag (m) w = weight per metre (N/m) l = span (m) T = tension (N)	4(a)

Table 5: Sag

6. Cables:

Title	Equation	Notation	Ref.
Stress in the dielectric	$g = \frac{Q}{2\pi\varepsilon_0\varepsilon_r\,x} = \frac{V}{x\,\ln(D/d)}$	Q = charge (C/m) ε_r = relative permittivity	2(c)
★ **Maximum and minimum stress**	$g_{max} = \frac{2V}{d\,\ln(D/d)},\ g_{min} = \frac{2V}{D\,\ln(D/d)}$	at the core and at the sheath	2(c)
★ **Cable capacitance**	$C = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)}$ F/m	single-core cable	2(c)
Cable charging current	$I_c = 2\pi f C V$	V = core-to-sheath voltage (V)	2(c)
★ **Insulation resistance**	$R = \frac{\rho}{2\pi l}\ln\frac{r_2}{r_1}$	ρ = resistivity ($\Omega\cdot$ cm) l = length (cm) r_1, r_2 = core and sheath radii	4(b)
★ **Most economical size**	$d = \frac{2V}{g_{max}},\ D = e\,d = 2.718\,d$	V = peak core-to-sheath voltage (kV)	6(a)
Layer stresses	$g_{1max} = \frac{Q}{\pi\varepsilon_0\varepsilon_1 d},\ g_{2max} = \frac{Q}{\pi\varepsilon_0\varepsilon_2 d_1}$	d_1 = outer diameter of layer 1 (cm)	4(d)
Two dielectrics, stresses	$\varepsilon_1 d\,g_{1max} = \varepsilon_2 d_1\,g_{2max}$	the same charge Q in both layers	4(d)
Two dielectrics, voltage	$V = \frac{g_{1max}\,d}{2}\ln\frac{d_1}{d} + \frac{g_{2max}\,d_1}{2}\ln\frac{D}{d_1}$	ε_1, ε_2 = relative permittivities of the inner and outer layers	4(d)
★ **Capacitance grading**	$V = \frac{g_{1max}\,d}{2}\ln\frac{d_1}{d} + \frac{g_{2max}\,d_1}{2}\ln\frac{d_2}{d_1} + \frac{g_{3max}\,d_2}{2}\ln\frac{D}{d_2}$	three layers; d_2 = outer diameter of layer 2 (cm)	2(d)
Equal maximum stress	$\varepsilon_1 d = \varepsilon_2 d_1 = \varepsilon_3 d_2$	every layer reaches the same g_{max}	2(d)
Graded voltage	$V = \frac{g_{max}}{2}\left[d\ln\frac{d_1}{d} + d_1\ln\frac{d_2}{d_1} + d_2\ln\frac{D}{d_2}\right]$	g_{max} = common maximum stress (kV/cm)	2(d)
★ **Murray loop balance**	$\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X},\ R_X = \frac{Q}{P+Q}R_{loop}$	P, Q = ratio arms (Ω) R_{loop} = faulty + sound core (Ω) R_X = resistance up to the fault (Ω)	1(b)

Table 6: Cables

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7. Economics of transmission:

Title	Equation	Notation	Ref.
Annual cost	$P_1 + P_2 a + \frac{P_3}{a}$	a = conductor area $P_2 a$ = interest and depreciation P_3/a = cost of energy lost	6(b)
★ **Kelvin's law**	$P_2 a = \frac{P_3}{a}$	the most economical conductor area	6(b)

Table 7: Economics of transmission

8. Stability:

Title	Equation	Notation	Ref.
★ **Power-angle equation**	$P_e = \frac{\lvert E'\rvert\lvert V\rvert}{X}\sin\delta = P_{max}\sin\delta$	E' = transient emf (pu) V = infinite-bus voltage (pu) X = transfer reactance (pu) δ = rotor angle	6(d)
Swing equation	$M\frac{d^2\delta}{dt^2} = P_m - P_e$	M = inertia constant P_m = mechanical input (pu)	6(c)
★ **Equal area criterion**	$A_1 = A_2$	A_1, A_2 = accelerating and decelerating areas; limit $\delta_{max} = 180° - \delta_1$	6(c)
★ **Transient emf**	$E' = V_t + jX'_dI$	V_t = terminal voltage (pu) X'_d = transient reactance (pu) I = current (pu)	6(d)
Transfer reactance (Y to Δ)	$X_{ab} = \frac{X_aX_b + X_bX_c + X_cX_a}{X_c}$	X_c = branch to the faulted node (pu)	6(d)

Table 8: Stability

9. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
GMR of a solid conductor	$r' = r\,e^{-1/4} = 0.7788\,r$
Unsymmetrical transposed line	$L = 2\times10^{-7}\ln\dfrac{D_{eq}}{r'} = 0.2\ln\dfrac{D_{eq}}{r'}$ mH/km
Equivalent spacing	$D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$
Composite conductor	$L_X = 2\times10^{-7}\ln\dfrac{D_m}{D_{sX}}$ H/m
Inductive reactance	$X_L = 2\pi f L$
Mutual inductance	$M = 2\times10^{-7}\ln\dfrac{D_{P_1T_2}\,D_{P_2T_1}}{D_{P_1T_1}\,D_{P_2T_2}}$ H/m
Capacitance to neutral	$C_n = \dfrac{2\pi\varepsilon_0}{\ln(D_{eq}/r)}$ F/m
Charging current	$I_c = 2\pi f\,C_n\,V_{ph}$
Critical disruptive voltage	$V_c = m_0\,g_0\,\delta\,r\,\ln\dfrac{d}{r}$ kV
Capacitance ratio	$K = \dfrac{C_1}{C}$
Unit voltages	$V_2 = V_1(1+K),\ V_3 = V_2 + K(V_1+V_2)$
String efficiency	$\eta = \dfrac{V}{n\,V_n}\times100\ \%$
Sag, level supports	$S = \dfrac{w\,l^2}{8\,T}$
Maximum and minimum stress	$g_{max} = \dfrac{2V}{d\,\ln(D/d)},\ g_{min} = \dfrac{2V}{D\,\ln(D/d)}$
Cable capacitance	$C = \dfrac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)}$ F/m
Insulation resistance	$R = \dfrac{\rho}{2\pi l}\ln\dfrac{r_2}{r_1}$
Most economical size	$d = \dfrac{2V}{g_{max}},\ D = e\,d = 2.718\,d$
Capacitance grading	$V = \dfrac{g_{1max}\,d}{2}\ln\dfrac{d_1}{d} + \dfrac{g_{2max}\,d_1}{2}\ln\dfrac{d_2}{d_1} + \dfrac{g_{3max}\,d_2}{2}\ln\dfrac{D}{d_2}$
Murray loop balance	$\dfrac{P}{Q} = \dfrac{R_{loop}-R_X}{R_X},\ R_X = \dfrac{Q}{P+Q}\,R_{loop}$
Kelvin's law	$P_2a = \dfrac{P_3}{a}$
Power-angle equation	$P_e = \dfrac{\lvert E'\rvert\lvert V\rvert}{X}\sin\delta = P_{max}\sin\delta$

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Title	Equation
Equal area criterion	$A_1 = A_2$
Transient emf	$E' = V_t + jX'_dI$

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10. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
ψ	flux linkage (Wb-T/m)	Inductance
I	current (A)	Inductance
D_1, D_2	distances of two points from the conductor (m)	External flux linkage
r'	geometric mean radius (m)	GMR of a solid conductor
D	equal spacing of the phases (m)	Symmetrical 3-phase line
D_{12}, D_{23}, D_{31}	the three phase spacings (m)	Equivalent spacing
D_m	mutual GMD between conductors X and Y (m)	Composite conductor
D_{sX}	self GMD of X (m)	Composite conductor
n, m	strands in X and Y	Mutual GMD
d	spacing of the sub-conductors (m)	Two-conductor bundle
f	frequency (Hz)	Inductive reactance
$D_{P_iT_j}$	distance from power wire i to telephone wire j (m)	Mutual inductance
E	voltage in the telephone line (V/m)	Induced voltage
q	charge (C/m)	Field of a line charge
x	distance from the conductor (m)	Field of a line charge
h	height above the earth (m)	Effect of the earth
V_{ph}	phase voltage (V)	Charging current
V	phase-to-neutral voltage (kV)	Field near the conductor
g_0	breakdown strength of air	Strength of air
m_0	irregularity factor (1 smooth, 0.80–0.87 stranded)	Critical disruptive voltage
r	conductor radius (cm)	Critical disruptive voltage
b	pressure (cm of Hg)	Air density factor
t	temperature (°C)	Air density factor
C_1	shunt capacitance to earth (F)	Capacitance ratio
C	self capacitance of a unit (F)	Capacitance ratio
V_1	voltage of the top (cross-arm) unit (V)	Unit voltages
n	number of units	String efficiency
V_n	voltage of the unit nearest the line (V)	String efficiency
S	sag (m)	Sag, level supports
w	weight per metre (N/m)	Sag, level supports
l	span (m)	Sag, level supports
T	tension (N)	Sag, level supports
Q	charge (C/m)	Stress in the dielectric
ε_r	relative permittivity	Stress in the dielectric
ρ	resistivity ($\Omega \cdot$ cm)	Insulation resistance
r_1, r_2	core and sheath radii	Insulation resistance

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Symbol	Meaning	First used in
d_1	outer diameter of layer 1 (cm)	Layer stresses
ε_1, ε_2	relative permittivities of the inner and outer layers	Two dielectrics, voltage
three layers; d_2	outer diameter of layer 2 (cm)	Capacitance grading
g_{max}	common maximum stress (kV/cm)	Graded voltage
P, Q	ratio arms (Ω)	Murray loop balance
R_{loop}	faulty + sound core (Ω)	Murray loop balance
R_X	resistance up to the fault (Ω)	Murray loop balance
a	conductor area	Annual cost
$P_2 a$	interest and depreciation	Annual cost
P_3/a	cost of energy lost	Annual cost
E'	transient emf (pu)	Power-angle equation
X	transfer reactance (pu)	Power-angle equation
δ	rotor angle	Power-angle equation
M	inertia constant	Swing equation
P_m	mechanical input (pu)	Swing equation
A_1, A_2	accelerating and decelerating areas; limit $\delta_{max} = 180° - \delta_1$	Equal area criterion
V_t	terminal voltage (pu)	Transient emf
X'_d	transient reactance (pu)	Transient emf
X_c	branch to the faulted node (pu)	Transfer reactance (Y to Δ)

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